C5: Physics of Atmospheres & Oceans

Course Notes: Thermodynamics and Radiation

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1 Overview & introduction to planetary climates

Hi everyone! Welcome to C5: Physics of Atmospheres & Oceans. Our agenda for our first class meeting today is the following:

1. Review the course format and structure (10 minutes)
2. Do a warm-up activity on fundamental planetary energy balance (20 minutes)
3. Broad-brush review of Earth's climate and other planetary climates (25 minutes)

I will make an agenda and notes handout for each class, listing the plans for the day as well as providing some further information that you can use for studying. These notes will not necessarily be comprehensive, and will instead be a summary of what is presented in class (occasionally it'll be the opposite, and they'll contain more information than we end up covering). I welcome feedback on these notes, as I consider them a living document that I will iterate on each year.

Before each lecture, I ask that you submit one (or more) questions that you have on the material either from the last lecture or the notes/reading from the upcoming lecture on Canvas. I will use these questions to better prepare each lecture, and I will often answer these questions directly during class when they are instructive for everyone.

The optional reading for Day 1: Pierrehumbert Principles of Planetary Climate (PPC), Chapter 1 (especially Sections 1.1-1.6). Please note that I am *not* following this textbook closely (or in the same order), but I think that it is the best possible reference for the material we will be covering.

1.1 Course Overview

The primary objective of this course is: *To learn how to use the basic building-blocks of physics to understand phenomena related to climate and weather on Earth and other planets, including the growing catalogue of exoplanets.* As a result, the focus of the course is to understand the fundamental physics that governs planetary climate. The course is largely Earth-centric, but these fundamental principles will be taught in a manner that can be generalised to apply to all other planets.

These course notes will cover the first fifteen lectures of C5. They are organised as follows:

1. Synopsis of planetary climates, big questions in planetary climate (1 lecture)
2. Thermodynamics of planetary atmospheres, both dry and moist ($\approx$ 5 lectures)
3. Radiative transfer of planetary atmospheres ($\approx$ 9 lectures)

Each section of these lecture notes corresponds to an individual lecture. Please note that these lecture notes do not cover the other modules of the course, including cloud physics (covered in Michelmas term), geophysical fluid dynamics (covered in both Hilary and Trinity), climate dynamics (covered in Hilary) and instrumentation (covered in Hilary).

1.1.1 Recommended textbooks

The course material for the thermodynamics and radiative transfer segment of C5 will largely follow the notes presented here rather than any individual textbook. That said, the primary recommended textbook for the thermodynamics and radiative transfer modules of C5 is Prof. Ray **Pierrehumbert's Principles of Planetary Climate (Cambridge University Press, 2010)**. Other recommended textbooks include:

1. Goody & Young, Atmospheric Radiation: Theoretical Basis (2nd edition), Oxford University Press, 1989 (highly recommended).

2. Liou, An Introduction to Atmospheric Radiation (2nd edition), Academic Press, 2002.

3. Peixoto & Oort, Physics of Climate, American Institute of Physics, 1992 (better for climate dynamics, but does include some fundamental thermodynamics and radiation).

4. Seinfeld & Pandis, Atmospheric Chemistry and Physics: From Air Pollution to Climate Change (2nd edition), Wiley, 2006 (more focused on chemistry and aerosols, but also includes relevant fundamental thermodynamics and radiation).

5. Sánchez-Lavega, An Introduction to Planetary Atmospheres, CRC Press, 2011 (good introduction to atmospheres throughout the Solar System).

6. Seager, Exoplanet Atmospheres, Princeton University Press, 2010 (comprehensive introduction to exoplanet atmospheres).

7. Heng, Exoplanetary Atmospheres: Theoretical Concepts and Foundations, Princeton University Press, 2017 (exoplanets text with more detailed descriptions of radiative transfer, scattering, and chemistry).

1.2 Energy balance activity

As a warm-up after the long summer break, please get into groups of two (or three) to work on the following questions.

1. *Why doesn't the Earth melt?* Earth receives a stellar flux of approximately $F_\star = 1360$ W m^{-2} from our Sun over a disk of area πa^2, where $a = 6371$ km is the radius of Earth. Assume Earth is an isothermal perfect sphere with a density of 5500 kg m^{-3} and a specific heat capacity of 840 J kg^{-1} K^{-1}. Write down a simple differential equation that governs the change in temperature of Earth with time, assuming that heat is mixed uniformly throughout the planet. How long would it take Earth to reach a temperature of 2500 K, approximately the melting point (liquidus) of rock?

   Answer:
   $$\frac{4}{3}\pi a^3 c_p \rho \frac{dT}{dt} = \pi a^2 F_\star \tag{1.1}$$
   Plugging in and integrating both sides, we find $\tau_{\text{heat}} = 2.3 \times 10^6$ yr $= 2.3$ Myr. This is quite short compared to the age of Earth (some ~ 4.5 Gyr). Clearly some process allows Earth to lose heat to space.

2. *What is Earth's radiatively stable temperature?* Radiation to space is the fundamental stabilising feedback that keeps Earth from melting and more generally keeps planetary temperatures at finite values. In reality, the Earth (and other planets) is in a top-of-atmosphere radiative balance between outgoing longwave radiation (OLR, i.e., thermal emission) and received stellar flux:

$$\pi a^2 F_\star (1 - \alpha) = 4\pi a^2 \sigma T_{\rm eff}^4, \tag{1.2}$$

where $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4} is the Stefan-Boltzmann constant, $T_{\rm eff}$ is the planetary emitting temperature, and α is the planetary albedo. Given the values from Question 1 above, plug in and solve for the radiative equilibrium temperature of Earth for both cases with zero albedo ($\alpha = 0$) and $\alpha = 0.3$.

Answer: The temperature with zero albedo $T_{\rm eff} = 279$ K, quite close to Earth's actual average surface temperature. Note that this estimate is a little high, as we have neglected Earth's albedo of ~ 0.3 – including the albedo, we would estimate a temperature of 255 K, which is colder than the triple point of water. In reality, Earth's atmosphere keeps the surface warmer than this no-atmosphere radiative equilibrium temperature through the greenhouse effect.

1.3 Big questions in planetary climate

There are myriad unsolved questions about planetary climate. Here are a few of the largest-scale open questions[1], to give you a sense of the scale of the problems we consider here (roughly in order of appearance in C5):

1. How do planetary atmospheres form, and what are they made of? Are atmospheres primarily inherited from the Solar nebula, built up from volatiles outgassed from the planetary interior, and/or due to the addition of volatiles from elsewhere?

2. What determines the thermal structure of planets, including the surface temperature, the vertical dependence of temperature on height, and the spatial variation of temperature with latitude and longitude?

3. How sensitive are climates to perturbations? What are the conditions under which climates can be unstable?

4. What controls the behaviour of clouds, and how do clouds affect planetary climate?

5. What are the possible atmospheric circulation regimes that can exist on a given planet, and what controls the boundaries between these regimes?

6. What are the mechanisms driving climatic structures (e.g., jet streams, Hadley cells) and weather (e.g., thunderstorms, tropical cyclones) in planetary atmospheres and oceans?

7. What can we learn about planetary climate from remote observations, including orbiting satellites (for Earth and Solar System planets) as well as telescopes (for exoplanets)?

[1] Credit: Adam Showman

1.4 Planetary climates in our Solar System and beyond

Below is a very brief survey of the climates of planets and moons in our Solar System, as well as a discussion of the potential climates of exoplanets.

1.4.1 Terrestrial bodies

There are four terrestrial bodies in our Solar System with a thick atmosphere: our own Earth, Venus, Mars, and Saturn's moon Titan. We briefly survey each of these climates as follows.

Earth

Earth is the largest terrestrial (rocky) planet in the Solar System. Some relevant quantities to note:

1. Earth mass $M_\oplus = 5.97 \times 10^{24}$ kg
2. Earth radius $R_\oplus = 6371$ km (Volumetric mean radius)
3. Rotation period, Solar (with respect to the Sun): $P_{\mathrm{rot},\odot} = 1$ day $= 86400$ sec
   Rotation period, sidereal (with respect to background stars): $P_{\mathrm{rot},\star} = 86164.1$ sec $=$ 23 hr 56 min 4.1 sec
4. Obliquity $\epsilon_\oplus = 23.5^\circ$
5. Average surface pressure $P_\oplus = 1$ bar $= 10^5$ Pa,
   average surface temperature $T_\oplus = 288.25$ K (2024)
6. Atmospheric composition: 78% N_2, 21% O_2, 0.9% Ar, 0.0425% CO_2 (August 2025), 0-1% H_2O (depending on location), 1933 ppb CH_4 (May 2025).

Earth is the only known planet with life that is affecting its present-day atmospheric composition, including the oxygen-rich nature of the atmosphere (which began ~ 2.5 Ga at the great oxygenation event, GOE, and underwent another large uptick associated with the Cambrian Explosion at ~ 550 Ma), as well as the increasing abundances of carbon dioxide and methane due to anthropogenic (human-induced) emissions.

The Earth has a temperature structure with many distinct layers, shown in Figure 1.1. This temperature structure has led to several standard definitions for atmospheric layers, which starting from the bottom are termed the *troposhere, stratosphere, mesosphere, and thermosphere*. Temperature decreases with height in the troposphere and mesosphere, and increases with height in the stratosphere and thermosphere. Curiously, the tropopause (the boundary between the troposphere and stratosphere) occurs at a pressure of ~ 0.1 bar on Earth and all Solar System planets with thick atmospheres (Robinson & Catling, 2014). Planetary scientists often also term the layer above the thermosphere the "exosphere," which is the layer where the atmosphere of a planet thins and begins to blend into space.

Figure 1.1 also shows the characteristic zonal-mean (east-west average) temperature distribution as a function of pressure and altitude. Notably, Earth has a large near-surface equator-to-pole temperature contrast due to the equator-to-pole contrast in received irradiation. This near-surface temperature contrast is the key driver of its tropospheric atmospheric

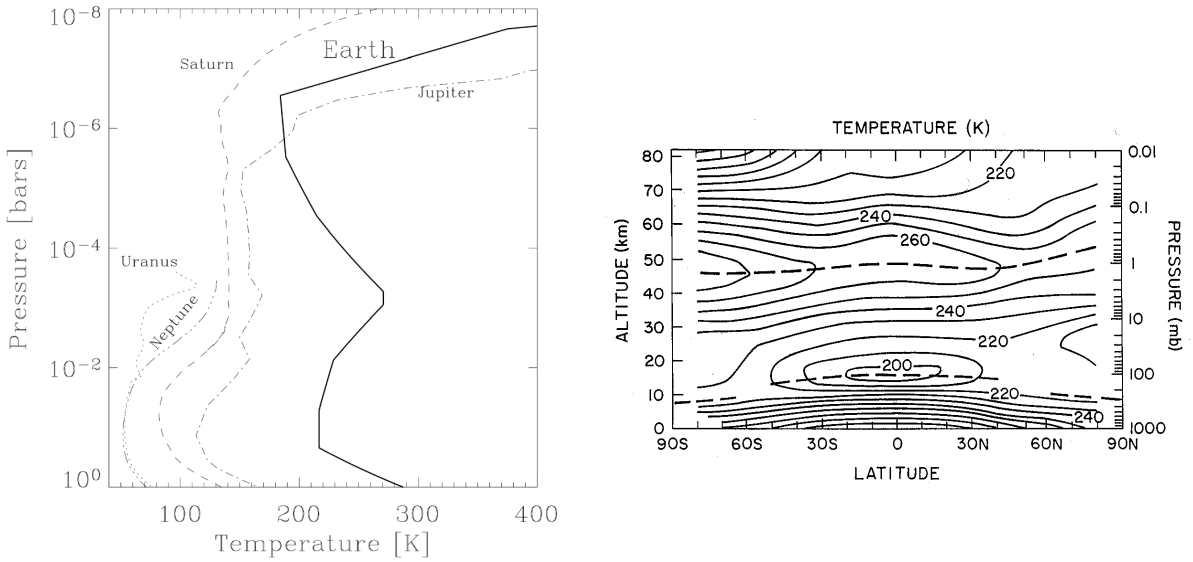


Figure 1.1: Left: Temperature-pressure profile of Earth compared to other Solar System objects with similar or cooler atmospheric temperatures. Right: Earth's zonal-mean thermal structure as a function of latitude and altitude/pressure (from Salby, 1996)

dynamics, which in turn mutes the equator-to-pole temperature contrast relative to its radiative equilibrium value – this will be covered in detail later in this course. In addition, note that Earth's troposphere is not uniformly thick – instead, the troposphere is ~ 15 km thick at the equator and ~ 8 km thick near the poles.

Figure 1.2 shows an image of the Earth taken by the DSCOVR mission from Lagrange point L1, 1.5×10^6 km from Earth. Earth is of course most notable for the "blue marble"

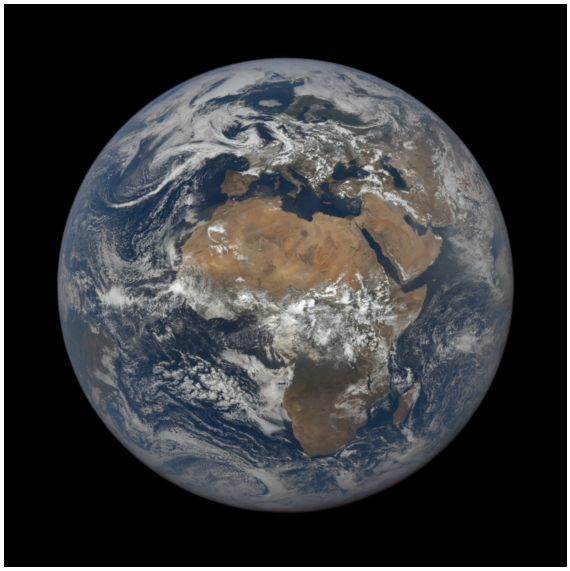


Figure 1.2: Earth from the DSCOVR mission (image taken by the EPIC instrument). Image taken from `https://epic.gsfc.nasa.gov`. One can clearly see the cloudy moist African tropics southward of the clear and dry Saharan sub-tropics.

appearance of its oceans from afar, but let's dig into this disk-resolved image in more detail. First, we can see that the tropics (especially over continental Africa) are cloudy relative to the surrounding regions (and a bit greener on the surface if you look closely). This implies a greater local amount of moisture relative to the surrounding regions (i.e., the sub-tropical

Sahara and South Africa). The reason for this is due to the Hadley circulation, in which warm moist air rises near the equator, rains out, and dries as it moves poleward, sinking and adiabatically warming at latitudes of $\sim 20 - 30^{\circ}$. The reason the Hadley cells do not extend further is because of Earth's relatively rapid rotation, as the Coriolis force prevents a parcel of gas being launched poleward from the equator to reach the pole. Instead, the Coriolis force deflects the parcel eastward (in either hemisphere), forming eastward sub-tropical jet streams in each hemisphere. In tandem, eddies (smaller-scale cyclones and anti-cyclones) form in the mid-latitudes, driven by the temperature contrast from the edge of the Hadley cell to the pole. These eddies and waves (especially Rossby waves) in turn generate a momentum flux that leads toward an eastward jet in the mid-latitudes. Earth just happens to have the correct rotation rate for these two jets (sub-tropical and eddy-driven) to merge (Kaspi & Showman, 2015), leading to Earth having one eastward mid-latitude jet in each hemisphere (see Figure 1.3).

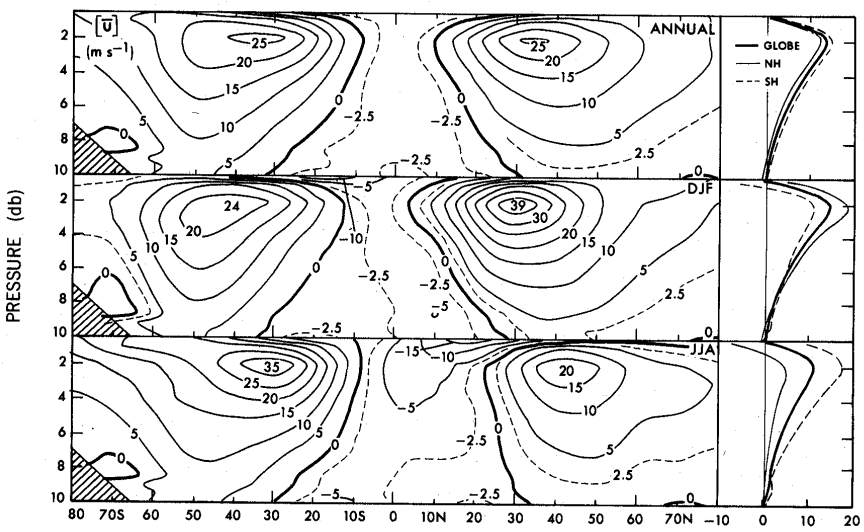


Figure 1.3: Zonal-mean zonal winds (east-west mean of the east-west winds) in Earth's atmosphere, on an annual average (top) and averaged over December-February (middle) and June-August (bottom). Figure adapted from Peixoto & Oort (1992).

Figure 1.4 shows the typical atmospheric features of an Earth-like exoplanet – note that Earth does not display all of these features, but planets like Earth with different planetary parameters (e.g., rotation rate, planetary radius, etc.) may. Specifically, the schematic displays a planet that is somewhat more slowly rotating than Earth, as it exhibits equatorial superrotation (winds at the equator are sufficiently fast that the atmosphere has greater eastward angular momentum than the solid-body rotation of the planet), similar to that in the atmospheres of the slower-rotating Venus and Titan (see the following sections for more). The Hadley cell is also slightly wider (extends to higher latitudes) than that of Earth itself, further indicative of somewhat slower rotation (Held & Hou, 1980). Finally, the subtropical and eddy-driven jets are distinct rather than merged into one, as the eddy-driven jet is preferentially formed by Rossby wave stirring at higher latitudes (Kaspi & Showman, 2015). Note also the distinction made in the diagram between the dynamical "tropics" and "extratropics." The dividing line between the tropics and extratropics typically corresponds to when the local Rossby number

$$\mathrm{Ro} = \frac{U}{fL} = 1, \tag{1.3}$$

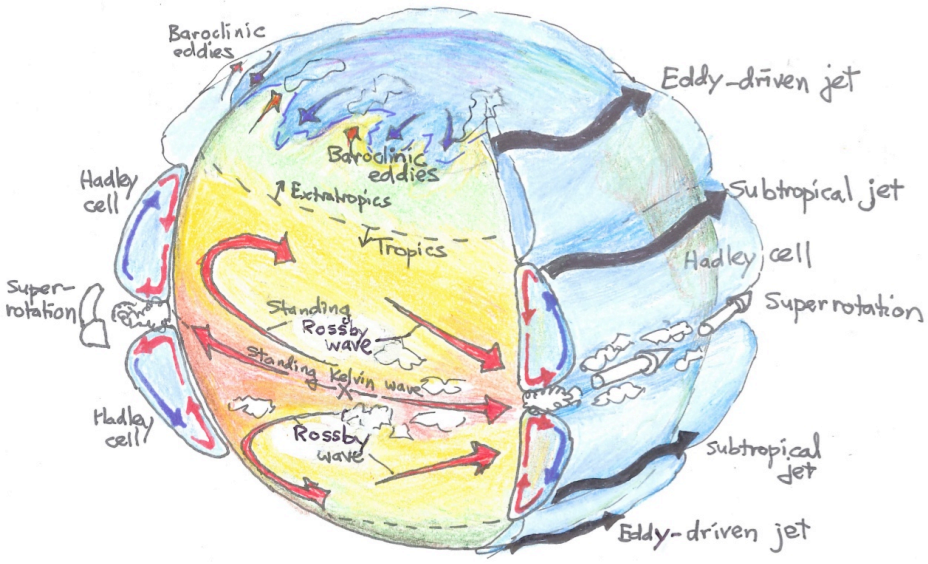


Figure 1.4: Schematic showing the key features of the atmospheric circulation of an Earth-like planet. Figure drawn by hand by Adam Showman, taken from Showman et al. (2013).

where the Rossby number represents the ratio of the advective to the Coriolis term in the momentum equation, with $f = 2\Omega \sin\phi$ the Coriolis parameter as a function of rotation rate Ω and latitude ϕ. Regions are dynamically "tropical" if Ro > 1 and the Coriolis force is weak relative to advection, and "extratropical" if Ro < 1 and the Coriolis force is strong relative to advection. Dynamically tropical regions are typically near the limit with weak temperature gradients (WTG) (Bretherton & Sobel, 2003), while the extratropics have large temperature contrasts that provide the energy to drive baroclinic eddies which in turn transport heat from the edge of the Hadley cell toward the pole.

Other terrestrial bodies: Venus, Mars, and Titan

Our Solar System has three other terrestrial bodies beyond Earth that have thick atmospheres – two are planets (Venus and Mars) and one is a moon of Saturn (Titan). Figure 1.5 shows the characteristic atmospheric properties of all three of these bodies including Earth. Venus is by far the hottest of the four – this is not simply because it is closest to the Sun (note that because of its high albedo, Venus has a *lower* equilibrium temperature than Earth). Instead, it is due to Venus having experienced a runaway greenhouse and extreme water loss (Ingersoll, 1969), as evidenced by its high atmospheric D/H ratio. Venus' atmosphere is much thicker than that of the other three objects, with a surface pressure of 92 bars and an atmosphere composed almost entirely of CO_2 (see Figure 1.6). Mars also has a CO_2-rich atmosphere, but it is much cooler than Venus due to its thin atmosphere with a surface pressure of 6 mbar. Mars has the most Earth-like meteorology of any planet in the Solar System, largely due to its similar rotation period of 24.6 hours and current obliquity of 25°. Titan has the closest atmospheric thickness and composition to Earth of any object in the Solar System, with a surface pressure of 1.47 bars and atmosphere comprised largely of N_2. However, Titan's atmosphere is much colder than Earth's due to its large mean separation of 9.5 au from the Sun, implying an equilibrium temperature sufficiently cold that its crust is comprised of water ice.

Each of these three terrestrial bodies have rich atmospheric phenomena, with entire subfields devoted to their study. For instance, Mars undergoes strong seasonal changes in its

Table 4.2 Basic atmospheric parameters for Venus, Earth, Mars and Titan.

Parameter	Venus	Earth	Mars	Titan	Reference
Mean heliocentric distance (AU)	0.723	1.000	1.524	9.543	1
Geometric albedo $A_{0,v}$	0.84	0.367	0.15	0.21	1, 3, 4
Bond albedo	0.75	0.306	0.25	0.20	1, 2, 3, 4, 5
Surface temperature (K)	737	288	215	93.7	1, 2, 3
Equilibrium temperature (K)	232	255	210	85	Calc.[a]
Exobase[b] temperature (K)	270–320	800–1250	200–300	149	2, 6, 7
Surface pressure (bar)	92	1.013	0.00636	1.47	1, 2, 3
Scale height at surface (km)	16	8.5	11	20	Calc.[c]

[a] Calculated with eq. (3.17) and $\epsilon = 1$. The globally and wavelength averaged emissivity for Earth is 0.96–0.98.
[b] A range of values is given for Venus, Earth and Mars, appropriate for a range of solar (low to high) activities.
[c] Calculated with eq. (4.2).
1: Yoder (1995), and http://nssdc.gsfc.nasa.gov/planetary/. 2: Chamberlain and Hunten (1987). 3: Fulchignoni *et al.* (2005). 4: Moroz (1983). 5: Hunten *et al.* (1984). 6: Waite *et al.* (2005). 7: Forbes *et al.* (2008).

Figure 1.5: Atmospheric physical properties of Venus, Earth, Mars, and Titan. Table taken from de Pater & Lissauer (2001).

atmospheric thickness due to the deposition (in winter) and sublimation (in spring) of CO_2 ice at its polar ice caps (see Figure 1.7). In addition, Mars regularly undergoes global-scale dust storms every few Martian years (and many more smaller-scale dust storms more regularly), as the planet is very dry given that it is too cold to presently have stable liquid water on its surface. Radiative feedbacks allow dust to interact with dynamics, as the dusty air absorbs over a broader range of wavelengths than clear air, leading to the dusty air being heated and imposing large temperature and pressure gradients that lift the dust and drive more heating. This positive feedback then can lead to rapid growth of planetary-scale dust storms.

Titan is shrouded in a layer of haze, preventing observation of the surface at visible wavelengths (see Figure 1.8). However, spacecraft observations have enabled more detailed study of the atmosphere of Titan, especially with the Cassini Saturn orbiter and Huygens entry probe. Most notably, Cassini/Huygens discovered that methane/ethane forms liquid lakes on Titan's surface, and more generally that these organics play the same role of water on Earth as a condensible species that is also stable as a liquid on the surface. Importantly, the chemical destruction timescale of methane by photolysis on Titan is predicted to be on the order of ~ 10 Myr (which in turn is expected to form the photochemical haze layer seen in visible-light images). This requires an unknown source replenishing this methane into the atmosphere, possibly deep-seated aquifers.

Though Venus is often thought of as Earth's twin, out of all the terrestrial bodies in the Solar System with thick atmospheres it has the atmosphere most dissimilar to that of Earth. Venus' atmosphere is covered in a thick layer of sulfuric acid clouds (see Figure 1.9), preventing study of the surface at visible light (like Titan). Venus further lies in a very different dynamical regime than Earth, with a thick atmosphere and a very slow (sidereal) rotation period of 243 Earth days (with a 117 Earth day Solar day). This slow rotation and thick atmosphere leads to very weak temperature gradients of only a few Kelvin in the deep atmosphere of Venus. In addition, Venus has very strong eastward (superrotating) winds at

Table 4.4 Atmospheric composition of Earth, Venus, Mars and Titan.[a]

Constituent	Earth[b]	Venus	Mars	Titan[c]	References
N_2	0.7808	0.035	0.027	~0.95	1, 2, 3, 4
O_2	0.2095	0–20 ppm	0.0013		1, 2, 5, 6
CO_2	345 ppm	0.965	0.953	10 ppb	1, 2, 4
CH_4	2 ppm		10–250 ppb	0.014–0.049[d]	2, 3, 4, 7, 8, 9
H_2O	<0.03[f]	30 ppm[e]	<100 ppm[f]	0.4 ppb	1, 10, 11
H_2O_2	1 ppb		18 ppb		2, 12
Ar	0.009	70 ppm	0.016	28 ppm	2, 3, 4, 8
CO	0.2 ppm	20 ppm[e]	700 ppm	45 ppm	2, 11, 13
O_3	~10 ppm[g]		0.01 ppm		1, 2
C_2H_2	8.7 ppb			189 ppm	14, 15
C_2H_6	13.6 ppb			121 ppm	14, 15
C_3H_8	18.7 ppb			0.6 ppm	14, 15, 16
C_2H_4	11.2 ppb			40 ppm	14, 15
C_3H_4				3.9 ppm	14
NO	<0.5 ppb		3 ppm		2
N_2O	0.35 ppm				1
SO_2	<2 ppb[f]	100 ppm			11, 15
H_2SO_4		1–2.5 ppm			17
H_2	0.5 ppm		10 ppm	0.004	2, 14
HCl		0.1 ppm[h]			11
HF		2 ppb[h]			11
COS		4 ppm			11
He	5 ppm	12 ppm			2
Ne	18 ppm	7 ppm	2.5 ppm	<0.01	2
Kr	1 ppm	0.2 ppm	0.3 ppm	<0.1 ppb	2, 5, 8
Xe	0.09 ppm	<0.1 ppm	0.08 ppm	<0.1 ppb	2, 8

Figure 1.6: Atmospheric composition of Venus, Earth, Mars, and Titan. Table taken from de Pater & Lissauer (2001).

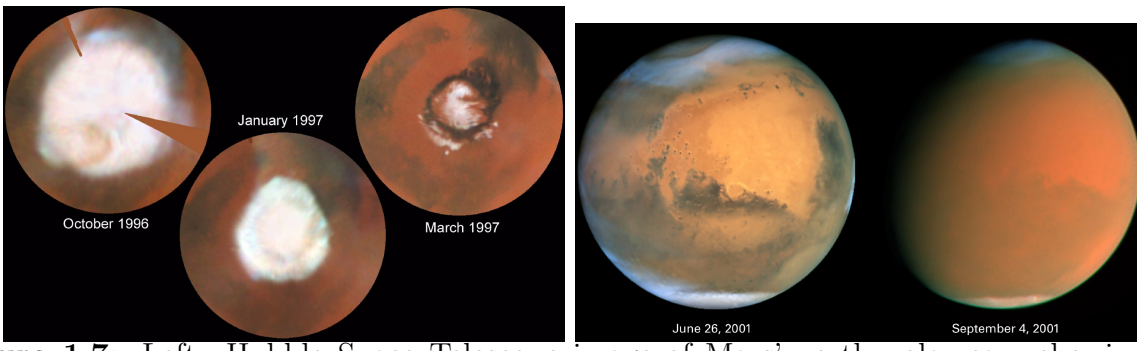


Figure 1.7: Left: Hubble Space Telescope image of Mars' north polar cap, showing its maximum and minimum extent as the CO_2 polar cap sublimates when winter transitions to spring. Right: The onset of a global dust storm on Mars in 2001, also observed with Hubble.

the cloud level, as shown by the "V feature" in the left-hand panel of Figure 1.9. Venus' atmospheric structure and circulation is (in my opinion) a bridge between terrestrial bodies with relatively thin atmospheres and more massive planets with thick atmospheres.

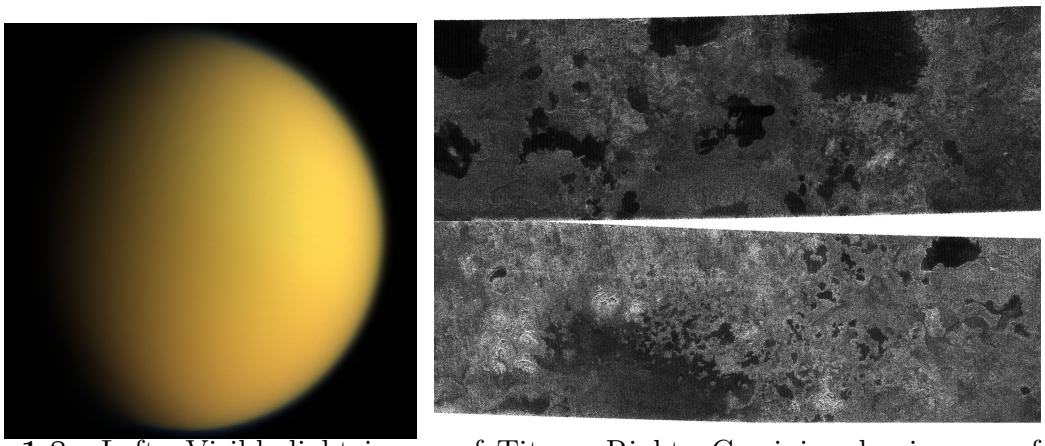


Figure 1.8: Left: Visible-light image of Titan. Right: Cassini radar images of Titan's surface, showing dark features that are likely lakes of ethane/methane on the surface.

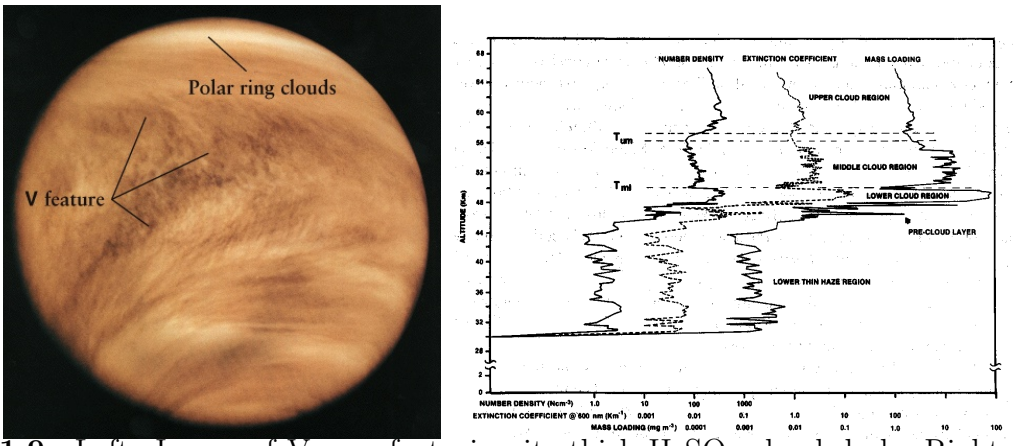


Figure 1.9: Left: Image of Venus, featuring its thick H_2SO_4 cloud deck. Right: Vertical structure of Venus' cloud decks, adapted from Esposito et al. (1983).

1.4.2 Gas and ice giants

Our Solar System has four planets beyond the asteroid belt which are completely fluid and have no observable solid surface: Jupiter ($11.2R_\oplus$), Saturn ($9.4R_\oplus$), Uranus ($4.0R_\oplus$), and Neptune ($3.9R_\oplus$). Figure 1.10 shows the characteristic atmospheric properties of the Solar System gaseous planets. The atmospheres of all these planets are dominated by H_2 and He inherited from the Solar nebula. However, each of these atmospheres are differentially enhanced in heavy elements with respect to our Sun.

Figure 1.11 shows elemental abundances of various species on Jupiter, Saturn, Uranus, and Neptune. One can see that Jupiter has the lowest abundance ratios of $\sim 3\times$ Solar for all species, the species observed for Saturn are $\sim 10\times$ Solar, and carbon on Uranus and Neptune (inferred from CH_4) is $\sim 100\times$ Solar. Note that these abundance ratios are often referred to as the "metallicity" with respect to Solar. Thus, though the atmospheres of all four planets are largely H_2, the bulk composition of Uranus and Neptune is distinct from that of Jupiter and Saturn. The enrichment of astrophysical "ices" (e.g., carbon-bearing, nitrogen-bearing, and oxygen-bearing species like CH_4, NH_3, and H_2O) in Uranus and Neptune (not just from the atmospheric abundances but also their densities and moments of inertia) is why they are

Parameter	Jupiter	Saturn	Uranus	Neptune	References
Mean heliocentric distance (AU)	5.203	9.543	19.19	30.07	1
Geometric albedo ($A_{0,v}$)	0.52	0.47	0.51	0.41	1
Geometric albedo ($A_{0,ir}$)	0.274 ± 0.013	0.242 ± 0.012	0.208 ± 0.048	0.25 ± 0.02	3
Bond albedo	0.343 ± 0.032	0.342 ± 0.030	0.290 ± 0.051	0.31 ± 0.04	3
Phase integral	1.25 ± 0.10	1.42 ± 0.10	1.40 ± 0.14	1.25 ± 0.10	3
Effective temperature (K)	124.4 ± 0.3	95.0 ± 0.4	59.1 ± 0.3	59.3 ± 0.8	2
Equilibrium temperature (K)	110	81	58	46	Calc.[b]
Temperature ($P = 1$ bar) (K)	165.0	134.8	76.4	71.5	5
Tropopause temperature (K)	111	82	53	52	5
Mesosphere temperature (K)	160–170	150	140–150	140–150	4
Exobase temperature (K)	900–1300	800	750	750	6, 7
Tropopause pressure (mbar)	140	65	110	140	5
Scale height (at 1 bar) (km)	24	47	25	23	Calc.[c]
Adiabatic lapse rate (K/km)	1.9	0.84	0.85	0.86	4
Energy balance[a]	1.67 ± 0.09	1.78 ± 0.09	1.06 ± 0.08	2.61 ± 0.28	2

[a] Ratio (energy radiated into space)/(solar energy absorbed).

[b] Calculated with eq. (3.17) and $\epsilon = 1$.

[c] Calculated with eq. (4.2).

1: Yoder (1995). 2: Hubbard *et al.* (1995). 3: Conrath *et al.* (1989b). 4: Chamberlain and Hunten (1987). 5: Lindal (1992). 6: Atreya (1986). 7: Bishop *et al.* (1995).

Figure 1.10: Atmospheric properties of Jupiter, Saturn, Uranus, and Neptune. Table taken from de Pater & Lissauer (2001).

often termed "ice giants" while the lower-metallicity Jupiter and Saturn are referred to as "gas giants."

Figure 2.3 in the following Chapter shows temperature-pressure profiles of the gas and ice giants compared to the terrestrial bodies with thick atmospheres. As for the terrestrial bodies, the gas and ice giants have tropospheres at depth that transition to stratospheres at a pressure of ~ 0.1 bar. Additionally, the gas and ice giants show relatively small latitudinal temperature variations at the cloud-tops of only a few K, unlike Earth and Mars which have large thermal variations due to the equator-to-pole irradiation contrast.

The circulation of the gas and ice giants is further distinct from one another. Figure 1.12 shows the zonal-mean zonal wind inferred from cloud tracking observations of each planet[2]. Jupiter and Saturn both show equatorial superrotation, while Uranus and Neptune are sub-rotating. Jupiter and Saturn's atmospheric circulation is banded, with alternating east-west jet streams. This banding on Jupiter corresponds to the alternating "belt and zone" structure due to less cloudy and darker belts and cloudier and brighter zone regions on Jupiter, where the clouds are largely comprised of NH_3, NH_4SH, and H_2O. Meanwhile, Uranus and Neptune have broad mid-to-high latitude jets, and their darker colour is largely due to the presence of CH_4 vapour with white CH_4 clouds overlain. The reason for the banded structure of these objects is due to their rapid rotation, which range from ~ 10 hours for Jupiter and Saturn

[2]Note that the rotation rate of each planet is typically inferred from spacecraft measurements of the time-variation of the magnetic field as it precesses, but the magnetic field of Saturn is well-aligned with its rotation axis, preventing a tight measurement of the rotation rate. Planetary scientists have used multiple methods to estimate the correct rotation rate of Saturn, leading to different resulting wind speeds which are measured with respect to the background rotation profile.

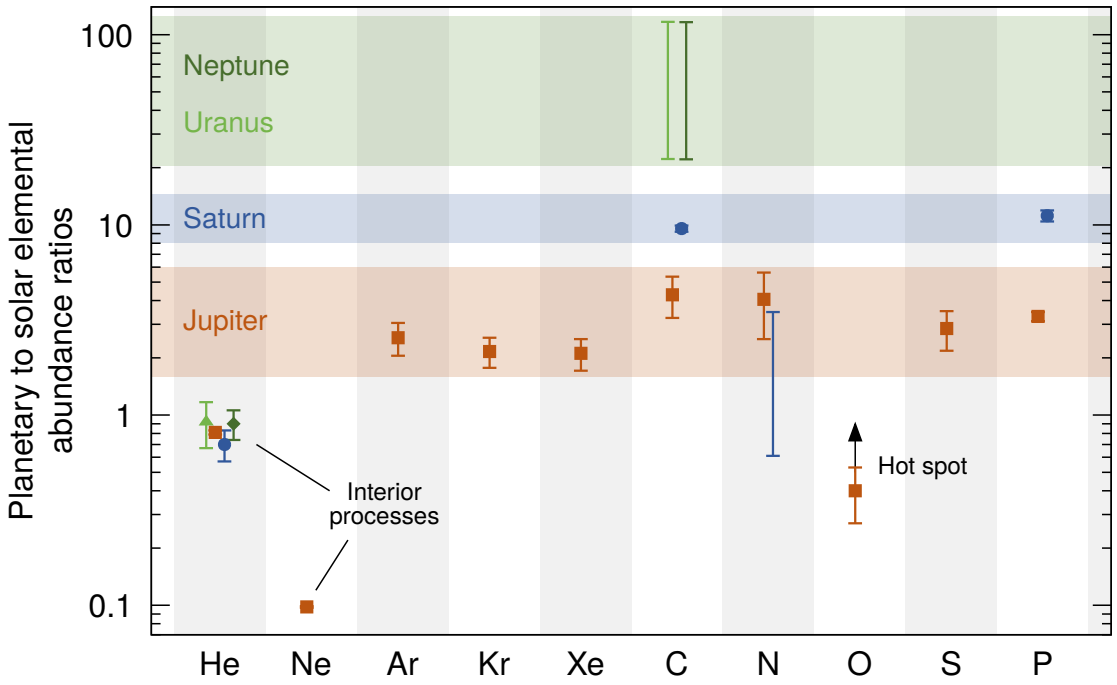


Figure 1.11: Elemental abundances of various species in the atmospheres of Jupiter, Saturn, Uranus, and Neptune. Measurements are with respect to Solar abundance of each species. Elemental abundances for Jupiter are largely from in-situ observations with Galileo, while those for Uranus and Neptune are solely through spectroscopic observations of CH_4. However, note that CH_4 condenses in the atmospheres of Uranus and Neptune, complicating this measurement and increasing its uncertainty. Figure taken from Mousis et al. (2018).

to $\sim$ 17 hours for Uranus.

1.4.3 Exoplanets

Our Milky Way Galaxy is teeming with planets. To date, astronomers have discovered 6,028 exoplanets (see Figure 1.13, value updated as of 11 Oct. 2025). Note that the population shown in Figure 1.13 is *not* the intrinsic population of exoplanets in our near. Instead, it is a biased sample due to the fact that it is easier for astronomers to detect certain types of exoplanets. For instance, the radial velocity and transit methods (the two most common methods with detections shown in the purple and teal colours in Figure 1.13) are both biased towards finding close-in and massive or large planets. In addition, the direct imaging method (taking an image of the planet by separating its light from that of the host star) is biased toward wide-separation and large planets. These three methods provide the detection of planets which are the best for follow-up characterisation – out of these, only direct imaging can (at present) find planets that are comparable to planets in our Solar System (Jupiter and Saturn in this case).

The climates of close-in planets are expected to be quite distinct from that of planets in our Solar System. This is because close-in planets with separations $\lesssim 0.1$ au are expected to be spun-down through tidal interactions with their host star to lie in a spin-synchronous state where their rotational period is equal to their orbital period (Guillot et al., 1996). These "tide-locked" planets thus have a permanent dayside which always faces their host

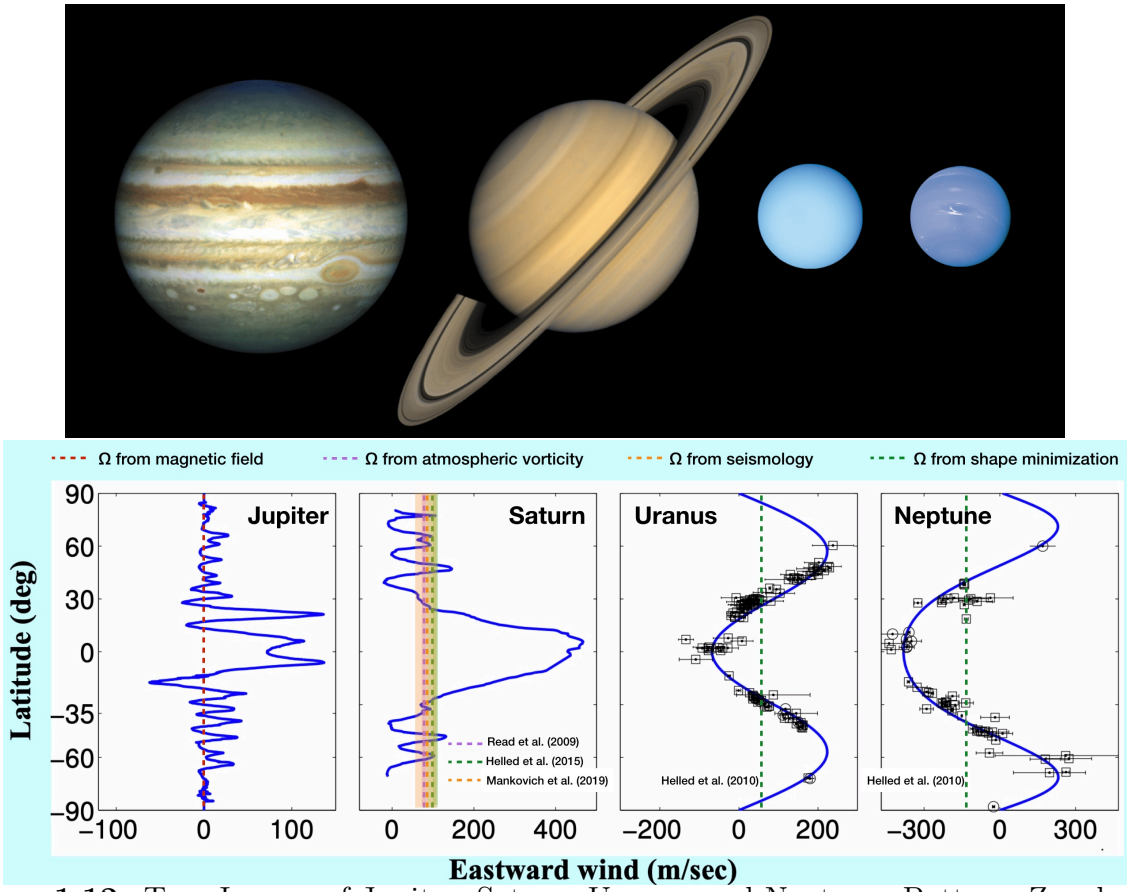


Figure 1.12: Top: Images of Jupiter, Saturn, Uranus, and Neptune. Bottom: Zonal-mean zonal winds (east west average of the east west wind) of the Solar System gas and ice giant planets from cloud tracking data. Jupiter and Saturn have superrotating equatorial regions with alternating east-west banded jets at higher latitudes, while Uranus and Neptune have sub-rotating equatorial regions and one broad eastward jet in each hemisphere. Figure taken from Guillot (2019).

star and a permanent nightside looking out on the darkness of space, which greatly affects their atmospheric circulation (for a detailed review, see Pierrehumbert & Hammond, 2019).

The hottest and largest class of transiting planets are termed "hot Jupiters," which by definition are gas giant planets with equilibrium temperatures $\geqslant 1000$ K. These are the best-observed close-in planets due to their hot temperatures and large radii enabling multiple types of observations to characterise their atmospheric properties. Figure 1.14 shows a schematic of the atmospheric circulation of these planets, alongside an early observation of the hot Jupiter HD 189733b compared to the first general circulation model (GCM) simulation of a hot Jupiter conducted in 2002. Hot Jupiters have large radiative equilibrium day-night temperature contrasts, which lead to large day-to-night pressure gradients that must be balanced by flow on the order of km/s (Showman & Guillot, 2002). This flow manifests in typical simulations of hot Jupiters as a planetary-scale standing wave pattern and superrotating equatorial jet (Showman & Polvani, 2011). The equatorial jet Doppler-shifts the planetary-scale wave pattern, leading to an eastward shift of the hottest point

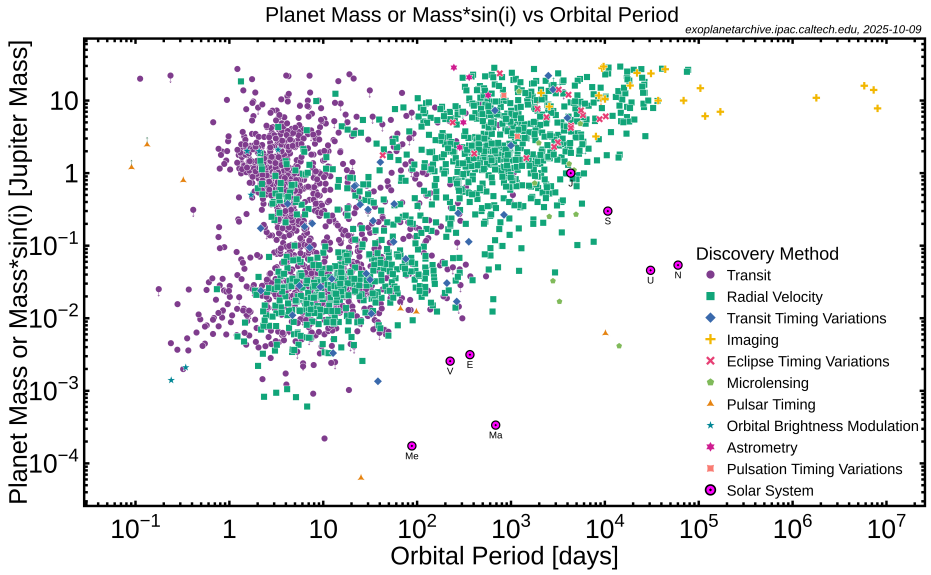


Figure 1.13: The exoplanet census as of October 2025. Note that the y-axis shows either the true planet mass, estimated planet mass, or true planet mass times the unknown sine of the inclination depending on the detection methods used to characterise a given planet (shown by the colour of each point). The Solar System planets are shown for comparison.

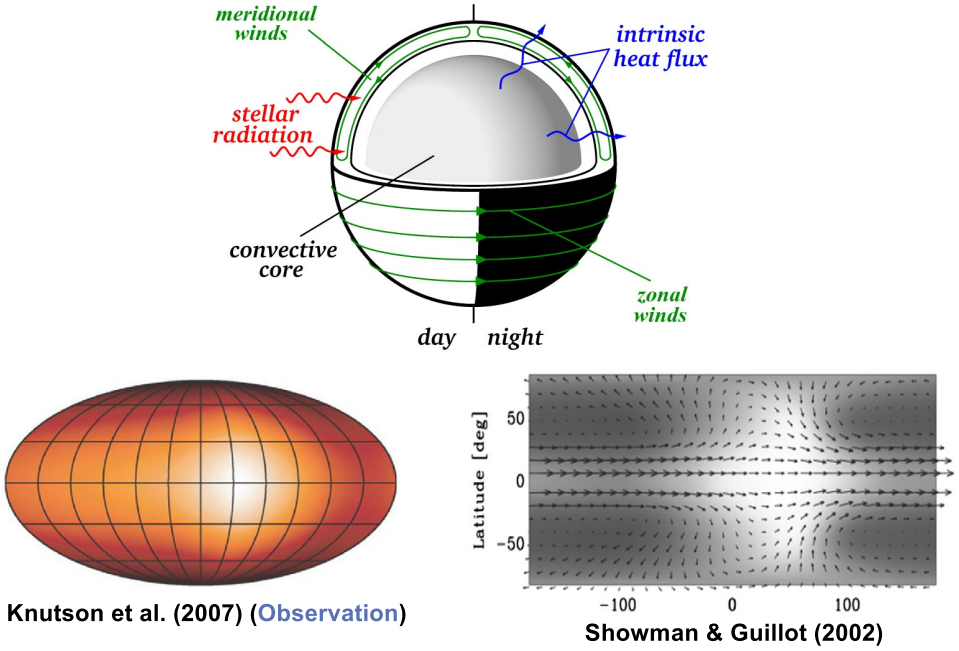


Figure 1.14: Top: Schematic of a hot Jupiter, with the host star on the left. Figure adapted from (Showman & Guillot, 2002). Bottom: Observed phase curve map of a hot Jupiter from the Spitzer Space Telescope (left) compared to a first-principles 3D general circulation model prediction of the atmospheric circulation of a hot Jupiter. Figure adapted from Showman et al. (2020).

downwind from the substellar point (Hammond & Pierrehumbert, 2018). This eastward hot spot offset has been observed for many hot Jupiters (Bell et al., 2021; Dang et al., 2025), indicative of strong atmospheric flow.

Another class of planet of interest are close-in rocky planets orbiting small, cool (M dwarf) stars. Due to the dimmer host stars, these close-in planets with orbital periods of days to tens of days can have equilibrium temperatures comparable to Earth. However, because of their close-in orbits, they are likely tide-locked to their host stars similarly to hot Jupiters. Figure 1.15 shows an example GCM simulation of a tide-locked temperate rocky planet compared to an equivalent simulation of a temperate rocky planet with a host star, orbit, and rotation period equivalent to Earth (but with no seasons). The key differences

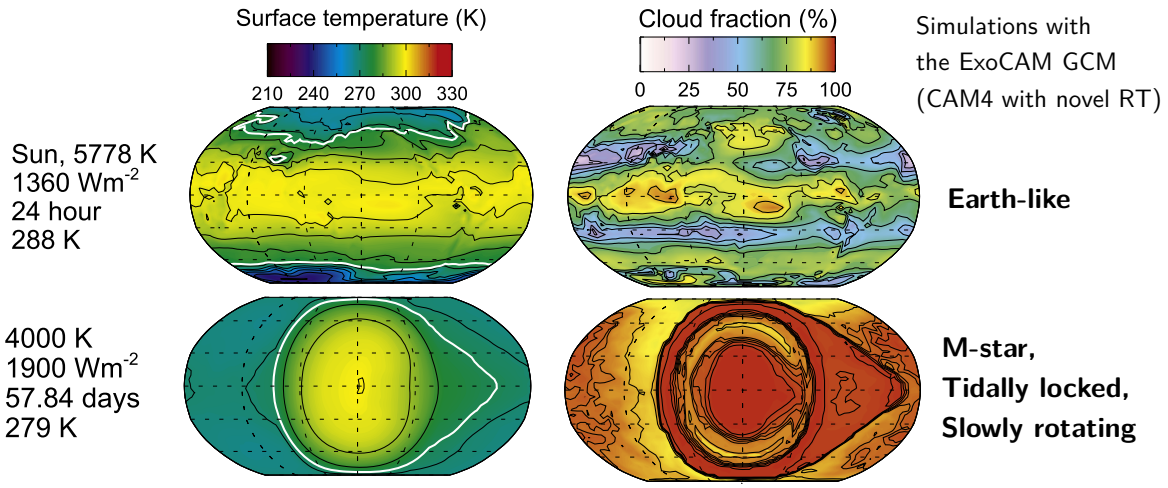


Figure 1.15: 3D general circulation model simulation of the atmosphere of an Earth-like planet with no seasons (top) compared to that of an otherwise equivalent planet with a tide-locked orbit around a smaller, cooler star with an effective temperature of only 4000 K. The left-hand panels show surface temperature, and the right-hand panels show cloud fraction. Figure adapted from (Kopparapu et al., 2017).

are that 1) in the tide-locked case, the radiative equilibrium temperature gradient driving the circulation is from day to night (sub-stellar to anti-stellar point) rather than equator to pole, 2) the rotation period of the tidally locked case is much longer, leading to a greater planetary Rossby number and weak influence of rotation on the flow. As a result, the tide-locked planet is dynamically all "tropics," with weak temperature gradients and vigorous convection throughout the dayside (rather than in an equatorial band as in the more Earth-like case shown on top). This leads to a high dayside cloud fraction in the tide-locked case, increasing the planetary albedo and cooling the global-mean temperature relative to the Earth-like case. It has been proposed that this stabilising cloud feedback on tide-locked planets could expand the habitable zones (i.e., the region around a given star where liquid water could be stable on the surface of an Earth-like planet) of M dwarf stars (Yang et al., 2013). Conversely, perhaps many rocky planets orbiting M dwarfs never underwent water condensation into surface oceans due to the bright pre-main-sequence phase of their host stars (Turbet et al., 2023).

2 Dry thermodynamics: atmospheric statics

Reading for Day 2: Pierrehumbert Principles of Planetary Climate (PPC), Sections 2.1-2.2, 2.5. Just a reminder that I am *not* following this textbook closely (or in the same order), but I think that it is the best possible reference for the material we will be covering. I think it is especially useful to have multiple viewpoints on the same topic – mine is more that of a planetary scientist[3], while Prof. Pierrehumbert has a deeper view encompassing the legacy of Earth climate alongside an understanding of the physical principles.

2.1 Thermodynamic description of air

2.1.1 Knudsen number

For most atmospheric phenomena of interest in this course, we are focused on scenarios where the relevant length scales are much larger than the typical mean free path of atmospheric species. This can be quantified with the Knudsen number,

$$\mathrm{Kn} = \frac{\lambda}{\mathcal{L}}, \tag{2.1}$$

where $\lambda = 1/(n\sigma)$ is the mean free path of the gas with n the number density and σ the elastic scattering (collision) cross section, and $\mathcal{L}$ is a typical lengthscale of interest. In our classroom, $n \sim 10^{25}\mathrm{m}^{-3}$ and typically $\sigma \sim 10^{-19}\mathrm{m}^2$ – as a result $\lambda \sim 10^{-6}$ m $= 1\mu$m. Compare this to any lengthscale we might be interested in considering atmospheric flows – whether the motions of clouds (that you'll learn from Prof. Stier), or large-scale atmospheric dynamics including jets and waves (that you'll learn from Prof. Woolings), and you will note that $\mathrm{Kn} \ll 1$. This implies that we can describe the atmosphere for our purposes with the continuum equations of mechanics and thermodynamics, and I will do so throughout these notes.

2.1.2 Equation of state

An equation of state is a relationship between thermodynamic variables, typically including pressure, temperature, and density. The ideal gas law generally suffices as an equation of state for atmospheres[4], which relates pressure p, temperature T, and number density (in moles per volume) as

$$p = nR_uT, \tag{2.2}$$

where $R_u = 8.314$ J mol^{-1} K^{-1} is the universal gas constant. Often, atmospheric scientists define a specific gas constant $R = R_u/\mu$, where μ is the mean molecular weight (mass per mole) of the atmospheric mixture of interest. Using the specific gas constant, we can write the ideal gas law as

$$p = \rho RT, \tag{2.3}$$

[3]Planetary scientists typically are trained to have a very diverse skillset encompassing planetary physics, chemistry, and geology. I am knowledgeable and able to hold discussions on many topics, but an actual expert in very few.

[4]Note that it does not suffice at sufficiently high pressures, but for the purposes of these notes we will differentiate the "atmosphere" from the "interior" through the use of standard approximations that can be used to study Earth's atmosphere.

Atmosphere	μ (kg/mole)	**R** (J/kg K)
Earth	0.029	287
Jupiter	0.0023	3600
CO_2	0.044	189

Table 1: Mean molecular weights and specific gas constants for common atmospheres.

where here ρ is the mass density (mass per volume) of the air. An equivalent form you may see in atmospheric contexts uses the specific volume $v = 1/\rho$,

$$pv = RT. \tag{2.4}$$

Table 1 shows the mean molecular weight and specific gas constant for some specific atmospheric cases of interest.

Note that individual components of air each obey their own ideal-gas type relationship

$$p_i = \rho_i R_i T, \tag{2.5}$$

where p_i is the partial pressure, which is the pressure that component i would have in isolation at the same density and temperature as the mixture, and ρ_i and R_i are similarly the density and specific gas constant of component i (and one can also define the number density of species i as $n_i = p_i/RT$). The total pressure is then simply the sum of all partial pressures,

$$p = \sum_i p_i. \tag{2.6}$$

2.1.3 Expressing concentrations

Because air is compressible, the absolute concentrations (i.e., densities, ρ_i or n_i) can change even if the number of moles of a given species within an air parcel is fixed[5]. As a result, we typically express the concentration of a species i relative to the total concentration.

The simplest way to express the relative concentration of a species is the mole fraction,

$$f_i \equiv \frac{n_i}{n}. \tag{2.7}$$

However, the mole fraction is not a useful quantity when taking into account condensation or other processes that change the relative abundances of species in air (e.g., when water vapour rains out on Earth, then the mole fraction of N_2, O_2, CH_4, CO_2, and all other trace species also changes). We can deal with this by using dry air as our reference point, rather than to the entire mixture. This is the mixing ratio, and we can express this either in mass

$$q_i = \frac{\rho_i}{\rho_d} = \frac{\mu_i}{\mu_d}\frac{n_i}{n_d} = \frac{\mu_i}{\mu_d}\frac{p_i}{p_d}, \text{ mass mixing ratio,} \tag{2.8}$$

where for the last equivalence we used the ideal gas law, or in volume (equivalently number)

$$\xi_i = \frac{n_i}{n_d}, \text{ volume mixing ratio,} \tag{2.9}$$

where in the above subscript d refers to the dry component of the mixture.

[5]Note that the number of moles could of course also change due to condensation or chemical reactions.

2.2 Hydrostatic balance

2.2.1 Derivation

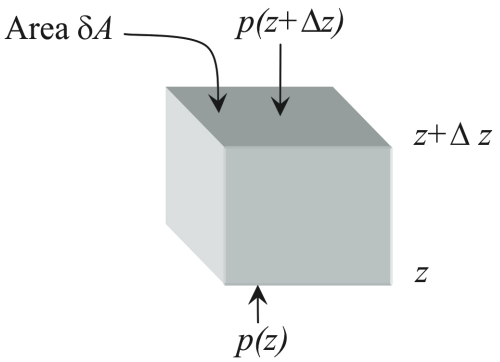


Figure 2.1: Schematic of a parcel of gas in hydrostatic equilibrium.

Figure 2.1 shows a parcel of gas held between z and $z+\Delta z$ at pressure $p(z)$ at the bottom and $p(z+\Delta z)$ at the top. Newton's second law ($F = ma$) implies that the change in pressure across Δz is the weight per area of the gas:

$$[p(z+\Delta z) - p(z)]\delta A = -\rho g \Delta z \delta A \ , \tag{2.10}$$

where $g = GM_p/(R_p+z)^2 \approx GM_p/R_p^2$ is the gravity, typically assumed to be constant in the thin-atmosphere approximation[6], and G is the universal gravitational constant. Thus,

$$\frac{p(z+\Delta z) - p(z)}{\Delta z} = -\rho g \ , \tag{2.11}$$

and given that

$$\lim_{\Delta z \to 0} \frac{p(z+\Delta z) - p(z)}{\Delta z} = \frac{dp}{dz} \ , \tag{2.12}$$

we can write the expression for hydrostatic balance

$$\frac{dp}{dz} = -\rho g \ . \tag{2.13}$$

Importantly, note that Equation (2.13) holds well to very good approximation *locally* on Earth as well as globally – this is important not just for its vertical structure but also geophysical fluid dynamics, as will be covered in Prof. Woolings' section of this course.

2.2.2 Implications of hydrostatic balance

Note that if we integrate hydrostatic balance (Equation 2.13) vertically assuming constant gravity, we find that the surface density (i.e., mass per area) in a given atmosphere column increases with pressure and decreases with gravity

$$\frac{\text{mass}}{\text{area}} = \Sigma = \frac{p}{g}. \tag{2.14}$$

[6]Note that this approximation breaks down for planets with thick envelopes, but is suitable for most models of Earth.

We can further substitute the ideal gas law (Equation 2.3) into the expression for hydrostatic equilibrium and integrate to find

$$\begin{aligned} \frac{1}{p}\frac{dp}{dz} &= -\frac{g}{RT} \\ \ln\left(\frac{p}{p_0}\right) &= -\int_{z_0}^{z}\frac{g}{RT}dz' \\ p = p_0\ \exp\left(-\int_{z_0}^{z}\frac{g}{RT}dz'\right) &= p_0\ \exp\left(-\int_{z_0}^{z}\frac{dz'}{H}\right), \end{aligned} \tag{2.15}$$

where $H = RT/g$ is the pressure scale height. For an isothermal atmosphere, the expression for pressure on height from hydrostatic equilibrium simplifies to (where here p_0 is the surface pressure where $z = 0$)

$$p = p_0 e^{-z/H}, \tag{2.16}$$

and thus the pressure (and density, for an isothermal atmosphere) decreases with increasing height over a characteristic e-folding distance of H. As a result, H is often termed the "isothermal" scale height, as it indeed serves as a characteristic e-folding distance in the vertical in the isothermal limit[7].

2.2.3 Hydrostatic balance activity: the limits of hydrostatic equilibrium

Please get in groups of two or three to work on the following activity thinking about what happens to a hydrostatic atmosphere at high z (or r in spherical geometry).

1. First, write down the limit as $z \to \infty$ from the expression of hydrostatic balance assuming constant gravity from Equation (2.16). What is wrong about using this expression at high z?

   Answer: $p \to 0$, but g varies with z.

2. Derive a more realistic expression for the dependence of pressure on height (or radius from the centre of the planet r, which is an easier quantity to use for this derivation) by not assuming constant gravity and instead taking $g = \frac{GM_p}{r^2}$ (still assuming an isothermal atmosphere). Take the limit of this expression as $r \to \infty$ – what do you find for the pressure at infinity? Does it make sense, and if not what physics might be missing from the problem?[8]

   Answer: Substituting for gravity into hydrostatic balance, we find

   $$d\ \ln p = -\frac{GM_p}{RT}\frac{dr}{r^2}. \tag{2.17}$$

   Integrating from p_0 to p and r_0 to r, we find that

   $$p = p_0 e^{\left[-\frac{GM_p}{RT}\left(1-\frac{r_0}{r}\right)\right]}. \tag{2.18}$$

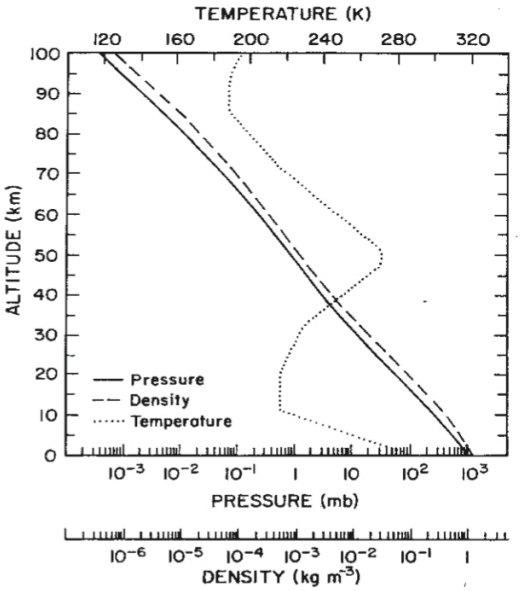
   As $r \to \infty$, $p \to p_0 e^{-\frac{GM_p}{RT}}$, not zero! The solution to this is to include a hydrodynamic (Parker) wind and solve for the full momentum equation.

[7] If one is considering a localized enough region, then the isothermal limit applies – hence why the local isothermal scale height can be a useful quantity.

[8] Note that we won't be dealing with this physics in C5 – it's more in the realm of astrophysics. However, I think it's important to realise the limits of the assumptions we make for atmospheres problems.

2.2.4 Vertical structure of planetary atmospheres

Figure 2.2 shows that as expected from our integration of hydrostatic balance above, in Earth's atmosphere the dependence of pressure and density on height is approximately exponential. The slight deviations from a pure exponential decrease are due to changes in the temperature with height, demonstrating that an isothermal approximation is insufficient to calculate the detailed pressure or density profile of a typical planetary atmosphere.



Global-mean pressure (solid), density (dashed), and temperature (dotted), as functions of altitude.
Source: U.S. Standard Atmosphere (1976).

Figure 2.2: U.S. standard atmosphere of Earth. Shown are pressure, density, and temperature profiles for typical Earth climate conditions. Note that the pressure and density x-axis is logarithmic.

However, the dependence of temperature on height is not so straightforward – for Earth there are multiple atmospheric levels, defined by the temperature gradient. The region near the surface where the temperature decreases with increasing height is known as the *troposphere*, with the region overlying that where the temperature increases with height known as the *stratosphere*[9]. Near-surface tropospheres and overlying stratospheres are ubiquitous in our Solar System, as shown in Figure 2.3. We'll next dive into the thermodynamic properties that control these vertical temperature profiles.

[9]One can further demarcate regions above this as mesosphere, thermosphere, exosphere, etc., but in C5 we largely focus on the troposphere and stratosphere given that these are the regions most relevant to weather prediction and present-day climate.

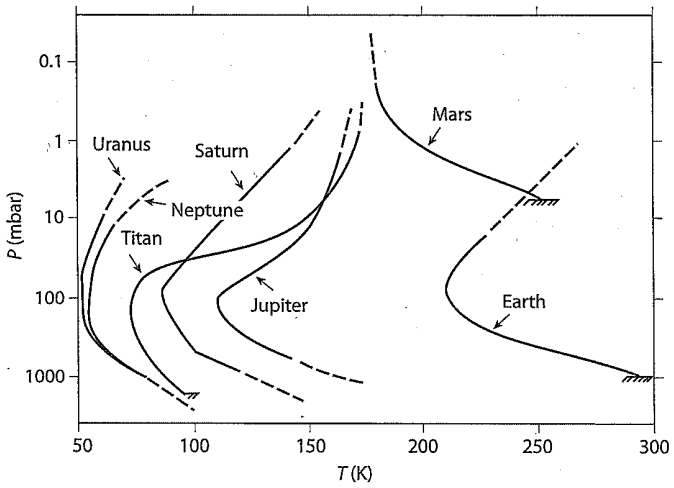


Figure 2.3: Atmospheric temperature-pressure profiles of planets in our Solar System with significant atmospheres (not including Venus), along with Titan. Figure adapted from Seager (2010).

3 Dry thermodynamics: first law, adiabatic processes

Reading for Day 3: Pierrehumbert Principles of Planetary Climate, Sections 2.3-2.4.

3.1 First law of thermodynamics

The first law of thermodynamics is a statement of energy conservation – namely, that the rate of change in the internal energy of a system is the sum of the working and heating rates

$$\frac{dU}{dt} = Q + W, \tag{3.1}$$

where U is the internal energy, Q is the heating rate (which represents interactions with environments at different temperature through processes such as conduction and radiation), and W is the working rate (which represents a mechanical exchange of energy with the environment). For an atmosphere, the working rate is dominated by the expansion work done *on* the system

$$W = -p\frac{dV}{dt}, \tag{3.2}$$

where V is the volume of the system. Thus, for an atmosphere we can write the first law of thermodynamics as

$$\frac{dU}{dt} = Q - p\frac{dV}{dt}. \tag{3.3}$$

For studies of atmospheres, an alternate version of the first law of thermodynamics is often used with enthalpy $H = U + pV$ instead of internal energy,

$$\frac{dH}{dt} = Q + V\frac{dp}{dt}. \tag{3.4}$$

Note that a constant p, constant T process has constant H but does *not* have constant U because work can be done on the gas.

3.2 Specific heats

The chain rule can be used to re-state the change in internal energy as

$$\frac{dU}{dt} = \frac{\partial U}{\partial T}\frac{dT}{dt} + \frac{\partial U}{\partial V}\frac{dV}{dt}. \tag{3.5}$$

We can define the heat capacity at constant volume C_v to be

$$C_v \equiv \left(\frac{\partial U}{\partial T}\right)_v, \tag{3.6}$$

which is a *specific heat* (c_p) when defined per mass or per mole. Note that for an ideal gas changes in internal energy due to changes in volume can be neglected[10], so we can write the change in internal energy as

$$\frac{dU}{dt} = C_v\frac{dT}{dt}. \tag{3.7}$$

[10]This is because the changes in internal energy caused by changes in volume would arise if inter-molecular interactions were important. In an ideal gas, inter-molecular interactions are by definition not important.

We can equivalently perform the chain rule on the change in enthalpy with time to express

$$\frac{dH}{dt} = \frac{\partial H}{\partial T}\frac{dT}{dt} + \frac{\partial H}{\partial p}\frac{dp}{dt}. \tag{3.8}$$

We can similarly define the heat capacity at constant pressure C_p to be

$$C_p \equiv \left(\frac{\partial H}{\partial T}\right)_p. \tag{3.9}$$

For an ideal gas, the enthalpy changes due to pressure changes are negligible, so we can write the change of enthalpy as

$$\frac{dH}{dt} = C_p\frac{dT}{dt}. \tag{3.10}$$

We can relate the heat capacities at constant pressure and volume as

$$C_p = C_v + \frac{\partial U}{\partial V}\frac{\partial V}{\partial T} + p\frac{\partial V}{\partial T}. \tag{3.11}$$

For an ideal gas, $\frac{\partial U}{\partial V} \equiv 0$. Thus, for an ideal gas the heat capacity and specific heat are related as

$$\begin{aligned} C_p &= C_v + NR_u, \\ c_p &= c_v + R_u, \end{aligned} \tag{3.12}$$

where N is the number of moles in the system. The heat capacity C_v per molecule is $k_B/2$ per degree of freedom, and note that the universal gas constant is related to the Boltzmann constant[11] as $R_u = N_A k_B$ with $N_A = 6.022 \times 10^{23}$ mol^{-1} Avogadro's number. Thus, the specific heat capacity c_v can also be expressed as $R_u/(2)$ per degree of freedom. For diatomic molecules, there are 3 translational and 2 rotational degrees of freedom, leading to $c_v = 5/2R_u$ and $c_p = 7/2R_u$. Two common combinations of specific heats are

$$\begin{aligned} \gamma &\equiv \frac{c_p}{c_v}, \\ \kappa &\equiv \frac{R}{c_p}, \end{aligned} \tag{3.13}$$

where for Earth air $\gamma \approx 1.4$ and $\kappa \approx 0.286$. The specific heat capacities and specific heat ratios of relevant atmospheric gases are shown in Figure 3.1.

3.3 Convective (in)stability

3.3.1 Adiabatic lapse rate: activity

Please get in groups of two or three to work on the following activity to derive the adiabatic lapse rate dT/dp as a function of ρ and c_p.

[11] Prof. Pierrehumbert's textbook even says "the Boltzmann constant should not be considered a fundamental constant of the Universe. It is just a unit conversion factor."

	H_2O	CH_4	CO_2	N_2	O_2	H_2	He	NH_3
Crit. point T	647.1	190.44	304.2	126.2	154.54	33.2	5.1	405.5
Crit. point p	221.e5	45.96e5	73.825e5	34.0e5	50.43e5	12.98e5	2.28e5	112.8
Triple point T	273.15	90.67	216.54	63.14	54.3	13.95	2.17	195.4
Triple point p	611.	.117e5	5.185e5	.1253e5	.0015e5	.072e5	.0507e5	.061e5
L vap(b.p.)	22.55e5	5.1e5	–	1.98e5	2.13e5	4.54e5	.203e5	13.71e5
L vap(t.p.)	24.93e5	5.36e5	3.97e5	2.18e5	2.42e5	??	??	16.58e5
L fusion	3.34e5	.5868e5	1.96e5	.2573e5	.139e5	.582e5	??	3.314e5
L sublimation	28.4e5	5.95e5	5.93e5	2.437e5	2.56e5	??	??	19.89e5
ρ liq(b.p.)	958.4	450.2	1032.	808.6	1141.	70.97	124.96	682.
ρ liq(t.p.)	999.87	??	1110.	??	1307.	??	??	734.2
ρ solid	917.	509.3	1562.	1026.	1351.	88.	200.	822.6
$c_p(0C/1bar)$	1847.	2195.	820.	1037.	916.	14230.	5196.	2060.
$\gamma(c_p/c_v)$	1.331	1.305	1.294	1.403	1.393	1.384	1.664	1.309

Table 2.1: Thermodynamic properties of selected gases. Latent heats of vaporization are given at both the boiling point (the point where saturation vapor pressure reaches $1bar$) and the triple point. Liquid densities are given at the boiling point and the triple point. For CO_2 the 'boiling point' is undefined, so the liquid density is given at $253K/20bar$ instead. Note that the maximum density of liquid water is $1000.00kg/m^3$ and occurs at $-4C$. Densities of solids are given at or near the triple point. All units are mks, so pressures are quoted as Pa with the appropriate exponent. Thus, $1bar$ is written as $1e5$ in the table.

Figure 3.1: Relevant thermodynamic data for common atmospheric gases. Table adapted from an early version of Pierrehumbert (2010) (see his Table 2.1).

1. One can derive the variation in temperature with pressure or height (i.e., the "lapse rate") of a convecting atmosphere from the first law of thermodynamics (Equation 3.4). With your partner, substitute in substitute in our relationship for enthalpy changes with time from Equation (3.10) into Equation (3.4) to derive an expression for the lapse rate in pressure coordinates dT/dp.

2. If you finish this quickly, then go on to derive the lapse rate in $d\ln T/d\ln p$ space from Equation (3.17) below and the lapse rate in height coordinates, dT/dz, from Equation (3.19) below.

Answer: First, we can substitute in our relationship for enthalpy changes with time from Equation (3.10) into Equation (3.4)

$$C_p \frac{dT}{dt} = Q + V\frac{dp}{dt}, \tag{3.14}$$

and then divide by mass to write our expression above as a function of specific heat capacity and density

$$c_p \frac{dT}{dt} = q + \frac{1}{\rho}\frac{dp}{dt}. \tag{3.15}$$

Then, noting that an adiabat implies $q = 0$ (no heating/cooling), and re-arranging, we can write

$$\frac{dT}{dp} = \frac{1}{\rho c_p}. \tag{3.16}$$

You can see that this is indeed a lapse rate, just in pressure coordinates.

3.3.2 Common expressions for the lapse rate

We can use the ideal gas law to convert the lapse rate dT/dp that you derived in the activity above to write down a simplified expression for the lapse rate in logarithmic temperature and pressure coordinates as

$$\frac{d\ln T}{d\ln p} = \frac{R}{c_p}. \tag{3.17}$$

Note that here and in the following statements of the adiabatic lapse rate c_p is per mass (in units of $\mathrm{JK^{-1}\ kg^{-1}}$) rather than per mol. Finally note that hydrostatic equilibrium states

$$dp = -\rho g dz, \tag{3.18}$$

and thus we can re-write the lapse rate in height coordinates as

$$\frac{dT}{dz} = -\frac{g}{c_p}. \tag{3.19}$$

You will see Equation (3.17) and Equation (3.19) *very* often – I recommend just remembering these expressions if you're interested in atmospheres beyond Earth. Typical values for the lapse rate in K/km on terrestrial bodies in the Solar System are -10.8 for Venus, -9.8 for Earth, -4.6 for Mars, and -1.3 for Titan.

3.3.3 Convective (in)stability criteria

Atmospheres with temperature profiles that decrease more sharply with height than the adiabatic profile are unstable to convection. This is because if we perturb a parcel at the same density as the surroundings, the parcel's density will change according to the adiabatic relation while keeping a pressure close to that of the surroundings. If the density is less than the environmental density for upward displacements (or greater for downward displacements), then the parcel will accelerate away from its initial position – this will initiate convection. Figure 3.2 shows a schematic of convectively unstable and stable temperature profiles compared to the adiabatic lapse rate. If the parcel is hotter than the surrounding

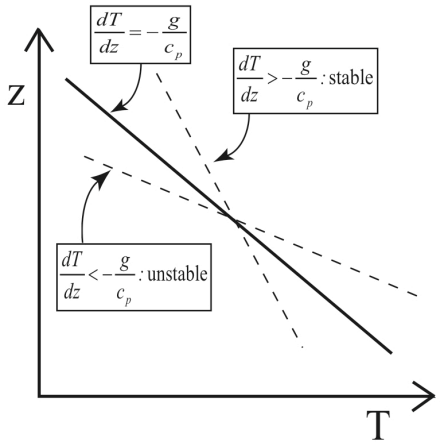


Figure 3.2: Schematic showing the convective lapse rate along with convectively unstable and stable temperature profiles. Figure courtesy Adam Showman.

air than it will be less dense (and if it is cooler, then it will be more dense). The atmosphere

will then be unstable to convection if temperature decreases with height faster than the adiabatic lapse rate $-g/c_p$ and stable if temperature decreases with height more slowly than the adiabatic lapse rate. Thus, we can express the convective stability of the atmosphere as

$$\begin{aligned}
\frac{dT}{dz} &< -\frac{g}{c_p} \text{ unstable,} \\
\frac{dT}{dz} &= -\frac{g}{c_p} \text{ marginally stable,} \\
\frac{dT}{dz} &> -\frac{g}{c_p} \text{ stable.}
\end{aligned} \tag{3.20}$$

Atmospheres that are forced to be unstable will adjust via convection to have lapse rates that are approximately stable, with $dT/dz \approx -g/c_p$. In general, planetary atmospheres are stable to convection except at deep levels, either near the surface for rocky planets or within the deep envelope and interior in gas giant planets.

3.3.4 Potential temperature

Note that potential temperature is a very useful quantity to determine stability of a planetary atmosphere. The potential temperature is defined such that

$$d\ln\theta \equiv d\ln T - \frac{R}{c_p} d\ln p, \tag{3.21}$$

and is equivalent to entropy in an atmospheric context. For constant $R/c_p \equiv \kappa$,

$$\theta = T\left(\frac{p_o}{p}\right)^{\kappa}. \tag{3.22}$$

The potential temperature θ is thus the temperature that a parcel of dry air at pressure p and temperature T would have it if were expanded or compressed adiabatically to a reference pressure[12] p_0. We can further relate the potential temperature variation with height $d\theta/dz$ to the lapse rate dT/dz as

$$\begin{aligned}
\frac{1}{\theta}\frac{d\theta}{dz} &= \frac{1}{T}\frac{dT}{dz} - \frac{\kappa}{p}\frac{dp}{dz}, \\
\frac{1}{\theta}\frac{d\theta}{dz} &= \frac{1}{T}\frac{dT}{dz} + \frac{\kappa}{p}\rho g, \\
\frac{1}{\theta}\frac{d\theta}{dz} &= \frac{1}{T}\left(\frac{dT}{dz} + \frac{g}{c_p}\right),
\end{aligned} \tag{3.23}$$

where we have used hydrostatic equilibrium in step 2 and the ideal gas law in step 3. Note that this implies that if $d\theta/dz = 0$, the lapse rate $dT/dz = -g/c_p$ (i.e., a dry adiabat). Thus, we can write an equivalent expression to Equation (3.20) using potential temperature:

$$\begin{aligned}
\frac{d\theta}{dz} &< 0 \text{ unstable,} \\
\frac{d\theta}{dz} &= 0 \text{ marginally stable,} \\
\frac{d\theta}{dz} &> 0 \text{ stable.}
\end{aligned} \tag{3.24}$$

[12]For Earth, this reference pressure is typically taken to be 1 bar.

Figure 3.3 shows an equivalent schematic to Figure 3.2 but now using potential temperature instead of temperature. An author's note is that I find potential temperature to be a

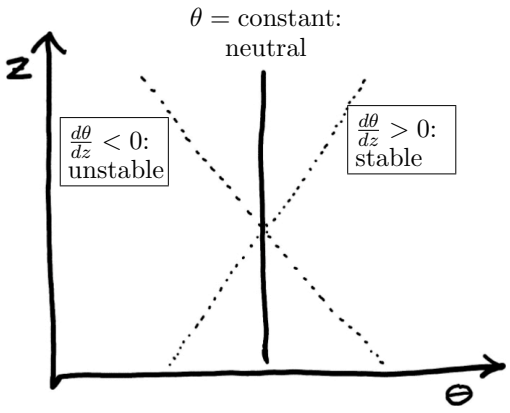


Figure 3.3: Schematic showing the convective lapse rate in potential temperature ($d\theta/dz = 0$) as well as convectively stable ($d\theta/dz > 0$) and unstable ($d\theta/dz < 0$) profiles. Figure courtesy Adam Showman.

much more intuitive way to diagnose whether an atmosphere is statically stable[13], and so I recommend using it where possible.

3.3.5 Convective available potential energy (CAPE)

Convective Available Potential Energy (CAPE) is a useful measure to determine the susceptibility of a given temperature (and moisture, but so far we consider dry cases alone) profile to deep convection (see Section 9.5.1 of Holton & Hakim, 2013). Specifically, CAPE is a measure of the maximum possible kinetic energy that a statistically unstable parcel can acquire, assuming no mixing with the environment and instantaneous adjustment to the local environmental pressure.

We can derive CAPE as follows. First, we write the acceleration of a parcel due to buoyancy (b) where the density of a parcel is ρ_p and that of the environment is ρ_e,

$$b = g\frac{\rho_e - \rho_p}{\rho_p} = g\frac{1/T_e - 1/T_p}{1/T_p} = g\frac{T_p - T_e}{T_e}, \tag{3.25}$$

where T_p is the temperature of the parcel and T_e is the temperature of the environment. We can then calculate the maximum kinetic energy of a parcel displaced vertically from the level of free convection (LFC, the vertical point where deep convection begins, see Figure 3.4) to the level of neutral buoyancy (LNB, the higher level above the LFC where convection is inhibited) as

$$\frac{w_{\max}^2}{2} = \int_{z_{LFC}}^{z_{LNB}} g\left(\frac{T_p - T_e}{T_e}\right) dz = \int_{p_{LNB}}^{p_{LFC}} R(T_p - T_e) d\ln p, \tag{3.26}$$

where the second equality uses the ideal gas law and hydrostatic balance. This is the definition of CAPE – thus, we can express CAPE as

$$\text{CAPE} \equiv \int_{p_{LNB}}^{p_{LFC}} R(T_p - T_e) d\ln p. \tag{3.27}$$

[13]Potential temperature is also often the thermodynamic quantity used in 3D circulation models of planetary atmospheres. It's especially helpful that the primitive equations of meteorology map onto the shallow-water equations in isentropic coordinates...but that's for much later!

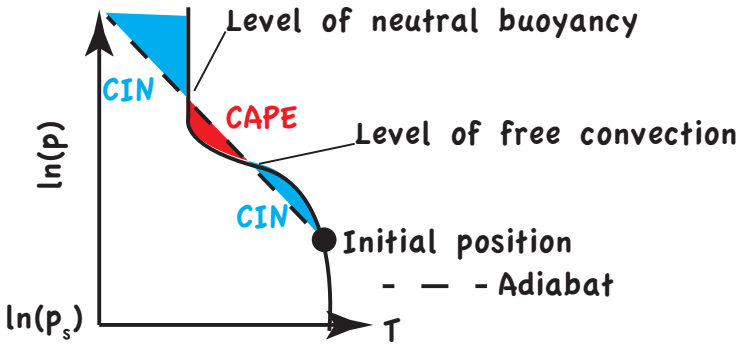


Figure 3.4: Schematic showing the trajectory of a parcel in T-ln(p) space alongside the LFC, LNB, and regions where CAPE is positive (red) and negative (blue). Figure courtesy Ray Pierrehumbert.

CAPE is thus positive in regions that are convective, and negative (termed convective inhibition, CIN) in regions in which convection is inhibited.

4 Moist thermodynamics: Phase diagrams, Clausius-Clapeyron relation

Reading for Day 4: Pierrehumbert Principles of Planetary Climate, Section 2.6.

Most atmospheres in the Solar System (and presumably beyond) have species that are *condensible* – that is, they can condense (change phase from gaseous to liquid) or undergo deposition (change of phase from gaseous to solid). We will use the more general term "condense" often throughout this course to refer to both processes. Condensibles in the Solar System include water vapour (on Earth, Mars, and the gas and ice giants), carbon dioxide (Mars), methane (Titan), nitrogen (Triton, Pluto), ammonia (gas and ice giants) and hydrogen sulphide (gas and ice giants). Note that condensation is a local process, and typically occurs because air parcels undergo changes in their local temperature and pressure either due to radiative processes or atmospheric motion that lead them to reach a phase boundary where condensation occurs.

4.1 Phase diagrams

Phase diagrams are used to delineate where in temperature-pressure space a given substance is stable as a gas, liquid, or solid. Figure 4.1 shows a generalised phase diagram for an atmospheric species. The phase boundaries are clear between gas (stable at high temperature, low pressure), liquid (stable at high temperature, high pressure), and solid (stable at low temperature, high pressure). There are also two key locations on the phase diagram: 1) the triple point, which is where solid, liquid, and gas can co-exist[14], and 2) the critical point, which is where the distinction between liquid and gas disappears[15]. See Figure 3.1 for critical

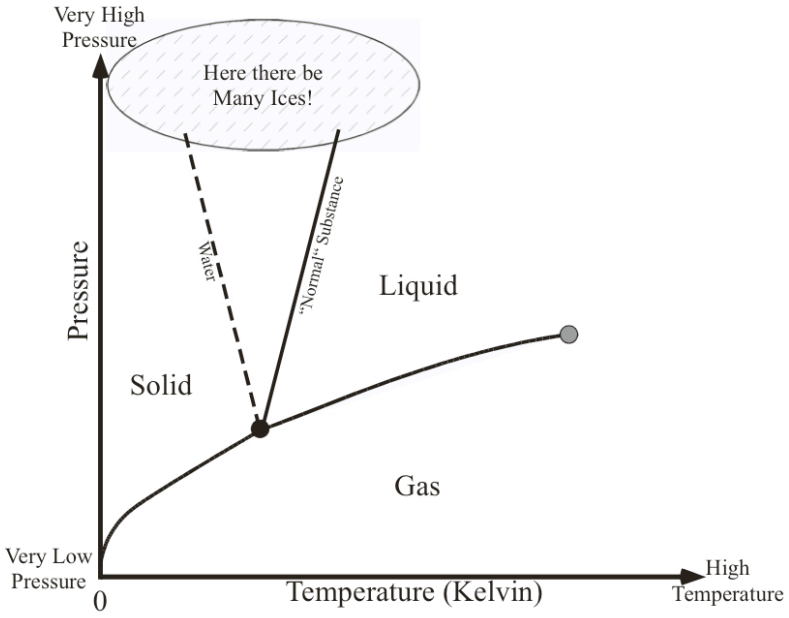


Figure 4.1: A general schematic form of a phase diagram (Figure 2.4 of Pierrehumbert (2010)). The triple point is a black circle, the critical point is a grey circle, the solid-liquid phase boundary for a normal substance (for which the solid phase is denser than its liquid phase) is shown as a solid curve, and the phase boundary for water (not a normal substance by this definition) is shown as a dashed curve.

[14]Notably, the temperature of Mars' atmosphere is close to the triple point of CO_2.

[15]Supercritical species beyond the critical point typically become miscible with other species, which is thought to prevent layering of the interiors of gas and ice giant planets.

and triple points of various atmospheric species – for water, the critical point occurs at a temperature of 647.1 K and pressure of 221 bars, clearly far beyond the conditions seen on Earth.

4.1.1 Latent heat

Latent heat (typically denoted L) is the amount of energy released when a unit of mass of a substance changes phase while holding temperature constant. Thus,

$$Q = mL, \tag{4.1}$$

where Q is the amount of energy released and m is the mass of a substance. The *latent heat of vaporization* (also sometimes termed the latent heat of condensation when in the reverse direction) corresponds to the heat required to vaporise a substance from liquid to gas (or the heat output when a substance condenses from gas to liquid). The *latent heat of fusion* corresponds to the heat required to change the phase of a substance from solid to liquid (or the heat output when a substance changes phase from liquid to solid). The *latent heat of sublimation* corresponds to the heat required to change the phase of a substance from solid to gas (or the heat output when a substance changes phase from gas to solid). Note that the sign convention is dependent on the direction of heat transfer, with positive referring to energy released and negative referring to energy input.

The latent heat can often reasonably be assumed to be constant over a broad range of temperature, but they do decrease with increasing temperature and of course go toward zero at the critical point. For example, the latent heat of vaporisation of water decreases from 2.493×10^6 J kg^{-1} at the triple point (273.15 K) to 2.255×10^6 J kg^{-1} at the critical point (647.1 K), a $\sim 9.5\%$ change – not large, but also not negligible. We will assume that the latent heat is approximately constant in the derivations that follow, but this should be relaxed for practical applications.

4.1.2 Activity – application of latent heat and phase diagrams: two thought experiments

Please get in groups of two or three to work on the following activity considering two thought experiments – the first is a direct calculation, and the second more of an estimation that has been conducted more rigorously in recent work.

1. Eventually as the Sun brightens on the main-sequence, Earth will undergo a runaway greenhouse (similar to Venus) where its oceans will boil and vaporise. Earth's oceans are approximately 0.023% of the total mass of the Earth (which is $M_\oplus = 5.97 \times 10^{24}$ kg). How long would it take Earth's oceans to boil, once their temperature reaches the boiling point of water? Assume for simplicity in this estimate that the received Solar flux on the surface (after taking into account albedo) is ≈ 1360 W m^2.

   Answer: The timescale to evaporate the oceans

   $$\tau_{\rm vap} = \frac{E_{\rm vap}}{P_\odot} = \frac{m_{\rm ocean} L_{\rm vap}}{F_\odot \pi R_\oplus^2} = \frac{2.3 \times 10^{-4} \times 5.97 \times 10^{24}\ {\rm kg} \times 22.55 \times 10^5\ {\rm Jkg}^{-1}}{1360\ {\rm W\ m}^{-2} \times \pi (6371 \times 10^3\ {\rm m})^2} \approx 556\,{\rm yr}. \tag{4.2}$$

   This is very fast! Effectively instantaneous from a geological perspective – phase changes are quite rapid. Let's not cause Earth to get to this point anytime soon.

2. The sub-Neptune K2-18b, with a mass of $7.2M_\oplus$ and a radius of $2.5R_\oplus$, implying a bulk density of 2.28 g cm^{-3}. Its atmosphere is expected to potentially have $\gtrsim 10\%$ H_2O by volume. Assuming for simplicity that the atmosphere of K2-18b is comprised only of water (i.e., a steam atmosphere), would you expect this steam atmosphere to sit atop 1) an ocean, with a discrete atmosphere/ocean interface?, or would it 2) be a totally fluid planet like Uranus and Neptune?

   Answer: This is still being debated in the community! Explain the hycean worlds vs. runaway greenhouse debate, and resulting prediction of habitable ocean planet vs. runaway greenhouse.

4.2 Clausius-Clapeyron relation

As you will learn more in the next bit of C5 from Prof. Stier, clouds can form in planetary atmospheres where species are thermodynamically favoured to undergo a phase transition from vapour to solid phases. Cloud condensation is thermodynamically allowed when the partial pressure of a species exceeds its saturation vapour pressure. The key point to stress is that the saturation vapour pressure is extremely sensitive to temperature, as we discuss below.

The saturation vapour pressure is set by the Clausius-Clapeyron relationship[16], which for an ideal gas can be expressed as

$$\frac{dp_{\rm sat}}{dT} = \frac{1}{T}\frac{L}{\rho_{\rm vap}^{-1} - \rho_{\rm cond}^{-1}}, \tag{4.3}$$

where $\rho_{\rm vap}$ is the density of the vapour phase, $\rho_{\rm cond}$ is the density of the condensed phase, and $p_{\rm sat}$ is the saturation vapour pressure. If $\rho_{\rm cond} \gg \rho_{\rm vap}$ (as is the case for most relevant atmospheric condensibles), then (using the ideal gas law to relate $\rho_{\rm vap}$ to p) Clausius-Clapeyron simplifies to

$$\frac{dp_{\rm sat}}{dT} \approx \frac{p_{\rm sat}L}{RT^2}, \tag{4.4}$$

where $p_{\rm sat}$ is the saturation vapour pressure and L is the latent heat of vaporisation/condensation, fusion, or sublimation of the species of interest. If we assume $L \approx$ constant, Equation (4.4) can be directly integrated to obtain the saturation vapour pressure curve

$$p_{\rm sat} = p_0 \exp\left[\frac{L}{R}\left(\frac{1}{T_0} - \frac{1}{T}\right)\right], \tag{4.5}$$

implying that the saturation vapour pressure scales exponentially with $-1/T$. As a result, the saturation vapour pressure of a given species decreases with decreasing temperature, as shown in Figure 4.2. Note that this decrease in saturation vapour pressure with decreasing temperature occurs for all species, but the exact slope depends on the condensible of interest. Figure 4.3 shows the saturation vapour pressure as a function of temperature for water vapour in particular (which as you can see from Figure 4.2 is a condensible not just on Earth but also cold gas giants like Jupiter). Given the steep decline of temperature with height for

[16] For those of you who completed their first three years at Oxford, you may recall deriving this in A1.

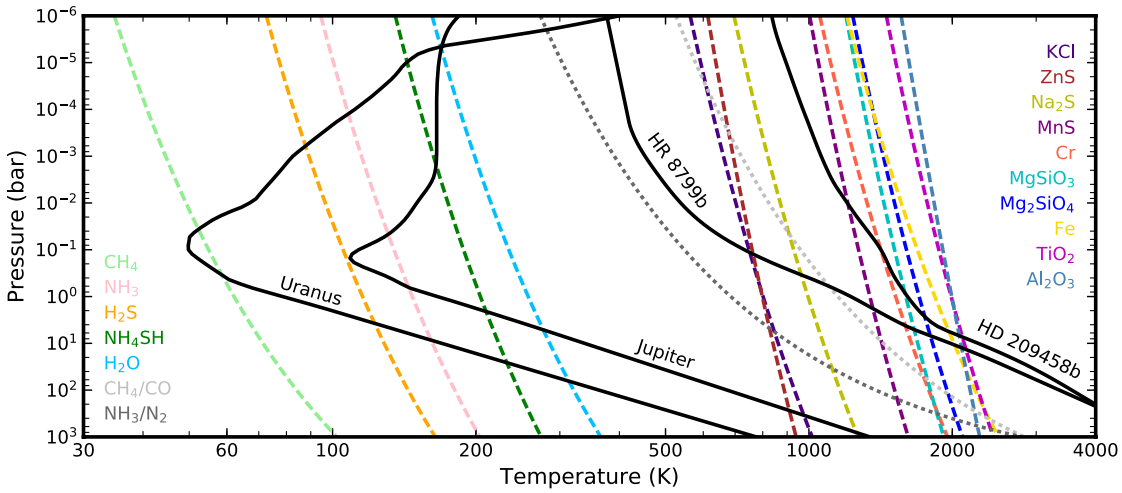


Figure 4.2: Temperature profiles of Solar System giant planets and exoplanets (black lines) compared with condensation curves of various species (dashed lines) and equilibrium chemistry equivalency curves (dotted lines). Figure adapted from Gao et al. (2021).

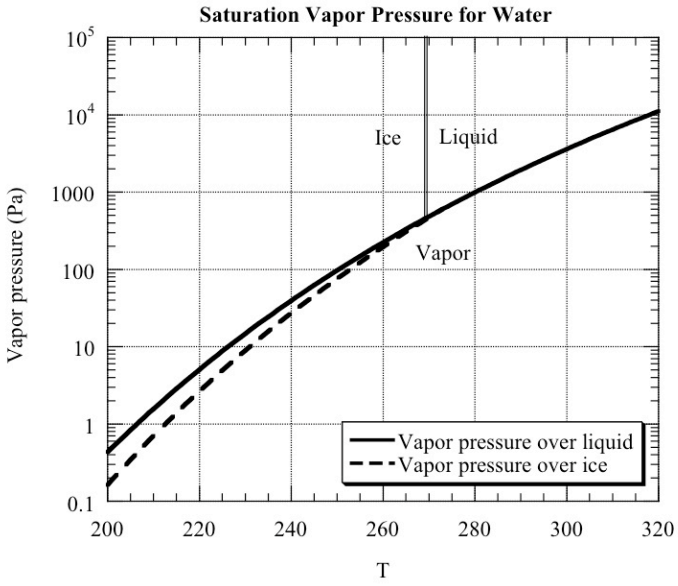


Figure 4.3: Saturation vapour pressure as a function of temperature for water vapour. Figure adapted from Pierrehumbert (2010)

an adiabatic atmosphere (derived earlier in Section 3.3), this implies that condensible water vapour mixing ratios are orders of magnitude smaller at the tropopause than at the surface. It also more generally implies that condensible water vapour mixing ratios are smaller in colder regions of the planet, e.g., implying a decrease in atmospheric moisture from the hotter equator toward the colder pole[17].

Using condensation temperature instead of saturation vapour pressure, one can equivalently state that condensation from gas to cloud (solid or liquid) is thermodynamically

[17]The low saturation vapour pressure for cold temperatures is why, for instance, one's skin can become relatively dry in the winter (this is even more of an issue for those in colder, drier places like Toronto, Boston, or Chicago than the relatively humid Oxford).

allowed at a given pressure when $T < T_{\rm cond}$, which then occurs when temperature profiles cross the condensation curves in Figure 4.2[18]. This is why clouds generally form at high altitudes in planetary atmospheres – because the temperature decreases with height in the troposphere, clouds form at altitudes where the temperature is cool enough to allow for condensation. The exception is when the local saturation vapour pressure is enhanced (e.g., due to mixing), increasing the partial pressure of a species to allow for condensation.

[18]As a planetary scientist, I personally prefer using condensation temperature over saturation vapour pressure as a rule-of-thumb because the variation in temperature between planets is typically much larger than the variation in the atmospheric pressure ranges of interest.

5 Moist thermodynamics: moist pseudo-adiabat

Reading for Day 5: Pierrehumbert Principles of Planetary Climate, Section 2.7.

5.1 Single-component moist (pseudo)-adiabat

The simplest case of a condensible atmosphere is one where the entire atmosphere is made up of the condensible species. There are no perfect examples of this in the Solar System, but Mars (CO_2-dominated) and Triton (N_2-dominated) have atmospheres that can be well approximated as having a single condensible, as likely can the atmospheres of some exoplanets (e.g., super-Earth or sub-Neptune waterworlds with steam-rich atmospheres). In this case, the saturation vapour pressure is then the total atmospheric pressure when condensation occurs. This implies that the temperature gradient we previously derived in Equation (4.4) from Clausius-Clapeyron is then simply the moist-adiabatic lapse rate of the atmosphere. More generally, note that this is termed the moist *pseudo*-adiabat rather than the moist adiabat – I will call it moist adiabat as shorthand, but this is slightly incorrect. The reason that this is a pseudo-adiabat rather than an adiabat is that we assume the condensate is instantaneously removed from the parcel, causing the process of condensation to not be reversible since the condensate mass has left the parcel. The moist pseudo-adiabat is approximately the true moist adiabat if the amount of retained condensate mass is small (i.e., if "rainout," that is the removal of condensate from the vapour, is a reasonable approximation), which is generally true for Earth.

For an atmosphere composed of a single condensible[19] component, the Clausius-Clapeyron relationship can be expressed as a single-component moist adiabatic lapse rate

$$\frac{d\ln T}{d\ln p} = \frac{RT}{L}, \tag{5.1}$$

where p here is both the total pressure and the saturation vapour pressure (because the gas species is always at saturation if it is the only vapour component).

Given that the dry adiabatic gradient is $d\ln T/d\ln p = R/c_p$, the moist adiabatic gradient is smaller than the dry adiabatic gradient if $L > c_p T$ – which is almost always satisfied for most condensible species of interest, including water, carbon dioxide, methane, and nitrogen (which typically have $L/(c_p T) \approx 2 - 6$, see Figure 3.1).

Note also that in the single-component limit if there is no rainout of condensate (i.e., condensate remains in the air parcel being considered), then moist-adiabatic perturbations to temperature are reversible. This means that if condensation of vapour occurs as air rises, then vaporisation/sublimation of condensate occurs as it sinks, and the resulting air parcel follows the moist-adiabatic lapse rate for both ascent and descent. However, in the real case of Earth this is incorrect, both because it is not comprised of a single-component atmosphere and because descending air in the mid-latitudes is typically drier than ascending air in the hot, moist tropics due to rainout of condensed water from air in the moist regions (see also our caveat above r.e. the "pseudo"-adiabat). In the case where descent occurs after the air

[19] Any gaseous species is a condensible if one considers cold enough temperatures! Even H_2 (which has a condensation temperature of 20.28 K).

has dried out, then the descending air follows the steeper dry adiabat[20].

Finally, note that in the case where $L \approx$ constant, we can invert our previously derived expression for the saturation vapour pressure as a function of temperature (Equation 4.5) to derive the moist adiabat:

$$T(p) = \frac{T_0}{1 - \frac{RT_0}{L} \ln\left(\frac{p}{p_0}\right)}. \tag{5.2}$$

Note that Equation (5.2) only governs the entire atmospheric temperature profile if the temperature at the surface is cold enough for condensation to occur all the way to the ground (an example of this is the pole of the winter hemisphere on Mars). In warmer conditions (e.g., the low latitudes of Mars), then the near-surface temperatures are too warm for condensation, and the near-surface atmosphere follows the dry adiabat. The atmospheric lapse rate then only follows the moist adiabatic lapse rate above the condensation level at which saturation finally occurs – see Figure 5.1 for an example applicable to the warmer regions of present-day Mars. In this case, the atmosphere follows a dry adiabat from the surface (at

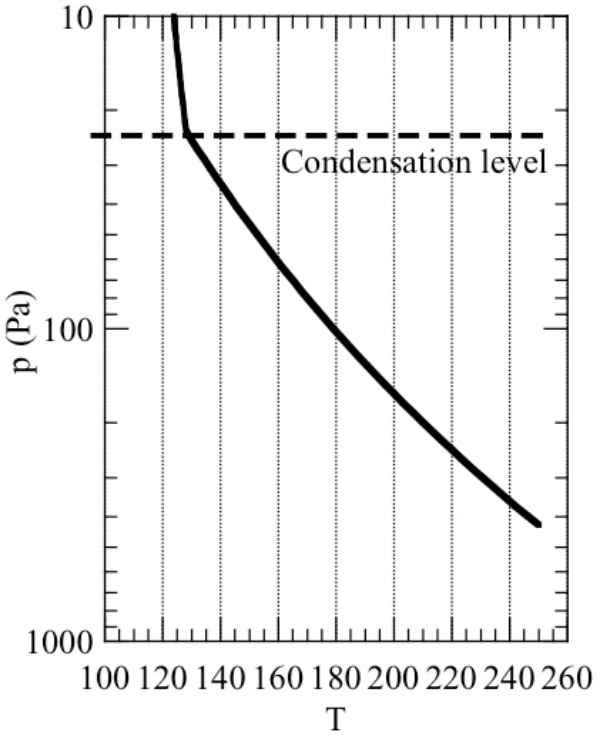


Figure 5.1: Adiabatic profile for a pure CO_2 atmosphere with a surface pressure and temperature similar to warm regions on present-day Mars. Figure adapted from Pierrehumbert (2010).

4.5 mbars) to the condensation level (0.25 mbars), above which it follows a moist adiabat. Note that as we expected, the moist adiabatic profile is shallower than the dry adiabat due to the latent heat of CO_2 exceeding the enthalpy.

[20]This is why Earth's sub-tropics near the descending branch of the Hadley cell are hot arid regions such as the Saharan and Sonoran deserts.

5.2 Two-component moist (pseudo)-adiabat

In reality, the vast majority of planetary atmospheres are composed of multiple components, including both condensible and non-condensible species (for the temperatures and pressures of interest). Here we consider one limiting case of interest of an atmosphere with one condensible species alongside a background non-condensible species. This is relevant most notably to H_2O on Earth, alongside CH_4 on Titan and several condensibles on the gas and ice giants (including NH_3 and H_2O on Jupiter and Saturn and additionally CH_4 on Neptune, in which cases the fact that the background H_2 is lighter than the condensibles is a complicating factor not considered in detail here).

We will derive the generalised moist adiabat for a single condensible by generalising the first law of thermodynamics to two species, one that can condense with changes in temperature and pressure and one that cannot condense. We make the following *assumptions:*

1. Rainout of condensate (i.e., condensate instantaneously is removed from the atmosphere upon formation), implying that we don't have to track the effect of condensate on the density or specific heat of an atmospheric parcel.

2. The atmosphere follows the ideal gas law, which is a reasonable approximation for the Earth-like pressures and temperatures of interest.

3. $\rho_{\rm sat} < \rho_{\rm condensate}$, consistent with our assumption of rainout above as well as our previous simplified form of Clausius-Clapeyron in Equation (4.4).

The first law of thermodynamics for a system with N_d moles of dry gas and $N_{\rm sat}$ moles of condensible gas at its saturation vapour pressure is

$$C_{p,d}dT + C_{p,c}dT - Vdp + LdN_{\rm sat} = \delta Q, \tag{5.3}$$

where $C_{p,d}$ is the heat capacity of the dry atmospheric component, $C_{p,c}$ is the heat capacity of the condensible vapour, V is the volume, and $dN_{\rm sat}$ is the number of moles of condensible vapour that condense when the system experiences a given temperature change dT and pressure change dp. Let us re-write Equation (5.3) using specific heat capacity (to be more consistent with the rest of our notes as well as the Pierrehumbert textbook), as in Pierrehumbert Eq. (2.28),

$$m_d c_{p,d}dT + m_c c_{p,c}dT - \frac{m_d}{\rho_d}dp_d - \frac{m_c}{\rho_c}dp_c + Ldm_c = (m_d + m_c)\delta q = 0, \tag{5.4}$$

where we have expanded the Vdp term for each individual species with $V = m/\rho$ and taken $\delta q = 0$[21] for an adiabat. Note that we have not included a term for heat storage in the condensed phase in the air parcel, consistent with our assumption of rainout throughout this and the previous single-component derivation.

We can then derive the two-component moist (pseudo)-adiabat by first substituting in the ideal gas law for $\rho = p/RT$ into Equation (5.4)

$$0 = m_d c_{p,d}dT + m_c c_{p,c}dT - m_d R_d T\frac{dp_d}{p_d} - m_c R_c T\frac{dp_c}{p_c} + Ldm_c, \tag{5.5}$$

[21] I apologise for using q as both specific energy and mass mixing ratio – this term will be set to zero so it's irrelevant.

and dividing by $m_d dT$ and collecting terms to reproduce Eq. (2.29) of Pierrehumbert (2010):

$$0 = c_{p,d} d\ln T - R_d d\ln p_d + q_c \left[c_{p,c} d\ln T - R_c d\ln p_c \right] + \frac{L}{T} dq_c, \tag{5.6}$$

where here I have defined $q_c = m_c/m_d$[22]. Note that the first two terms on the right-hand side represent the contribution of the dry air to the non-condensing entropy of the mixture, while the second two collected terms are the contribution of the moist air to this non-condensing entropy. The final term $L dq_c/T$ introduces the change in entropy due to condensation.

We can now make the assumption that the atmosphere is saturated when it begins to lie on the moist pseudo-adiabat – this implies that we can use Equation (4.4) to replace p_c by $p_{c,\mathrm{sat}}$ (i.e., the saturation vapour pressure of the condensible) as well as q_c by q_{sat} in Equation (5.6). Thus, $d\ln p_{\mathrm{sat}} = \frac{L}{R_c T} d\ln T$, $q_{\mathrm{sat}} = \frac{p_{c,\mathrm{sat}}}{p_d} \frac{\mu_c}{\mu_d}$, with $p_{c,\mathrm{sat}}$ given by Clausius-Clapeyron. The change in saturation mass mixing ratio is then (Pierrehumbert, 2010 Eq. 2.31)

$$dq_{\mathrm{sat}} = \frac{\mu_c}{\mu_d} d\left(\frac{p_{c,\mathrm{sat}}}{p_d} \right) = \frac{\mu_c}{\mu_d} \frac{p_c}{p_d} d\ln \frac{p_c}{p_d} = q_{\mathrm{sat}} \left(d\ln p_{c,\mathrm{sat}} - d\ln p_d \right). \tag{5.7}$$

Substituting in our expressions for $d\ln p_{\mathrm{sat}}$, q_{sat}, and dq_{sat} to Equation (5.6), we find (similar to Pierrehumbert, 2010 Eq. 2.32):

$$0 = d\ln T \left[c_{p,d} + q_{\mathrm{sat}} \left(c_{p,c} - \frac{L}{T} + \frac{L^2}{R_c T^2} \right) \right] - d\ln p_d \left[R_d + \frac{L}{T} q_{\mathrm{sat}} \right]. \tag{5.8}$$

We can now divide through to obtain the two-component moist adiabatic lapse rate in specific (per mass) units (equivalent to Eq. 2.33 of Pierrehumbert, 2010):

$$\frac{d\ln T}{d\ln p_d} = \frac{R_d + \frac{L}{T} q_{\mathrm{sat}}}{c_{p,d} + q_{\mathrm{sat}} \left(c_{p,c} - \frac{L}{T} + \frac{L^2}{R_c T^2} \right)}. \tag{5.9}$$

Note that if $q_{\mathrm{sat}} \to 0$ Equation (5.9) reduces to the dry adiabat (Equation 3.17), as expected.

5.3 Dilute two-component moist adiabat

An important simplified form of the two-component moist adiabat is the *dilute* two-component moist adiabat. This is given below in the case of one dilute condensible species in per mole rather than per mass units that is somewhat simpler than the expression in Equation (5.9) above (but equivalent to it in the dilute limit):

$$\frac{d\ln T}{d\ln p} = \frac{R_u + \frac{L\xi}{T}}{c_p + \frac{L^2 \xi}{R_u T^2}}. \tag{5.10}$$

In Equation (5.10), $\xi = p_{\mathrm{sat}}/p_d$ is the number (or volume) mixing ratio of the condensible gas, i.e., the ratio of the saturation vapour pressure to the partial pressure of the dry gas component p_d. Note that L and c_p in Equation (5.10) are written per mole, rather than per mass – if they were per mass, the expression would include the molecular mass ratio of the condensible and dry species.

[22] I use q rather than r as in the Pierrehumbert (2010) text because q is often used to represent mass mixing ratio in the planetary atmosphere literature.

5.3.1 Activity: deriving the dilute two-component moist adiabat

Please get in groups of two or three to work on the following activity to derive Equation (5.10) from the first law of thermodynamics.

1. Derive Equation (5.10) starting from the generalised first law of thermodynamics in Equation (5.6), but neglecting the terms in the brackets regarding the contribution of the moist air to the non-condensing entropy. This is the traditional derivation of the dilute two-component moist adiabat. You can first derive an equivalent version of Equation (5.10) in per mass units, and then convert.

   Answer: The simplified dilute version of the first law of thermodynamics is

   $$0 = c_{p,d} d\ln T - R_d d\ln p_d + \frac{L}{T} dq_c. \tag{5.11}$$

   As before, we can make the assumption that the atmospheric is saturated when it begins to lie on the moist pseudo-adiabat, implying as above that

   $$dq_{\rm sat} = q_{\rm sat}(d\ln p_{\rm c,sat} - d\ln p_d) = q_{\rm sat}\left(\frac{L}{R_c T} d\ln T - d\ln p_d\right). \tag{5.12}$$

   We can then express

   $$0 = c_p d\ln T - R_d d\ln p_d + \frac{q_c L^2}{R_c T^2} d\ln T - \frac{L q_c}{T} d\ln p_d. \tag{5.13}$$

   Rearranging, we find the dilute moist adiabat

   $$\frac{d\ln T}{d\ln p} = \frac{R_d + \frac{q_c L}{T}}{c_p + \frac{q_c L^2}{R_c T^2}}. \tag{5.14}$$

   We can convert from q to ξ using $q = \xi \frac{\mu_c}{\mu_d}$. Doing so, we can recover Equation (5.10).

5.3.2 Discussion of dilute two-component moist adiabat

As stated above, Equation (5.10) is now the *dilute* limit of the moist (pseudo)-adiabat, where we have taken $d\ln p_d \to d\ln p$ (i.e., we assumed that there is a small enough abundance of condensible such that its partial pressure does not have a significant impact on the total atmospheric pressure), and assumed that $q_{\rm sat}$ is small (this is equivalent to neglecting the two terms regarding the contribution of the moist air to the non-condensing entropy in Equation 5.6). This dilute limit is reasonable for Earth, but not for atmospheres which are composed of a condensible with high mass mixing ratio (e.g., steam atmospheres on exoplanets with a significant background non-condensible species).

Let's check the limits of Equation (5.10). First, note that as $\xi \to 0$, $\frac{d\ln T}{d\ln p} \to R/c_p$, which is the dry adiabat. Conversely, as $\xi \to \infty$, $\frac{d\ln T}{d\ln p} \to RT/L$, which is the expression for the one-species moist (pseudo)-adiabat that we derived in Equation (5.1).

Figure 5.2 (Figure 2.7 of Pierrehumbert, 2010) shows the moist adiabat for water vapour in Earth air for a 1 bar background atmosphere comparable to that of present-day Earth. As

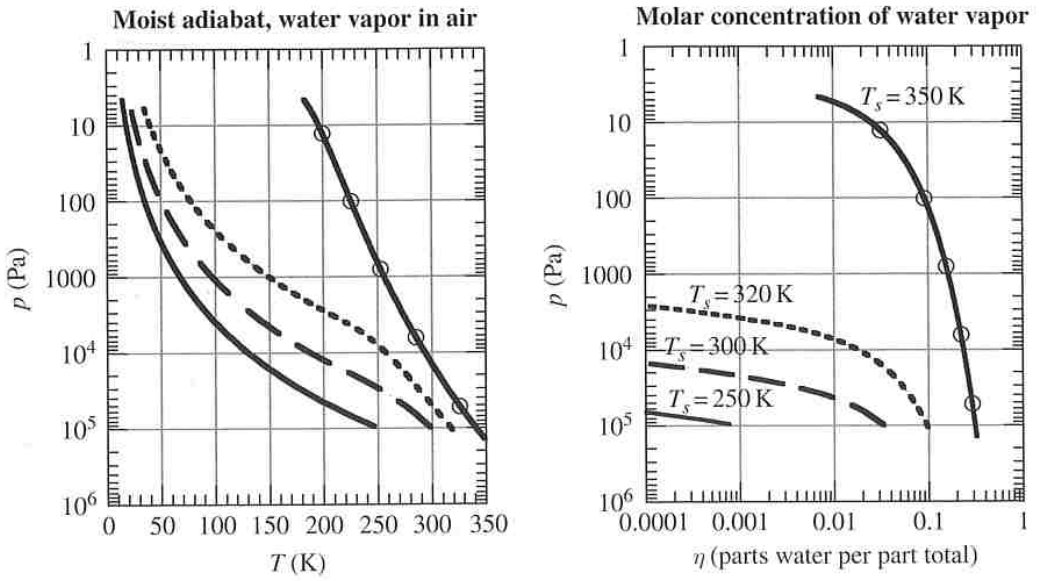


Figure 5.2: Moist adiabat for dilute saturated water vapour mixed with Earth air, where the dry component has a partial pressure of 1 bar at the surface. The left-hand panel shows the temperature profile, and the right shows the molar concentration of water vapour (ξ in these notes). Results are shown for varying surface temperature (linestyles). Figure adapted from Pierrehumbert (2010).

the surface temperature increases, the moist adiabat becomes shallower due to the additional latent heat release as the saturated parcels with higher molar concentration rise and condense. Note that the dilute limit begins to break down for sufficiently high surface temperatures, as water vapour is well-mixed with high molar ratios $\gtrsim 0.1$ throughout the troposphere.

Let's also compare the two-component moist adiabatic lapse rate to the dry adiabatic lapse rate $\frac{d\ln T}{d\ln p} = R/c_p$. We find that the moist adiabatic lapse rate is shallower than the dry adiabatic lapse rate if

$$\frac{R_u + \frac{L\xi}{T}}{c_p + \frac{L^2\xi}{R_u T^2}} < R_u/c_p,$$

$$\text{equivalently,} \tag{5.15}$$

$$\frac{1 + \frac{L\xi}{R_u T}}{1 + \frac{L^2\xi}{c_p R_u T^2}} < 1.$$

In the limit of large L, we find that this criterion implies $L/(Tc_p) > 1$, exactly the same condition as we had for the single-component moist adiabat.

Finally, note that the moist adiabat starts to deviate from that of a single-component gas when the dry terms in the numerator and denominator (R_u and c_p, i.e., not involving ξ) become important. Using the numerator alone, this deviation from a single-component moist adiabat occurs when $\xi \lesssim \frac{RT}{L}$ – for Earth, this occurs when $\xi \sim 0.05$. This is roughly an order of magnitude higher than the typical mixing ratio of water vapour on Earth, implying that the moist adiabat on Earth cannot be well represented by the single-component limit. Note that this also implies that the moist adiabat depends on ξ itself, which in turn depends

strongly on pressure because ξ is exponentially sensitive to temperature through Clausius-Clapeyron. Thus, the moist-adiabatic gradient will converge toward a dry adiabatic gradient in the upper troposphere as $\xi \to 0$ due to the exponential decrease of saturation vapour pressure with decreasing temperature.

6 Moist thermodynamics: moist convection

There is no new reading for today – we are wrapping up moist thermodynamics before heading onto radiative transfer. Please note that you will learn much more detail about moist convection in the following portion of C5 focused on clouds taught by Prof. Stier, who is a world expert in the topic. This and the past few lectures are meant to be a primer rather than the finishing coat.

6.1 Moist stability criterion: conditional instability

We of course expect the atmosphere to be stable to convection if the environmental lapse rate dT/dz has a greater magnitude (is more negative) than the adiabatic lapse rate $(dT/dz)_{\text{ad}}$. However, we have derived two "adiabatic" lapse rates – the dry adiabat, as well as the moist pseudo-adiabat. In reality, the stability of a parcel of gas depends on the environmental lapse rate relative to both the dry and moist adiabatic lapse rates, as well as whether the air in that parcel of gas is saturated.

Figure 6.1 shows the three possibilities for atmospheric stability given an environmental lapse rate $\Gamma = dT/dz$. These can be summarised as follows:

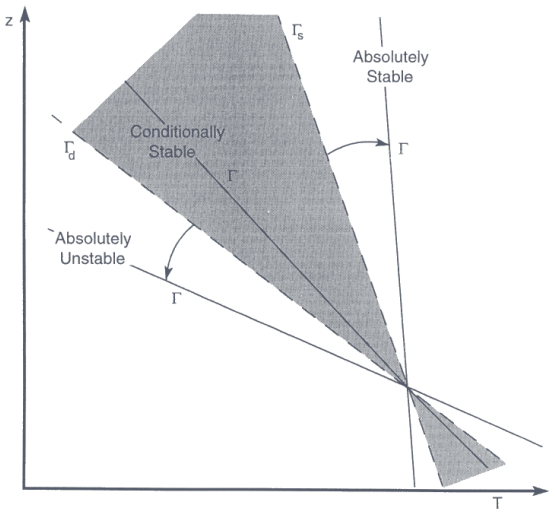


Figure 6.1: Vertical stability of moist air as a function of temperature and height for given values of the environmental lapse rate Γ. Figure adapted from Salby (1996).

1. $\Gamma < \Gamma_{\text{dry}}$: If the lapse rate is steeper (more negative) than the dry adiabat, the atmosphere is unstable to convection (i.e., adiabatic perturbations) regardless of its relative humidity. If the air is unsaturated, the air will follow the dry adiabat. If it is saturated, the air will follow the moist adiabat.

2. $\Gamma > \Gamma_{\text{moist}}$: If the lapse rate is larger (less negative) than the moist adiabat, then the atmosphere is absolutely stable to convection regardless of the relative humidity.

3. $\Gamma_{\text{dry}} < \Gamma < \Gamma_{\text{moist}}$: In this case, the atmosphere is stable against dry convection but unstable to moist convection. This implies that unsaturated air parcels will return to their original position if perturbed along the dry adiabat. However, saturated air parcels (following the moist adiabat) will become buoyant and convect. This scenario is termed conditional instability, with the atmosphere considered to be *conditionally stable*.

The lapse rate in Earth's troposphere lies between the moist and dry adiabatic gradients, implying that our troposphere is conditionally stable. Meanwhile, stratospheres (by definition) are stable to convection as in case (b) above.

The fact that Earth's atmosphere as a whole is conditionally unstable is governed through multiple processes. In the tropics, a key process is moist convection itself (leading to localised regions of rapid ascent), as well as mass conservation requiring descending (and resulting relatively dry) regions that can cool via radiation to space (see Figure 6.2). Because the

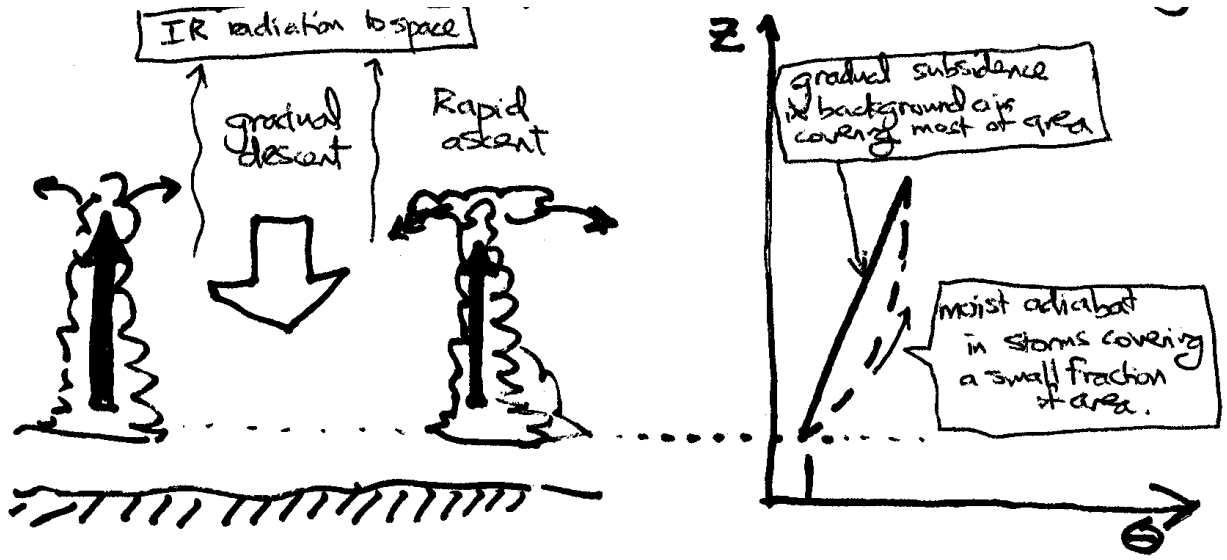


Figure 6.2: Schematic showing how moist convection interacts with the larger-scale (drier) environment, producing a conditionally unstable profile. Left: schematic showing smaller regions of strong moist convection (clouds) and larger drier descending regions radiatively cooling to space. Right: schematic potential temperature-height profile showing the typical profile in a moist region (dashed) and that in a dry region (solid). Figure courtesy Adam Showman.

background air radiates to space, its potential temperature θ decreases during subsidence, leading to a conditionally unstable profile. The difference between this profile and the moist adiabat implies that ascending saturated moist convective plumes have large buoyancy, leading to rapid ascent on the order of hours (compared to the subsidence over the course of days to weeks in dry regions). This causes the convection on Earth to be highly asymmetric, with fast updrafts covering a small fraction of the total area, even in the relatively moist tropics.

6.2 Describing moist convection: LCL, LFC, LNB, moist CAPE

Figure 6.3 shows a typical temperature-altitude profile for a moist parcel (dashed line) compared to the dry adiabat (dotted line) and a typical background (environmental) profile (solid line). The point at which the parcel first condenses and moves from the dry adiabat (at lower altitudes) to the moist pseudo-adiabat (at higher altitudes) is termed the lifting condensation level, or LCL. This is where a cloud deck can begin to form (see sketch on the

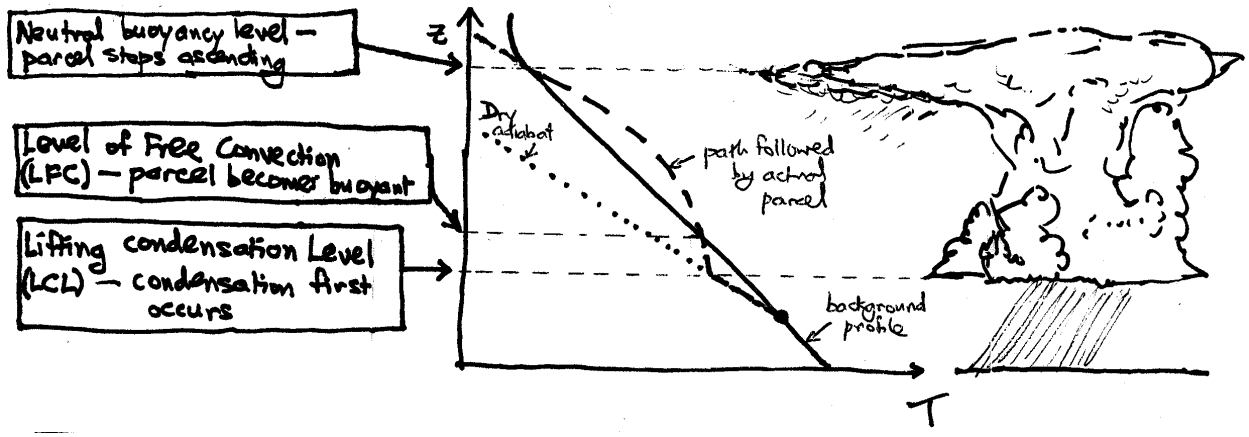


Figure 6.3: Annotated temperature-height profile showing the thermodynamic path followed by a moist parcel of gas as it ascends and condenses, forming a cloud (shown on the right). Figure courtesy Adam Showman.

right), with corresponding rainout of moisture below the cloud deck. The saturated parcels then become buoyant (with respect to the background environmental profile) at the level of free convection, or LFC. At this point, the saturated parcels undergo rapid ascent up to the region where they dry out and the parcel's temperature profile crosses the environmental profile once again – this is the level of neutral buoyancy (LNB), where the parcel stops ascending.

Finally, note that moist CAPE can be calculated in much the same way as dry CAPE (Equation 3.27). However, the key difference is that the parcel's temperature for moist CAPE is calculated with respect to the moist rather than the dry adiabat. Similarly, the definition of moist LFC and LNB are equivalent to the dry version, except again the parcel is following the moist rather than the dry adiabat. The key difference in the moist case is that if the parcel begins unsaturated at low altitudes, it will then rise following the dry adiabat up to the LCL (as in Figure 6.3), above which it is lifted along the saturated moist adiabat until it reaches the LNB.

6.3 Some moisture-related definitions to know

6.3.1 Humidity metrics

There are two key humidity metrics that you will see used regularly – know the differences between them!

1. Specific humidity: This is the *specific concentration* of water vapour. The specific concentration is defined as the ratio of the mass of a given substance to the total mass of a parcel,

$$q_A \equiv \frac{\rho_c}{\rho_{\rm tot}} = \frac{\mu_c}{\bar{\mu}} \frac{p_c}{p_{\rm tot}} = \frac{\mu_c}{\bar{\mu}} \xi_c, \tag{6.1}$$

where $\bar{\mu}$ is the mean molecular weight of the mixture, $\rho_{\rm tot}$ is the total mass density of the parcel, and the subscript c represents the condensible species of interest (water vapour in this case).

2. Relative humidity: This is the degree of subsaturation, equivalent to the *saturation ratio*

$$\mathrm{RH} = \frac{p_c}{p_{c,\mathrm{sat}}}, \tag{6.2}$$

where p_c is the partial pressure of the condensible (water vapour in this case) and $p_{c,\mathrm{sat}}$ is the saturation vapour pressure of the condensible (given by Clausius-Clapeyron). Note that the relative humidity is equal to the ratio of the volume or number mixing ratio of the condensible in a given mixture to the mixing ratio the substance would have if it were saturated. This is then of course different to the ratio of specific humidity to saturation specific humidity, because specific humidity is defined using mass rather than number or volume mixing ratio.

Another common quantity regarding moisture (sub)saturation that you may see (especially in more public/popular contexts) is the "dew point" or "frost point" depression, defined as $T - T_{\mathrm{dew,frost}}(p_c)$. In this case, $T_{\mathrm{dew,frost}}$ is the condensation temperature following Clausius-Clapeyron (Equation 5.2).

6.3.2 Virtual temperature

Virtual temperature is a quantity used to represent the temperature dry air needs to have the same pressure and density as moist air[23]. We know from ideal gas that dry air follows $p_d = \rho_d R_d T$, while that of the condensible follows $p_c = \rho_c R_c T$. The virtual temperature leverages this to write ideal gas relationship for moist air (the combination of dry and water vapour in this case) as

$$p = \rho R_d T_v, \tag{6.3}$$

where T_v is the virtual temperature. Using the ideal gas relationships for each component of the moist air, we can express virtual temperature as (Equation 2.64 of Holton & Hakim, 2013)

$$T_v = \frac{T}{1 - \frac{p_c}{p}\left(1 - \frac{R_d}{R_c}\right)}. \tag{6.4}$$

Note that $T_v \geqslant T$, and the difference is typically quite small – on the order of a few K. Regardless, these small differences can be critical to the conditional instability of a moist parcel.

[23] I *never* use this quantity because I find it confusing to mix multiple equations of state for an imaginary yet sometimes useful quantity, but you will see this quite often in papers and texts and should be familiar with how it is defined.

7 Radiative transfer I : Blackbody radiation

Reading for Day 7: Pierrehumbert Principles of Planetary Climate, Section 3.1-3.3.

7.1 Description of radiation

7.1.1 Solid angle

Let us consider a ray (i.e., a unit vector pointing in a given direction). This ray can be described by spherical coordinates (θ, ϕ), and contains information about direction but not length. A solid angle Ω is a measure of a set of rays pointing in a range of directions. For example, if we integrate over all directions, $\Omega = 4\pi$, and if we integrate over all upward directions $\Omega = 2\pi$. Figure 7.1 shows the difference between solid angle and the more typical plane angle.

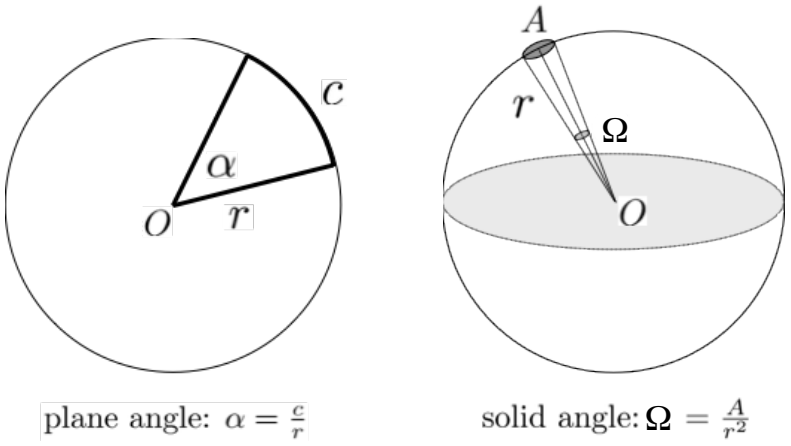


Figure 7.1: Difference between a planar angle (left) and solid angle (right). Figure courtesy Ray Pierrehumbert.

Figure 7.2 shows the differential solid angle, $d\Omega = \sin\theta d\theta d\phi$. We can also express $d\Omega = -d\mu d\phi$, where $\mu \equiv \cos\theta$. Thus, the projected solid angle $d\Omega_\perp = \cos\zeta d\Omega$, where ζ is the *zenith angle* (i.e., the angle between the direction of a ray and the normal to the surface that the ray intersects).

7.1.2 Radiance

Radiance, typically written as I_ν (or in the case of blackbody radiance B_ν) is the rate at which radiation energy flows per unit frequency (or wavelength) per unit solid angle per unit time per unit area of a surface perpendicular to the direction identified by the vector $\vec{\mathcal{L}}$ in Figure 7.2. Radiance is sometimes also called "intensity." The units of radiance are W m^{-2} Hz^{-1} steradian^{-1} (replace Hz with length, e.g., μm if using the per wavelength form). Radiance is positive by definition. Thus, $I_\nu d\nu d\Omega dA$ is the radiation energy between frequency ν and $\nu + d\nu$ and directions within a solid angle $d\Omega$ of the vector $\vec{\mathcal{L}}$ that flows per unit time through the area dA perpendicular to the vector $\vec{\mathcal{L}}$.

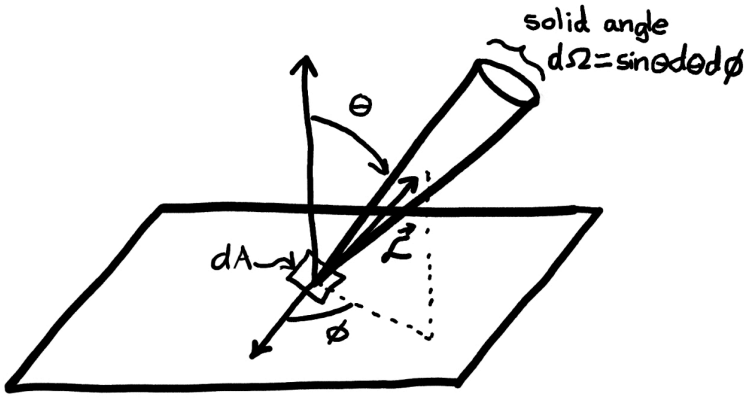


Figure 7.2: Schematic showing radiance through a surface perpendicular to some direction identified by the vector $\vec{\mathcal{L}}$. Figure courtesy Adam Showman.

7.1.3 Flux

Flux (typically denoted F_ν) is the total radiation energy per unit frequency (or wavelength) per unit area, of all directions, that passes through some surface. Thus, we can relate flux to radiance as

$$F_\nu = \int_{4\pi} I_\nu \cos\theta d\Omega = \int_0^{2\pi} \int_0^{\pi} I_\nu(\phi,\theta) \cos\theta \sin\theta d\theta d\phi. \tag{7.1}$$

We can further consider upward and downward fluxes when integrating only over thee upward or downward directions:

$$F_\nu^+ = \int_0^{2\pi} \int_0^{\pi/2} I_\nu(\phi,\theta) \cos\theta \sin\theta d\theta d\phi, \tag{7.2}$$

$$F_\nu^- = -\int_0^{2\pi} \int_{\pi/2}^{\pi} I_\nu(\phi,\theta) \cos\theta \sin\theta d\theta d\phi. \tag{7.3}$$

We have written this such that F_ν^+ and F_ν^- are both positive and $F_\nu = F_\nu^+ - F_\nu^-$. The total flux is the the integral over frequency or wavelength,

$$F \equiv \int_0^\infty F_\nu d\nu = \int_0^\infty F_\lambda d\lambda. \tag{7.4}$$

Note that for *isotropic* radiation, $F_\nu^+ = F_\nu^- = \pi I_\nu$, and thus $F_\nu = 0$.

7.2 Blackbody radiance

A detailed understanding of planetary climate requires an understanding of the way light is transferred, absorbed, and scattered through a planetary atmosphere. This requires a full wavelength-dependent understanding of these processes, which is necessarily complicated. However, we can first gain understanding by using a simpler class of models that consider only the bulk energy balance of the planet, assuming that emission follows that of a blackbody (i.e., the radiation field inside a fixed-temperature enclosed region in thermodynamic equilibrium).

The radiance of a blackbody depends only on temperature and frequency (or wavelength), and is given by the Planck function

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT-1}} = \frac{2k_B^3T^3}{h^2c^2}\frac{u^3}{e^u-1}, \tag{7.5}$$

where h is Planck's constant, c is the speed of light in a vacuum, k_B is Boltzmann's constant, T is temperature, and ν is the frequency of radiation. u is a dimensionless quantity equal to half the ratio of the photon energy to the thermal energy in one degree of freedom of matter in thermodynamic equilibrium with radiation, and is given by

$$u = \frac{h\nu}{k_BT} = \frac{hc}{\lambda k_BT}. \tag{7.6}$$

We can equivalently write the Planck function in terms of wavelength as

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda k_BT}-1} = \frac{2k_B^5T^5}{h^4c^3}\frac{u^5}{e^u-1}. \tag{7.7}$$

Note that Equation (7.5) is *not* equivalent to Equation (7.7) when written equivalently using u rather than ν or λ. This implies that the shape and peak of the blackbody curve is dependent on whether it is written in wavelength or frequency space. Thus, the peak of the blackbody curve has no intrinsic physical meaning, as it depends on the units – see more below in Section 7.2.1.

Example blackbody radiance curves are plotted in Figure 7.3, over the same range of wavelength and frequency. Note first that cooler curves have smaller peak frequencies and

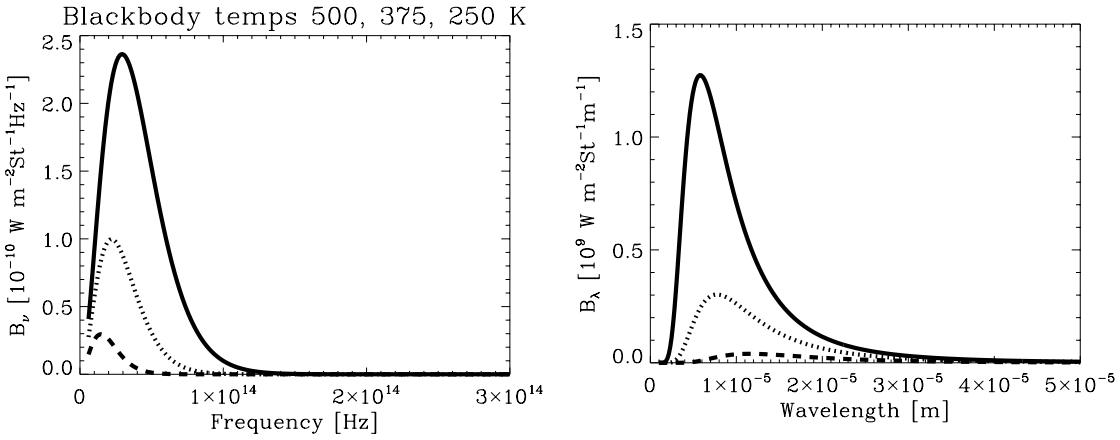


Figure 7.3: Blackbody curves for varying temperature emitters. Note how curves for different temperatures do not cross, and the shape (and peak) of the blackbody curves differs in frequency vs. wavelength space.

longer peak wavelengths. Additionally, cooler curves have smaller radiances, with curves for different temperatures never crossing. Finally, note that blackbodies of different temperature all peak at the same value of u.

We can relate the blackbody radiance B (which is the radiation energy per second meter squared per steradian of solid angle per unit of frequency or wavelength) to the energy flux F (which is the radiation energy per second per meter squared per unit of frequency or wavelength) by taking $F = \pi B$, that is

$$F_\nu(T) = \frac{2\pi h\nu^3}{c^2}\frac{1}{e^{h\nu/kT-1}} = \frac{2\pi k_B^3 T^3}{h^2c^2}\frac{u^3}{e^u-1}, \tag{7.8}$$

and

$$F_\lambda(T) = \frac{2\pi hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda k_B T}-1} = \frac{2\pi k_B^5 T^5}{h^4c^3}\frac{u^5}{e^u-1}. \tag{7.9}$$

7.2.1 Wien's law

The peak of the blackbody curve in frequency or wavelength space (i.e., the frequency where $\partial B/\partial\nu = 0$ or the wavelength where $\partial B/\partial\lambda = 0$) is given by Wien's law:

$$\nu_{\max} = 5.87\times10^{10}\frac{T}{\mathrm{K}}\ \mathrm{Hz}, \lambda_{\max} \quad = 2897\frac{\mathrm{K}}{T}\ \mu\mathrm{m}. \tag{7.10}$$

An alternate way to characterise blackbody radiation more consistently is not by the peak frequency or wavelength, but by the *median emission frequency* that splits the energy flux such that half of it is contained at greater frequencies and half at lesser frequencies. This quantity does not depend on units, and so the median emission frequency or wavelength will occur at the same value of $u = 3.503$, intermediate between the peaks of B_ν and B_λ.

7.2.2 Stefan-Boltzmann law

If one integrates F_ν or F_λ over all frequencies or all wavelengths, respectively, one obtains the total flux emitted by a blackbody, known as the Stefan-Boltzmann law:

$$F(T) = \int_0^\infty \pi B_\nu(T)d\nu = \int_0^\infty \pi B_\lambda(T)d\lambda = \left[\frac{2\pi k_B^4}{h^3c^2}\int_0^\infty \frac{u^3}{e^u-1}du\right]T^4 = \sigma T^4. \tag{7.11}$$

In the equation above, $\sigma = 5.67\times10^{-8}$ W m^{-2} K^{-4} is the Stefan-Boltzmann constant. Note that the Stefan-Boltzmann constant only depends on two constants determining the non-classical nature of the universe (h and c) through the combination h^3c^2 – thus, in principle these two constants could trade off to provide a classical universe ($h\to0$, $c\to\infty$) while h^3c^2 stays fixed, allowing well-behaved blackbody radiation[24].

7.3 Simple energy balance models

We can use blackbody radiation to develop very idealised energy balance models to estimate planetary surface temperature, both ignoring and including the infrared radiative effect of an overlying atmosphere.

7.3.1 Equilibrium temperature

We previously derived the equilibrium temperature of Earth in our activity from Day 1. More generally, we can write a statement of energy balance for an isothermal object with no internal heat source as the power emitted equal to the power absorbed, or

$$f\pi R_p^2\epsilon\sigma T_{\mathrm{eq}}^4 = (1-\alpha)F_{\mathrm{inc}}\pi R_p^2, \tag{7.12}$$

[24]This was noted by Ray in his 2024 slides.

where R_p is the radius of the planet (or other object), f is a factor regarding how well heat is redistributed across the planet ($f = 4$ in the case of full redistribution across the sphere and $f = 1$ in the case of no redistribution whatsoever), ϵ is the emissivity, σ is the Stefan-Boltzmann constant, $T_{\rm eq}$ is the equilibrium emitting temperature, α is the albedo, and $F_{\rm inc}$ is the incoming (stellar) flux. We can thus solve for the equilibrium temperature as

$$T_{\rm eq} = \left[\frac{(1-\alpha)F_{\rm inc}}{f\epsilon\sigma}\right]^{1/4} . \tag{7.13}$$

Note that in the case of $f = 4$ one typically expresses $T_{\rm eq}$ as the "full-redistribution equilibrium temperature," while in the case of $f = 1$ one titles $T_{\rm eq}$ as the "irradiation temperature" or "zero-redistribution equilibrium temperature."

Using the inverse square law, we can express

$$F_{\rm inc} = F_\star = \frac{L_\star}{4\pi r_{\rm orb}^2}, \tag{7.14}$$

where $L_\star = 4\pi R_\star^2 \sigma T_\star^4$ from Stefan-Boltzmann and $r_{\rm orb}$ is the planet-star separation. Thus, for mature planets that have cooled off from formation (or more generally, planets that receive much more incident stellar power than their intrinsic cooling luminosity), one can approximate the planetary temperature by the equilibrium temperature (assuming here $\epsilon = 1$):

$$T_{\rm eq} = T_\star \left[\frac{(1-\alpha)}{f}\right]^{1/4} \sqrt{\frac{R_\star}{r_{\rm orb}}} . \tag{7.15}$$

In Equation (7.15), α is the Bond albedo, which is the fraction of total energy incident on the planet that is not absorbed and re-radiated (i.e., the amount that is scattered/reflected by clouds, haze, ice, and gas).

7.3.2 One-layer greenhouse & two-layer atmosphere

Please get in groups of two or three to work on the following activity deriving the surface temperature for a one-layer atmosphere, alongside the upper-atmospheric temperature in thermal energy balance.

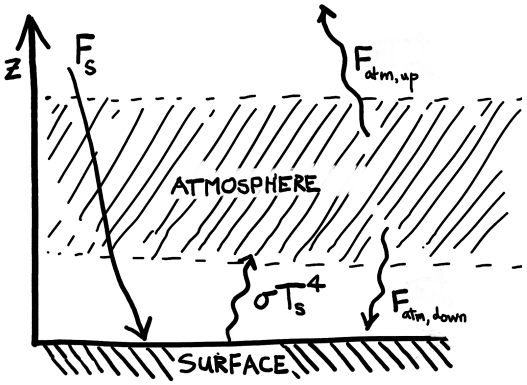


Figure 7.4: Schematic of the one-layer atmospheric greenhouse problem, courtesy Adam Showman.

1. Consider the one-layer atmosphere model shown in Figure 7.4. Assume that the atmosphere is completely transparent to shortwave radiation, and completely opaque to longwave radiation. Write down an expression for the surface energy balance, and solve it to find the surface temperature as a function of F_s and $F_{\text{atm,down}}$. How does this compare to the case without an atmosphere (i.e., the equilibrium temperature discussed above)?

2. Let us now consider what happens at very low pressures, where the gas becomes nearly transparent to the upwelling infrared radiation from below with $\epsilon_{\text{IR}} \ll 1$. Again neglect shortwave absorption in this region, so that the energy balance in this low pressure region is still between absorption and emission of IR. Given that the flux of IR upwelling radiation from the upper troposphere (below the nearly transparent low pressure region) is σT_e^4, write down another energy balance expression for the low-pressure region – to do so you should assume that the fraction of radiation absorbed (the absorptivity) is equal to the fraction emitted (the emissivity) – this is Kirchoff's law. Solve this to find the temperature of this low pressure region, which is termed the "skin temperature" for reasons we will discuss.

3. Now that we have simple models for the troposphere (where the greenhouse effect is strong) and the stratosphere (where the atmosphere converges to the skin temperature), put the two pieces together to develop a two-layer model of the temperature profile in a planetary atmosphere. Assume that the lapse rate in the troposphere is $dT/dz = \Gamma$.

8 Radiative transfer I: Absorption and emission

Reading for Day 8: Pierrehumbert Principles of Planetary Climate, Section 3.5-3.6.

We will now go beyond the blackbody approximation and consider absorption and emission of radiation through an idealised slab (i.e., patch of atmosphere) to derive the fundamental equation governing (non-scattering) radiative transfer.

8.1 Lambert's law and absorption

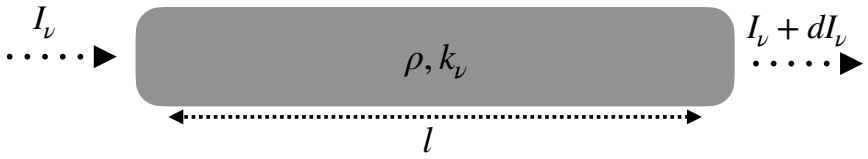


Figure 8.1: Purely absorbing slab with a density ρ, length l, and absorption coefficient k_ν (or k_λ).

Consider radiation with an initial radiance at a given wavelength I_λ (or frequency I_ν) that impinges upon an absorbing and/or emitting slab that has mass density ρ, absorption coefficient k_ν (units $\mathrm{m}^2\ \mathrm{kg}^{-1}$), and thermal radiance B_ν (in the case where it is emitting).

Let us first consider the case of a purely absorbing slab, with $B_\nu = 0$. Lambert's law is an empirical relationship which shows that the fractional change in radiance due to absorption is proportional to the mass per area in the slab. Thus, Lambert's law states that this slab absorbs radiation, causing a decrease in radiance leaving the slab (at distance dl) of

$$dI_\nu = -I_\nu k_\nu \rho dl \quad \text{Lambert's law.} \tag{8.1}$$

We can further define the *absorptivity* at any given frequency, which is the fraction of incident radiance of a given frequency absorbed by the material. In the case of an infinitesimal slab, $a_\nu = k_\nu \rho dl$, implying that we can express Lambert's law as

$$dI_\nu = -I_\nu a_\nu. \tag{8.2}$$

8.2 Kirchhoff's law and emission

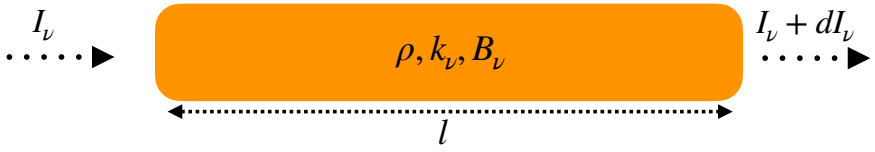


Figure 8.2: Emitting slab with a density ρ, length l, blackbody radiance B_ν, and (in the case of both an absorbing and emitting slab) an absorption coefficient k_ν (or k_λ).

In the last lecture, we introduced the *emissivity* ϵ_ν – this is the ratio of the radiance emitted by a substance to the blackbody radiance B_ν. Thus, blackbodies have $\epsilon_\nu = 1$, while real substances which are imperfect emitters rather than blackbodies have $\epsilon_\nu < 1$.

Kirchhoff's law states that substances in thermodynamic equilibrium emit as efficiently as they absorb, and thus

$$\epsilon_\nu = a_\nu \quad \text{Kirchoff}'\text{s law.} \tag{8.3}$$

Note that Kirchhoff's law is only valid if the substance is in thermodynamic equilibrium. This is a consequence of the second law of thermodynamics – for instance, if an object with $\epsilon \neq a$ is placed in a parcel of gas molecules emitting as blackbodies, conduction will attempt to equalise the temperatures, while radiation will try to make them different. This would lead to a violation of the second law of thermodynamics, as thermal energy would be transferred from the cold to the hot region – thus, $\epsilon = a$ (both at specific frequencies or wavelengths and integrated over all frequencies or wavelengths).

Therefore, the emission from an infinitesimal slab is $dI_\nu = \epsilon_\nu B_\nu$ which from Kirchhoff's law must be $dI_\nu = a_\nu B_\nu$. Replacing $a_\nu = k_\nu \rho dl$, the change in emitted radiation must be

$$dI_\nu = B_\nu k_\nu \rho dl \quad \text{emitting} - \text{only slab.} \tag{8.4}$$

8.3 Schwarzschild's equation

We can sum Lambert's and Kirchhoff's laws together to get the total change in intensity across the slab. Doing so and dividing through by dl, we can write Schwarzschild's equation for radiative transfer ignoring scattering:

$$\frac{dI_\nu}{dl} = \rho k_\nu (B_\nu - I_\nu) \quad \text{Schwarzschild}'\text{s equation.} \tag{8.5}$$

This equation tells us how the radiance from the slab is affected by a combination of absorption and emission, and we will provide solutions from it under certain idealisations in the following lecture.

For now, let us turn to discuss two related non-dimensional quantities (optical path and optical depth) which we will use throughout the radiative transfer module.

8.4 Optical path and optical depth

We can define the optical path $\tilde{\tau}$ as

$$\tilde{\tau}_\nu = \int_0^l \rho k_\nu dl'. \tag{8.6}$$

This is a convenient measure of the amount of absorption/emission along a path followed by incident radiation, and as you will see it represents the e-folding distance for absorption/emission.

Our definition of τ allows us to re-write Schwarzschild's equation as

$$\frac{dI_\nu}{d\tilde{\tau}_\nu} = B_\nu - I_\nu. \tag{8.7}$$

We can define a related quantity to optical path, known as the optical depth τ, which is related to the optical path as

$$d\tau = -\mu d\tilde{\tau}, \tag{8.8}$$

where $\mu \equiv \cos\theta$. We can thus express Schwarzschild's equation ignoring scattering in optical depth coordinates as

$$\mu \frac{dI_\nu}{d\tau} = I_\nu - B_\nu. \tag{8.9}$$

Note that optical depth τ is essentially a vertical coordinate, and it increases downward (from the top of the atmosphere). This is because for upward-oriented radiances $\mu > 0$, and for downward-oriented radiances $\mu < 0$.

8.4.1 Photosphere pressure

It's clear from the above derivations that the optical depth (equivalent to the optical path if $\mu = 1$) is a critical parameter for estimating radiative properties of planetary atmospheres. If the optical depth $\tau \gg 1$, then the region is optically thick and $I_\nu \approx B_\nu$. If the optical depth is $\tau \ll 1$, the region is optically thin and $I_\nu \approx I_\nu(0)$. However, if $\tau \approx 1$, then *both* incoming starlight is absorbed and radiation escapes to space – this special region of the atmosphere is called the "photosphere."

We can estimate the photosphere using the definition of optical depth,

$$\tau \approx 1 = \int_z^\infty k_\nu \rho dz = k_\nu \frac{p}{g}, \tag{8.10}$$

where we have integrated assuming hydrostatic balance ($-\rho dz = dp/g$).

If we assume that the absorption coefficient (equivalently, the opacity) is constant we can estimate the photosphere pressure where $\tau = 1$:

$$p_{\tau=1} \approx \frac{g}{k_\nu}. \tag{8.11}$$

For a typical hot Jupiter exoplanet with $g = 10$ m s^{-2} and $k_{\rm IR} = 4 \times 10^{-3}$ cm^2 g^{-1}, $p_{\rm IR} \approx 0.25$ bars. Jupiter itself has an absorption coefficient of $\sim 10^{-3}$ m^2 kg^{-1}, and a gravity of 26 m s^{-2}, implying $p_{\tau \approx 1} \sim 0.3$ bars. Typically the thermal photosphere is at the ~ 100 mbar $-$ 1bar pressure level for irradiated planets.

8.5 Probing the thermal structure

The temperature-pressure profile can be directly probed by measuring spectral features in emission spectra of planets. Figure 8.3 demonstrates how the temperature-pressure profile is coupled to spectral features. In non-inverted atmospheres where the temperature decreases with increasing pressure, spectral features appear as dips in the emission spectrum at the wavelengths where the opacity from absorbers is large. This is because the radiatively active molecules absorb the thermal emission from below and re-emit it at their (cooler) local temperature, causing a reduction in the planetary flux at that wavelength. In this case, molecule 2 has a larger relative dip (i.e., a larger absorption feature) because it is optically thick at lower pressures than molecule 1.

Conversely, in thermally inverted atmospheres there is a bump, or emission feature, rather than an absorption feature. This is because the molecules now are re-emitting the radiation they absorb at hotter local temperatures, causing an increase in the planet flux in the wavelengths at which they are opaque.

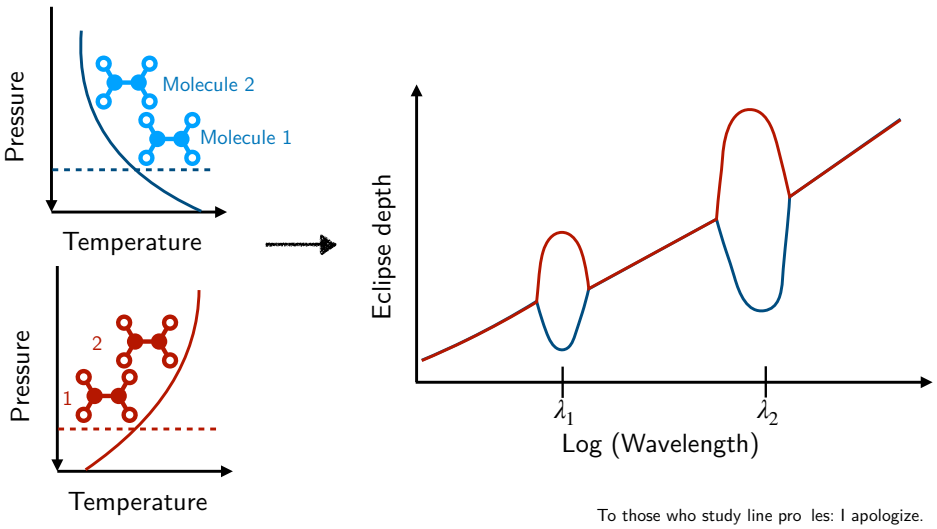


Figure 8.3: Cartoon demonstrating how absorption and emission features arise from non-inverted and inverted temperature pressure profiles, respectively. Note that here the y-axis of the spectrum shows the "eclipse depth" relevant for exoplanets, which is the planetary flux normalised by the flux of another blackbody (the star in this case).

The fundamental ideas above can be generalised when knowing the full wavelength-dependence of the absorption coefficients for various gases to translate the observed thermal emission of a planet into an estimated temperature-pressure profile. Models which solve for this inverse problem are known as "retrievals," as they are benchmarked from Earth-observing satellites (as well as planetary missions) and now applied more broadly to exoplanets.

9 Radiative transfer I: Schwarzschild's equation

Reading for Day 9: Pierrehumbert Principles of Planetary Climate, Section 4.1-4.2.

Note: we will mostly be covering the Day 8 notes in Lecture 9, and will cover these notes as time allows.

9.1 Schwarzschild's equation recap

Recall from our previous day's notes that we can write Schwarzschild's equation for radiative transfer ignoring scattering as

$$\frac{dI_\nu}{dl} = \rho k_\nu (B_\nu - I_\nu) \text{ Schwarzschild}'\text{s equation} \tag{9.1}$$

in length coordinates, or equivalently in optical path coordinates as

$$\frac{dI_\nu}{d\tilde{\tau}} = B_\nu - I_\nu, \tag{9.2}$$

or optical depth coordinates as

$$\mu \frac{dI_\nu}{d\tau} = I_\nu - B_\nu, \tag{9.3}$$

where $\mu = \cos\theta$, with θ the zenith angle between the light path and the vertical in the atmosphere. Moving forward, we'll use optical path coordinates and drop the tilde for simplicity (and I'll be clear when we're using optical depth instead).

9.2 Solutions to Schwarzschild's equation

9.2.1 Pure absorption: Beer's law

In the case where the medium is purely absorbing, $B_\nu = 0$ and we can integrate Schwarzschild's equation to solve for the radiance as a function of optical path as

$$I_\nu = I_\nu(0) e^{-\tau} \quad \text{Beer}'\text{s law.} \tag{9.4}$$

Thus, in the case of pure absorption the optical path τ can be interpreted as the e-folding scale of the radiance. If $\tau \gg 1$, then the region is optically thick and $I_\nu \approx 0$. If $\tau \ll 1$, then the region is optically thin and $I_\nu \approx I_\nu(0)$.

Beer's law has a wide range of applications, especially regarding the propagation of sunlight through Earth's atmosphere, as the thermal radiance of Earth is quite small in the optical (and thus we can well approximate $B_\nu \approx 0$ in Schwarzschild's equation). We discuss one such application to the slant geometry in Section 9.3.

9.2.2 Isothermal atmosphere

We previously considered atmospheres with no emission to derive Beer's law for transmission, but now we will consider atmospheres with emission. First, let us consider an isothermal medium where B_ν is constant. We can change variables to simplify our derivation, using

$$\begin{aligned} \chi &= I_\nu - B_\nu \\ d\chi &= dI_\nu. \end{aligned} \tag{9.5}$$

Thus, we can write Schwarzschild's equation with the change of variable as

$$\frac{d\chi}{d\tau} = -\chi. \tag{9.6}$$

Integrating, we find

$$\chi = \chi_0 e^{-\tau}. \tag{9.7}$$

Plugging back in for our change of variables, we find the isothermal solution of Schwarzschild's equation:

$$I_\nu = B_\nu + (I_\nu(0) - B_\nu)\, e^{-\tau}. \tag{9.8}$$

Let's consider some limits of this isothermal solution. When the optical path is very small ($\tau \ll 1$), the radiance is the same before and after entering the slab ($I_\nu \approx I_\nu(0)$), as in the case of Beer's law above. When the optical path is very large ($\tau \gg 1$), $I_\nu \approx B_\nu$ – that is, an infinitely opaque isothermal medium acts as a blackbody. Finally, if $I_\nu(0) = 0$, then $I_\nu = B_\nu(1 - e^{-\tau})$, and the radiance e-folds toward a blackbody with increasing optical thickness.

9.2.3 General solution

There of course is also a general solution to Schwarzschild's equation ignoring scattering. To derive it, we can use an integrating factor e^τ such that

$$\frac{d}{d\tau}(e^\tau I_\nu) = e^\tau B_\nu, \tag{9.9}$$

and thus

$$e^{\tau'} I_\nu|_0^\tau = \int_0^\tau e^{\tau'} B_\nu d\tau'. \tag{9.10}$$

One can then rearrange to find the general solution to Schwarzschild's equation,

$$I_\nu(\tau) = I_\nu(0)e^{-\tau} + \int_{\tau=0}^{\tau} e^{\tau'-\tau} B_\nu d\tau. \tag{9.11}$$

Note that the first half of the right hand side is simply Beer's law, and the second half is the additional contribution from thermal emission.

9.3 Application of Beer's law: slant vs. normal optical path

Beer's law can be applied to transmission geometry (i.e., the propagation of slant rays through an atmosphere – this is relevant both for exoplanet observations as well as stellar occultations of Earth and Solar System objects), as in most cases we can ignore the atmospheric emission contribution to the transit depth. This then allows us to relate the observed flux deficit to an optical path, and thus a region in the atmosphere that is probed via transmission spectroscopy, given that regions with $\tau > 1$ are opaque to transmitted starlight.

Let us consider a *slant* optical path through the atmosphere τ – this slant optical path is significantly larger than the typical optical depth. Figure 9.1 shows the geometry of the slant optical path (in the direction of x) relative to the vertical optical depth (in the direction of z). We will now follow the derivation in Fortney (2005)[25]. Recall that an isothermal atmosphere

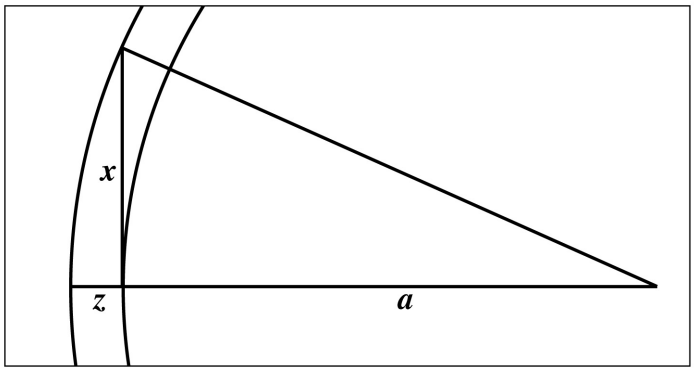


Figure 9.1: Geometry used to derive the slant optical path of transmission through an atmosphere. Figure adapted from Fortney (2005).

in hydrostatic equilibrium has a dependence of pressure p and equivalently number density n on height above a given level z_0 as

$$\begin{aligned} p(z) &= p(z_0)\ \exp\left(-\frac{(z-z_0)}{H}\right), \\ n(z) &= n(z_0)\ \exp\left(-\frac{(z-z_0)}{H}\right), \end{aligned} \tag{9.12}$$

where $H = RT/g$ is the pressure scale height, which is equal to the density scale height for an isothermal atmosphere. From Figure 9.1, we can note that

$$a^2 + x^2 = (a+z)^2 = a^2 + 2az + z^2. \tag{9.13}$$

If $2az \gg z^2$ (i.e., if the atmosphere is thin relative to the planetary radius), then $z \approx x^2/2a$. As a result, we can write the dependence of number density on x as

$$n(x) = n_0\ \exp\left(-\frac{x^2}{2aH}\right). \tag{9.14}$$

If we integrate this dependence over x to find the integrated density N_H from horizon to horizon, we find

$$N_H = \int_{-\infty}^{\infty} n(x)dx = n_0 \frac{1}{2}\sqrt{\pi}\sqrt{2aH}\ \mathrm{erf}\left(\frac{x}{\sqrt{2aH}}\right)\Big|_{-\infty}^{\infty} = n_0\sqrt{2\pi aH}. \tag{9.15}$$

The ratio of the horizontally integrated density to the vertically integrated density is equal to the ratio of the horizontal to vertical optical path. This is

$$\frac{N_H}{N_V} = \frac{\tau_H}{\tau_V} = \sqrt{\frac{2\pi a}{H}}. \tag{9.16}$$

[25]This is derived specifically in the context of to exoplanets, but can be applied to any slant optical path, also relevant for Earth observation.

Because $a >> H$, this value is always much larger than one – for Earth it is ~ 75, and for Jupiter it is ~ 128. Generally speaking, due to the high slant optical path, transit spectra probe low pressures of ≈ 1 mbar, with higher resolution observations probing even lower pressures in absorption lines. This also causes condensate clouds and hazes to have a large impact on the depth of transmission spectral features, as they move the continuum (deepest region that can be probed) to lower pressures (Fortney, 2005).

10 Radiative transfer catch-up: Solutions to Schwarzschild's equation

We are almost exactly one lecture behind, so for today we will mostly cover the Day 9 notes for today alongside the below activity on transmission spectra. Some notes on transmission spectroscopy are also included in case you are interested in learning more.

10.1 Practical application: transmission spectroscopy

10.1.1 Transmission spectroscopy

Transmission spectroscopy probes the atmospheres of exoplanets by studying the transmission (or filtering) of light from the host star through the limb of an exoplanet that appears to occult the host star. Note that the same physics also applies to occultations in our Solar System, either Solar occultations (where one probes the filtered sunlight through the object's limb) or stellar occultations (where one probes filtered starlight from a more distant star).

Figure 10.1 shows the geometry of a transit event, along with the geometry of the secondary eclipse that is used to derive planetary emission spectra (that we discussed in the last class). Transmission spectra are generally measuring small deviations (at most on the orders

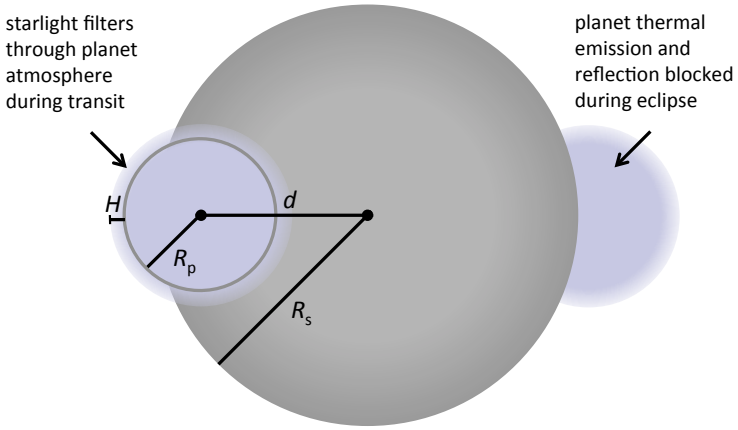


Figure 10.1: Geometry of transmission and emission observations. Here the stellar radius is R_s, planet radius is R_p, the separation of centers of planet and star is d, and the pressure scale height of the planet's atmosphere is H. Figure adapted from Kreidberg (2017).

of parts per thousand) on top of the larger transit signal. Similarly to the transit method itself, transmission spectral signatures are larger for planets orbiting smaller stars. However, there is also a strong dependence of the transmission spectra signal on the atmospheric composition, temperature, and gravity of the orbiting planet, which we derive below.

The transmission spectral features are a small atmospheric contribution imprinted on top of the total transit depth. Note that the transit depth for the solid-body planet is $\delta = (R_p/R_s)^2$ (this can be derived by considering the ratios of the projected solid angles of the planet and host star). The transit depth including the atmospheric contribution is then

$$\delta_\lambda = \frac{(R_p + A_{H,\lambda})^2}{R_\star^2}, \tag{10.1}$$

where $A_{H,\lambda}$ is the apparent atmospheric height which is being probed. We can write the apparent atmospheric height as a number of scale heights, i.e.,

$$A_{H,\lambda} = nH = n\frac{RT}{g} \tag{10.2}$$

where H is the (isothermal) pressure scale height. As a result, we can isolate the transmission spectral feature amplitude (i.e., the contribution due to the atmosphere alone) as

$$\delta_{\lambda,atm} = \frac{(R_p + nH)^2}{R_\star^2} - \frac{R_p^2}{R_\star^2} \approx \frac{2R_p nH}{R_\star^2} \approx \frac{2R_p nRT}{gR_\star^2}. \tag{10.3}$$

Typically, $n \approx 2$ for low spectral resolution observations of cloud-free atmospheres (Kreidberg, 2017). As a result, we expect the transmission spectral feature amplitudes to be larger for hotter and lower-gravity planets, which have a larger scale height. We also expect transmission spectral features to be larger for low mean molecular weight atmospheres, as decreasing mean molecular weight increases R, which also increases the scale height. Note that our derivation does not include the effects of clouds, which also mute spectral features by increasing the optical depth of the atmosphere at low pressures.

10.1.2 The case of WASP-43b

Figure 10.2 shows a transmission spectrum observed with the Hubble Space Telescope/Wide Field Camera 3 (HST/WFC3) (bottom) of the hot Jupiter WASP-43b. This

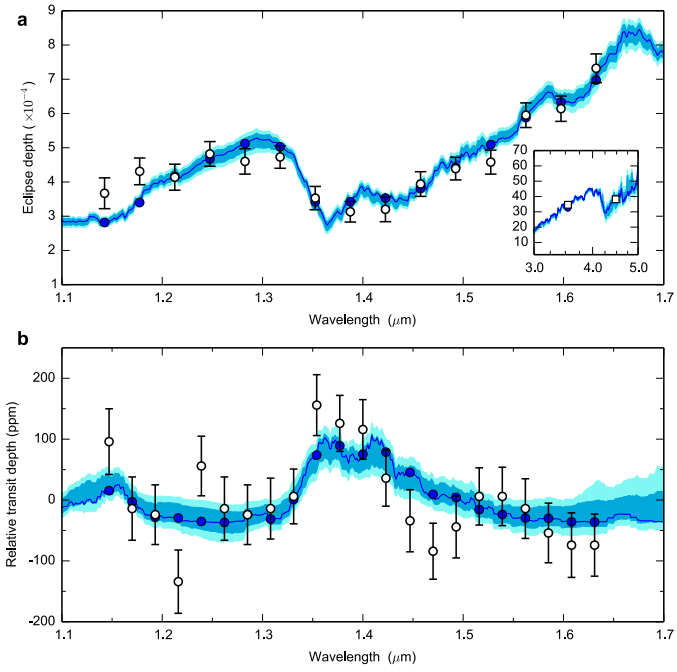


Figure 10.2: Emission spectrum (top) and transmission spectrum (bottom) of WASP-43b observed with the Hubble Space Telescope Wide Field Camera 3 instrument (inset on top shows observations with the Spitzer Space Telescope). Note the spectral feature (bump) in the transmission spectrum at 1.4 μm due to water vapour absorption, which also causes an absorption feature (dip) in the emission spectrum. Figure adapted from Kreidberg (2017).

shows the change in transit depth as a function of wavelength, i.e., isolating only the atmospheric contribution A_H to the total transit depth. There is a bump in the spectrum at a wavelength of 1.4 μm. This bump corresponds to an increase in the effective transit depth, i.e., an increase in the effective size of the planet at 1.4 μm. As a result, this is an

absorption feature, as the planet absorbs more light at 1.4 μm than at other wavelengths in the instrumental range. Note that this bump in the transmission spectrum at 1.4 μm aligns well with a dip in the emission spectrum (top) at the same wavelength. Note that this dip in emission spectra similarly corresponds to an absorption feature due to the same species causing the bump in the transmission spectrum.

Given a transmission spectral feature at a given wavelength, astronomers can then infer the species that is causing this feature by comparing with model atmospheres. The wavelength-dependence of the atmospheric absorption is determined by the absorption cross section of a given species σ, which is related to opacity (equivalently absorption coefficient) κ as

$$n\sigma_\lambda = \kappa_\lambda \rho, \tag{10.4}$$

where subscripts indicate a wavelength-dependence. Figure 10.3 shows the dominant cross-sections in a typical hot Jupiter atmosphere in chemical equilibrium at Solar metallicity, a temperature of 1500 K, and a pressure of 300 mbar.

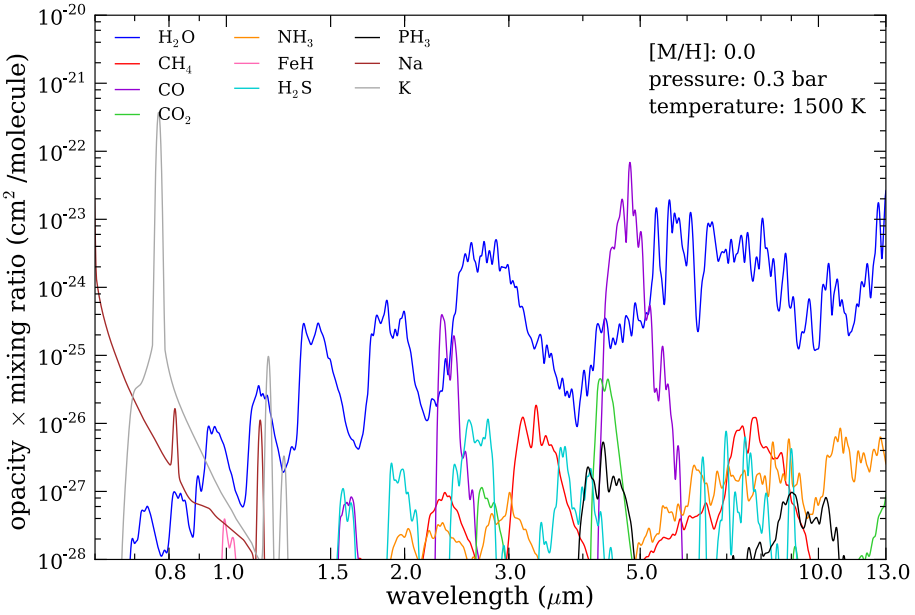


Figure 10.3: Cross sections of various species in a Solar composition atmosphere of a hot Jupiter at a pressure of 300 mbar. Note the strong features due to water vapor and carbon monoxide in the near-infrared. Figure adapted from Kreidberg (2017).

10.1.3 Transmission spectra activity

Please get in groups of two or three to work on the following activity on transmission spectroscopy.

1. Recall that there is one clear absorption feature in both the transmission and emission spectrum of WASP-43b, centred at a wavelength of 1.4 μm. What species causes this absorption feature?

2. WASP-43b has a mass of 565.7 $M_\oplus$, radius of 10.424 $R_\oplus$, and a full-redistribution equilibrium temperature of 1427 K. Estimate the scale height of the atmosphere of WASP-43, assuming that its atmosphere is made entirely of hydrogen and helium, with a specific gas constant $R = 3600$ J kg^{-1} K^{-1}. Compare this to the scale height of Earth, assuming for Earth that the atmospheric temperature is the global-mean temperature (288 K).

3. Estimate the effective height of the largest spectral features of WASP-43b in the HST/WFC3 bandpass, in kilometers. How many scale heights does this feature probe? Note: to answer this, you will need the radius of the host star WASP-43, which is $R = 0.757 R_\odot = 5.27 \times 10^5$ km.

4. Is WASP-43b a good transmission spectroscopy target relative to other gas giant planets *at a similar equilibrium temperature*?

11 Radiative transfer II: The two-stream equations

My treatment will diverge from Prof. Pierrehumbert's book for this and the next lecture to go into more detail in the two-stream equations and various approximations therein. That said, for Day 11 it may be useful to read Pierrehumbert Principles of Planetary Climate, Section 4.3.

11.1 Why should I care? The effect of radiation on thermal structure

Though we have so far only considered the effect of the atmosphere on the radiation field, radiation also affects the atmosphere by causing heating and/or cooling. Consider a slab of atmosphere shown in Figure 11.1, which has thickness Δz and an impinging radiative flux $F(z)$ coming in to the bottom and flux $F(z + \Delta z)$ emerging from the top. The differences

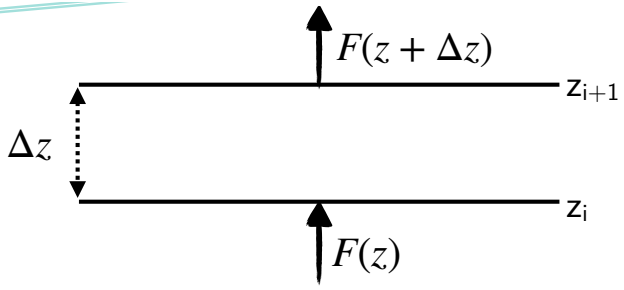


Figure 11.1: Schematic showing the changing flux passing vertically through a layer of atmosphere, which as shown in the notes is connected to the heating rate.

in flux will cause heating/cooling, as

$$-\left[F(z+\Delta z) - F(z)\right] = q\rho\Delta z, \tag{11.1}$$

where q is the thermodynamic heating/cooling per mass and ρq is the heating/cooling per volume. Thus, the volumetric heating

$$q\rho = -\frac{F(z+\Delta z) - F(z)}{\Delta z} = -\frac{\partial F}{\partial z}. \tag{11.2}$$

Usually, the dominant gradient of the flux is in the vertical alone, implying that Equation (11.2) well approximates the total radiative/cooling. However, more generally, for full 3D radiation the volumetric radiative heating/cooling is

$$\rho q = -\nabla \cdot \mathbf{F}, \tag{11.3}$$

where $\mathbf{F}$ is the full 3D flux vector. Thus, radiative heating/cooling is driven by flux convergence/divergence.

11.1.1 Radiative equilibrium

The specific state of radiative equilibrium occurs when

$$\nabla \cdot \mathbf{F} = 0 \quad \text{radiative equilibrium}, \tag{11.4}$$

which implies there is no net radiative flux and thus not heating/cooling. Typically atmospheres are *not* in a state of *local* radiative equilibrium, even though they are generally

close to an overall top-of-atmosphere energy balance where absorbed sunlight equals outgoing longwave radiation to space. The reason the atmosphere is not in a local radiative equilibrium is because atmospheric motions typically occur driven by the radiative equilibrium state. Note that radiative equilibrium is still a very useful zeroth-order model of an atmosphere, especially treatments that do not include dynamics. Radiative equilibrium also is the state to which the atmosphere is constantly being forced by the atmosphere, while dynamics acts to take it away from a state of radiative equilibrium.

11.2 Approximate radiative transfer equations

Solving the generalised Schwarzschild's equation (e.g., Equation 9.11) in 3D space encompassing all directions (various optical paths) and over all wavelengths or frequencies is very challenging – it is generally not feasible analytically except in certain very simplified limits which we have previously covered. In this and the following lectures we will discuss approximations to the radiative transfer equation to enable useful solutions. We will use these idealised solutions to then solve for the atmospheric temperature structure in cases of increasing complexity.

11.2.1 Treatment of Solar/stellar flux

Incoming stellar radiation comes in two forms: 1) the direct stellar beam, which is focused over a small range of solid angles; 2) diffuse stellar radiance associated with scattered sunlight and infrared thermal emission, which occurs over a broad range of solid angles. It will simplify the setup of the stellar radiative fluxes to separate out the radiance associated with the direct stellar beam from the diffuse radiance.

First, the stellar flux is

$$F_{\nu,\star} = I_{\nu,\star} d\Omega_\star, \tag{11.5}$$

where $d\Omega_\star$ is the range of solid angles subtended by the stellar beam. Given that the stellar direct beam is much stronger than the diffuse radiance, we can then neglect thermal emission or scattering of light (either stellar or thermal) into this direct stellar beam.

Thus, extinction (including both absorption of light in the beam and scattering of light out of the beam) dominates the change in stellar flux with distance through the atmosphere. Thus, we can write the stellar flux as

$$F_{\nu,\star} = F_{\nu,\star}(0) e^{\int \frac{e_\nu \rho}{\mu_\star} dz'}, \tag{11.6}$$

where $e_\nu = a_\nu + s_\nu$ is the extinction coefficient, equal to the absorption coefficient plus the scattering coefficient[26], and $\mu_\star = \cos\theta_\star$ is the zenith angle of the star. Note that the integration in Equation (11.6) is done from higher in the atmosphere (e.g., the top of atmosphere) to lower in the atmosphere (or even the surface), implying that the change in z is negative and thus the flux of the direct stellar beam will decrease due to extinction.

Now that we have specified the direct stellar beam, we only need to solve Schwarzschild's equation for the diffuse component of the flux. Ignoring scattering, there is no diffusive visible-wavelength radiation and thus we only need to solve the radiative transfer equation at infrared wavelengths where planetary thermal emission is important.

[26]Note that we are still neglecting scattering at this point.

11.2.2 The two-stream equations

The two-stream equations are an approximate form of the radiative transfer equation (Equation 9.2, referred to as Schwarzchild's equation in the case of no scattering) that track the bulk fluxes (and thus the bulk heating rate) rather than studying the details of how the radiance depends on direction. To derive these equations, we can make three primary assumptions:

1. The plane-parallel assumption: We assume that the radiances and fluxes depend only on height z and not on the horizontal directions θ, ϕ. Typically, the horizontal length scales of atmospheres greatly exceed the vertical length scales, implying that $\nabla \cdot \mathbf{F}$ is dominated by the $\partial F/\partial z$ term. This then allows us to reduce the number of spatial dimensions to one (z alone) and further assume a Cartesian geometry rather than using the full spherical geometry.

2. The two-stream approximation: We assume that the upward and downward radiances I_ν^+ and I_ν^- are individually constant with direction but are not equal to one another. Thus, we need only to keep track of two directions – upward and downward radiances (and fluxes).

3. The semi-isotropic assumption: We need to make some assumption about the directionality of the radiances in order to calculate I_ν^+ and I_ν^-. We will assume that the distribution of radiance is approximately isotropic, allowing us to derive a approximate angle-averaged forms of the radiative transfer transfer equation in each direction (see more below).

First, recall that we can write the upward and downward fluxes as

$$F_\nu^+ = \int_{\Omega^+} I_\nu(\vec{\omega})\ \cos\theta d\Omega, \tag{11.7}$$

$$F_\nu^- = \int_{\Omega^-} I_\nu(\vec{\omega})\ \cos\theta d\Omega, \tag{11.8}$$

where $\vec{\omega}$ is the ray pointing in the direction of the radiance. We can thus re-express Schwarzschild's equation for the upward radiance in optical depth coordinates (Equation 9.3) in terms of radiative flux as

$$\mu \frac{dF_\nu^+}{d\tau} = \int_{\Omega^+} I_\nu(\vec{\omega}) d\Omega - 2\pi B_\nu. \tag{11.9}$$

Now, we can make the *semi-isotropic* assumption by assuming that I_ν is isotropic, leading the integral $\int_{\Omega^+} I_\nu(\vec{\omega})d\Omega \approx 2F_\nu^+$ and implying that the mean zenith angle $\bar{\mu} = \cos\bar{\theta} = \cos 60^\circ = \frac{1}{2}$. Thus, we can write the semi-isotropic upward flux as

$$\frac{1}{2}\frac{dF_\nu^+}{d\tau} = \frac{dF_\nu^+}{d\tau^*} = F_\nu^+ - \pi B_\nu, \tag{11.10}$$

where we have defined $\tau^* = \tau/\cos\theta = \tau/\bar{\mu}$, where again $\bar{\mu} = 1/2$ is the mean zenith angle under the isotropic approximation. We can equivalently write an expression for the semi-isotropic downward flux as

$$\frac{dF_\nu^-}{d\tau^*} = -F_\nu^- + \pi B_\nu. \tag{11.11}$$

Thus, the semi-isotropic approximation has led to a reduced form of the radiative transfer equation using an appropriate angle-averaging of the equations for the upward and downward radiance. Note that the choice of $\bar{\mu} = 1/2$ here by no means unique – we could have equivalently chosen to use the average of $(\cos\theta)^2$ instead, which would have led to $\bar{\mu} = 2/3$ (and thus a factor of 1.5 in our conversion from τ^* to τ rather than a factor of 2) – more generally, often τ^* is taken to be $\tau^* = \frac{5}{3}\tau$. In principle, $\bar{\mu}$ can be chosen for a specific problem in order to best fit an angle-resolved calculation in a particular regime. However, we will continue to use $\bar{\mu} = 1/2$ where needed throughout these notes for simplicity (and consistency with Pierrehumbert, 2010).

The two-stream Equations (11.10) and (11.11) above can be recast in terms of the *net* diffuse flux $F_\nu = F_\nu^+ - F_\nu^-$ and the *mean* radiance

$$\bar{I}_\nu = \frac{I_\nu^+ + I_\nu^-}{2}, \tag{11.12}$$

by adding and then subtracting the two-stream equations to find two separate equations

$$\begin{aligned} -\frac{d}{d\tau^*}(F_\nu^+ - F_\nu^-) + F_\nu^+ + F_\nu^- &= 2\pi B_\nu, \\ \frac{d}{d\tau^*}(F_\nu^+ + F_\nu^-) - (F_\nu^+ - F_\nu^-) &= 0. \end{aligned} \tag{11.13}$$

Rearranging and noting that given $F_\nu^+ = \pi I_\nu^+$ and $F_\nu^- = \pi I_\nu^-$ implies $F_\nu^+ + F_\nu^- = 2\pi\bar{I}_\nu$, we find the typical expression of the two stream equations

$$-\frac{dF_\nu}{d\tau^*} + 2\pi\bar{I}_\nu = 2\pi B_\nu, \tag{11.14}$$

$$2\pi\frac{d\bar{I}_\nu}{d\tau^*} - F_\nu = 0. \tag{11.15}$$

We now continue to solve these two-stream equations to study their consequences for the atmospheric thermal structure.

11.2.3 Solution of the two-stream equations in the case of a semi-grey atmosphere with zero visible opacity

We start with the simplest possible case of a solution of the two stream equations. This is a one-dimensional atmosphere overlying a surface, where the atmosphere is completely transparent to sunlight and absorbs thermal radiation with an absorption coefficient that is independent of wavelength. This atmosphere with absorption in the infrared independent of wavelength is termed semi-grey[27].

[27] This type of atmosphere is often just called grey for simplicity, even though in principle it is not fully grey because the absorption coefficient is zero in visible wavelengths.

Let us further assume that the atmosphere and surface are both in radiative equilibrium with $\frac{\partial F}{\partial z} = 0$. This implies that the net thermal flux F_T and the net incoming stellar flux F_S are in balance, i.e.,

$$F_T - F_s = 0. \tag{11.16}$$

We can now use the two-stream Equations (11.14) and (11.15) to represent the diffuse thermal (infrared) flux. We can first use the definition of $\bar{I}_\nu$ to write Equation (11.15) as

$$\frac{d}{d\tau^*}(F^+ + F^-) - F_T = 0, \tag{11.17}$$

and then integrate it to find

$$F^+ + F^- = F_T\tau^* + \text{constant}. \tag{11.18}$$

The upper boundary condition is $F^+(\tau^* = 0) = F_T$ and $F^-(\tau^* = 0) = 0$ – this then implies that the constant is F_T (from evaluating at $\tau^* = 0$). Thus, we have a relationship between the upward and downward fluxes, the net thermal flux, and scaled optical depth

$$\boxed{F^+ + F^- = F_T(1 + \tau^*)}. \tag{11.19}$$

We now need to link this to the thermal blackbody radiance and thus temperature. Given that the net thermal flux is constant in our approximations, we know that $\frac{dF_T}{d\tau^*} = 0$, and thus we can express Equation (11.14) as

$$2\pi\bar{I} = F^+ + F^- = 2\pi B. \tag{11.20}$$

We can now combine Equations (11.19) and (11.20) to find

$$\boxed{\pi B = \frac{(F^+ + F^-)}{2} = \frac{F_T}{2}(1 + \tau^*).} \tag{11.21}$$

Given that this is a grey atmosphere, $\pi B = F_{\rm BB} = \sigma T^4$ from Stefan-Boltzmann. Thus, $T = \left(\frac{\pi B}{\sigma}\right)^{1/4}$. We can then use Equation (11.21) to find the dependence of temperature on scaled optical depth as

$$\boxed{T = \left[\frac{F_T}{2\sigma}(1 + \tau^*)\right]^{1/4}.} \tag{11.22}$$

We can further combine Equations (11.19) and (11.21) to evalulate the upward and downward fluxes,

$$F^+ = \frac{F_T}{2}(2 + \tau^*), \tag{11.23}$$

$$F^- = \frac{F_T}{2}\tau^*. \tag{11.24}$$

The above solutions are for the atmosphere – let us now calculate the fluxes and temperature at the ground. Let us assume that the surface is a blackbody that lies at a total optical depth $\tau^*_{\rm surf}$ and temperature T_g. Note that just above the ground, we know that the

atmospheric radiative balance implies $\pi B = \frac{F_T}{2}(1 + \tau^*_{\rm surf})$ and $F^+ = \frac{F_T}{2}(2 + \tau^*_{\rm surf})$. Given that this flux must be coming from the ground, then

$$F^+_{\rm ground} = \frac{F_T}{2}(2 + \tau^*_{\rm surf}) = \pi B(T_g) = \sigma T_g^4. \tag{11.25}$$

Solving for the ground temperature, we find

$$\boxed{T_g = \left[\frac{F_T}{2\sigma}(2 + \tau^*_{\rm surf})\right]^{1/4}}. \tag{11.26}$$

Comparing with the air temperature from Equation (11.22), we find that the ground temperature *exceeds* the air temperature just above the ground. Thus, there will be a discontinuity between the surface air and ground temperatures. One can see this discontinuity in the numerical solutions presented in Figure 11.2, which further show the dependence of the thermal flux and temperature on scaled optical depth. Note the following results from our solutions

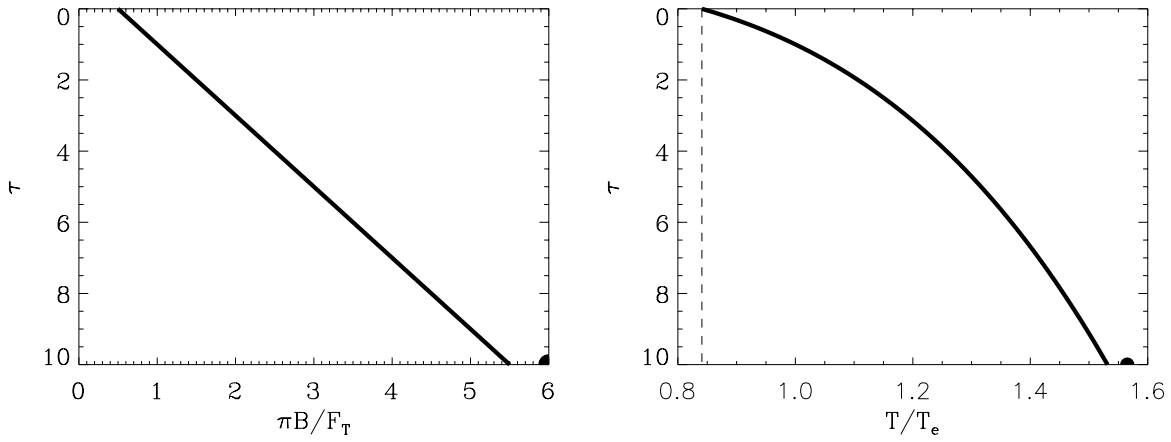


Figure 11.2: Solutions of the grey two-stream equations with no visible opacity assuming $\tau^\star_{\rm surf} = 10$.

above:

1. $B \propto \tau^*$ while $T \propto (1 + \tau^*)^{1/4}$. Thus, at high optical depth the near-surface and ground temperature are much larger than the effective temperature $T_e = \left(\frac{F_T}{\sigma}\right)^{1/4}$. This is the greenhouse effect.

2. Note also that the atmosphere approaches T_e at the region where $\tau^* \approx 1$, which agrees with our discussion of the photosphere previously and the resulting fact that we expect emission to space to occur from the $\tau^* \approx 1$ level with an emitted flux of σT_e^4.

3. The temperature at the top of the atmosphere ($\tau^* \to 0$) goes to a constant value of $T = \left(\frac{F_T}{2\sigma}\right)^{1/4}$. This value is equivalent to our previous derivation of the "skin temperature" using a simple idealised two-layer atmosphere model.

Our solutions above are dependent only on the scaled optical depth τ^*. If we would like to convert this to pressure, we need to make an assumption for how the absorption coefficient depends on pressure. We know from the definition of optical depth (here ignoring the scaling factor) that $d\tau = -k\rho dz = k\frac{dp}{g}$, where here k is the grey absorption coefficient. This then implies that

$$\frac{d\tau^*}{dp} = \frac{k}{\bar{\mu}g}. \tag{11.27}$$

We now consider three separate cases for how the absorption coefficient may depend on atmospheric properties.

First, let us consider the case of a *constant* absorption coefficient with pressure. Thus, integrating Equation (11.27), we find

$$\tau^* = \frac{k}{\bar{\mu}g}p. \tag{11.28}$$

We can thus re-express the scaled optical depth in terms of the scaled optical depth at the surface and pressure at the surface as

$$\tau^* = \tau^*_{\rm surf}\left(\frac{p}{p_{\rm surf}}\right) \quad \text{constant } k. \tag{11.29}$$

Second, let us consider a *pressure-broadened* case, where absorption coefficients linearly increase with pressure as

$$k = k_0\frac{p}{p_{\rm surf}}. \tag{11.30}$$

Then, integrating we find that the scaled optical depth

$$\tau^* = \frac{k_0}{2\bar{\mu}gp_{\rm surf}}p^2, \tag{11.31}$$

which can be written as a power-law from the surface as

$$\tau^* = \tau^*_{\rm surf}\left(\frac{p}{p_{\rm surf}}\right)^2 \quad \text{pressure} - \text{broadened } k. \tag{11.32}$$

Finally, let us consider a general case where the scaled optical depth depends on pressure as a power-law,

$$\tau^* = \tau^*_{\rm surf}\left(\frac{p}{p_{\rm surf}}\right)^\alpha. \tag{11.33}$$

Substituting this in to our expression for atmospheric temperature from Equation (11.22), we find a general expression for temperature as a function of pressure given α:

$$\boxed{T(p) = \left[\frac{F_T}{2\sigma}\left(1 + \tau^*_{\rm surf}\left(\frac{p}{p_{\rm surf}}\right)^\alpha\right)\right]^{1/4}}. \tag{11.34}$$

Here $\alpha = 1$ in the case of a constant absorption coefficient and $\alpha = 2$ in the case of pressure broadening.

Note that we can consider another case, where $\alpha = \frac{d\ln p_{\rm gg}}{d\ln p}$ represents the ratio of the atmospheric to greenhouse gas scale height. Here $p_{\rm gg}$ is the partial pressure of some (condensible) greenhouse gas such as H_2O that decreases in mixing ratio with height such that $p_{\rm gg} = p_{\rm gg,0}\left(\frac{p}{p_0}\right)^\alpha$. This then implies that if the absorption coefficient of the greenhouse gas is constant and there are no other greenhouse gases that $\tau^* = \tau^*_{\rm surf}\left(\frac{p}{p_{\rm surf}}\right)^\alpha$. In the case of water vapour on Earth, $\alpha \approx 4$.

Figure 11.3 shows solutions to Equation (11.34) for various values of $\alpha = 1, 2, 4$. We find

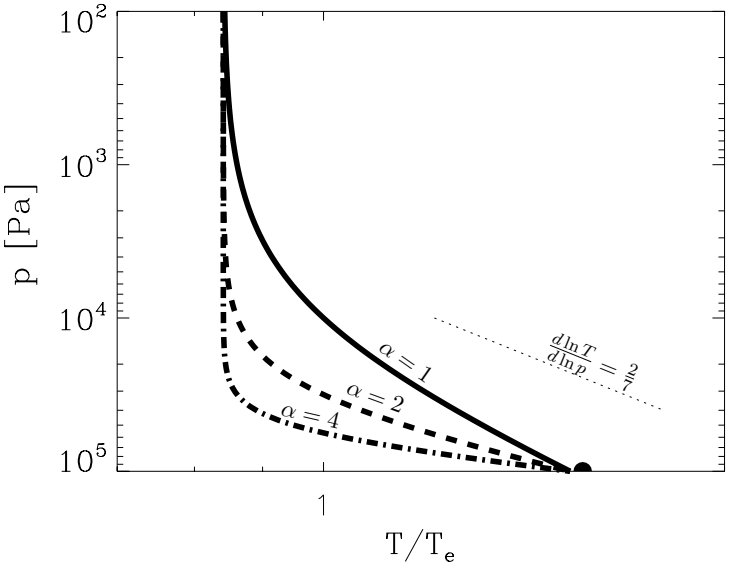


Figure 11.3: Solutions of the grey two-stream equations with no visible opacity for variable greenhouse gas scale heights $\alpha = d\ln p_{\rm gg}/d\ln p$ and surface optical depth $\tau^*_{\rm surf} = 10$.

that the value of α changes the slope of the deep atmospheric thermal profile, with higher α leading to steeper T-P profiles at depth. Note that all profiles converge to the same skin temperature at low pressures, as this is governed only by F_T. The dry adiabat is also shown on Figure 11.3 for comparison – note that the temperature-pressure profile becomes unstable to convection at depth for the cases of $\alpha = 2$ and $\alpha = 4$. In general, the profile near the surface is unstable in the optically thick limit if $\frac{4R}{\alpha c_p} < 1$. This is well satisfied for $\alpha \gtrsim 2$, implying a deep troposphere on Earth and other planets with pressure broadening and/or greenhouse gases that decay with height.

Note that in general, the above realisation that our cases with high α are unstable to convection implies that our purely radiative model of an atmosphere breaks down in those regimes. In reality, we must instead solve for radiative-convective equilibrium (RCE), including both the deep adiabat and the radiative-equilibrium solution above that. This is in principle an iterative process, as the deep tropospheric structure then affects the radiating temperature (and vice versa), requiring numerical solutions for accuracy.

12 Radiative transfer II: Double-grey solution to the two-stream equations

The assigned reading for today is a classic exoplanet paper that applies Earth climate ideas to hot Jupiters: Guillot (2010), `https://arxiv.org/abs/1006.4702`. This will show how we can put together what we've learned from the grey model and apply it any climate, including those far different than Earth's.

12.1 Semi-grey atmosphere with non-zero visible opacity (double-grey)

The primary issue with our initial semi-grey derivation above is that we totally neglected the absorption of light coming from the star. Real atmospheres absorb starlight in both the visible and ultraviolet[28], and if the absorption is sufficiently strong (e.g., O_3 absorption of UV Sunlight in Earth's stratosphere, absorption of visible Sunlight by dust lofted by dust storms in Mars' atmosphere), then a thermal inversion layer can form.

We will now consider a modified semi-grey two-stream plane-parallel atmosphere where we allow for absorption at both incoming stellar (visible and UV) and thermal (infrared) wavelengths. We assume for simplicity that the absorption coefficient in the visible wavelength k_v is constant across the spectrum of incoming starlight, while the absorption coefficient at infrared wavelengths k_IR is constant across the IR spectrum. This model is called double-grey because we assume that the atmosphere is grey in both the absorption of incoming starlight and absorption of outgoing thermal diffuse emission, but with different absorption coefficients in the two regions. We will again ignore scattering for simplicity.

As before, we start with a statement of top-of-atmosphere energy balance,

$$F_\mathrm{tot}(z) = -F_\mathrm{v}(z) + F_\mathrm{IR} = 0, \tag{12.1}$$

where F_v is the net stellar direct beam flux and F_IR is the net diffuse IR flux. Because we are treating the direct beam alone (i.e., neglecting scattering), we can express the direct beam stellar flux in the visible band as

$$F_\mathrm{v} = F_\mathrm{s} e^{-\tau_\mathrm{v}/\bar{\mu}_\mathrm{v}}, \tag{12.2}$$

where F_s is the incoming top-of-atmosphere stellar flux, τ_v is the optical depth at visible wavelengths, and $\bar{\mu}_\mathrm{v}$ is the cosine of the mean angle from the vertical at which sunlight enters the atmosphere.

We can now start with the two-stream equations for the diffuse thermal IR flux, written in grey form as:

$$-\frac{dF}{d\tau^*} + 2\pi\bar{I} = 2\pi B, \tag{12.3}$$

$$2\pi\frac{d\bar{I}}{d\tau^*} = F. \tag{12.4}$$

Now, if we take the derivative of Equation (12.3) with respect to τ^*,

$$-\frac{d^2F}{d\tau^{*2}} + 2\pi\frac{d\bar{I}}{d\tau^*} = 2\pi\frac{dB}{d\tau^*}, \tag{12.5}$$

[28] As well as infrared, especially for planets orbiting small, cool M-dwarf stars.

we can plug it into Equation (12.4) to make one second order ordinary differential equation from the two two-stream equations:

$$-\frac{d^2F}{d\tau^{*2}} + F = 2\pi\frac{dB}{d\tau^*}. \tag{12.6}$$

Importantly, because F_v is specified we *already know* what F is, as it's

$$F = F_\mathrm{IR} = F_\mathrm{v} = F_\mathrm{s}e^{-\tau_\mathrm{v}/\bar{\mu}_\mathrm{v}}. \tag{12.7}$$

However, to use the expression above for F we need to convert the visible optical depth τ_v to the infrared optical depth τ^*. We can use the definition of optical depth in each channel,

$$d\tau_\mathrm{v} = -\bar{\mu}_\mathrm{v}k_\mathrm{v}dz, \tag{12.8}$$

$$d\tau^* = -\bar{\mu}k_\mathrm{IR}dz, \tag{12.9}$$

to relate τ_v to τ^* as

$$\tau_\mathrm{v} = \frac{\bar{\mu}_\mathrm{v}k_\mathrm{v}}{\bar{\mu}k_\mathrm{IR}}\tau^*. \tag{12.10}$$

Thus, we can write the IR flux as

$$F_\mathrm{IR} = F_\mathrm{s}e^{-\tau_\mathrm{v}/\bar{\mu}_\mathrm{v}} = F_\mathrm{s}e^{-k_v\tau^*/(\bar{\mu}k_\mathrm{IR})}. \tag{12.11}$$

We can now plug Equation (12.11) into Equation (12.6), finding

$$-\frac{k_\mathrm{v}^2}{\bar{\mu}^2k_\mathrm{IR}^2}F_\mathrm{s}e^{-k_v\tau^*/(\bar{\mu}k_\mathrm{IR})} + F_\mathrm{s}e^{-k_v\tau^*/(\bar{\mu}k_\mathrm{IR})} = 2\pi\frac{dB}{d\tau^*}. \tag{12.12}$$

Rearranging, we can write

$$\pi\frac{dB}{d\tau^*} = \frac{F_\mathrm{s}}{2}\left(-\frac{k_\mathrm{v}^2}{\bar{\mu}^2k_\mathrm{IR}^2} + 1\right)e^{-k_v\tau^*/(\bar{\mu}k_\mathrm{IR})}. \tag{12.13}$$

We can now integrate this from 0 to τ (dropping the asterisk) to find

$$\begin{aligned}\pi\left[B(\tau) - B(0)\right] &= \frac{F_\mathrm{s}}{2}\left(-\frac{k_\mathrm{v}^2}{\bar{\mu}^2k_\mathrm{IR}^2} + 1\right)\left(-\frac{\bar{\mu}k_\mathrm{IR}}{k_\mathrm{v}}\right)\left[e^{-k_v\tau^*/(\bar{\mu}k_\mathrm{IR})}\right]_0^\tau \\ &= \frac{F_\mathrm{s}}{2}\left(\frac{k_\mathrm{v}}{\bar{\mu}k_\mathrm{IR}} - \frac{\bar{\mu}k_\mathrm{IR}}{k_\mathrm{v}}\right)\left[e^{-k_\mathrm{v}\tau/(\bar{\mu}k_\mathrm{IR})} - 1\right].\end{aligned} \tag{12.14}$$

We can now use use the top boundary condition to determine B(0). The top boundary condition (at $\tau = 0$) is $F^+ = F_\mathrm{s}$ (outgoing diffuse IR flux equals incoming stellar flux) and $F^- = 0$ (no incoming diffuse IR flux). We can now relate this to B using the first grey two-stream Equation (12.3),

$$-\frac{dF}{d\tau} + F^+ + F^- = 2\pi B, \tag{12.15}$$

and substituting in for F

$$F_s\frac{k_\mathrm{v}}{\bar{\mu}k_\mathrm{IR}}e^{-k_\mathrm{v}\tau/(\bar{\mu}k_\mathrm{IR})} + F^+ + F^- = 2\pi B. \tag{12.16}$$

Evaluating Equation (12.16) at $\tau = 0$, we find

$$F_{\rm s}\frac{k_{\rm v}}{\bar{\mu}k_{\rm IR}} + F_{\rm s} = 2\pi B(0), \tag{12.17}$$

and thus

$$\pi B(0) = \frac{F_{\rm s}}{2}\left(1 + \frac{k_{\rm v}}{\bar{\mu}k_{\rm IR}}\right). \tag{12.18}$$

Now plugging in our expression for $\pi B(0)$ into Equation (12.14), we find

$$\pi B(\tau) = \frac{F_{\rm s}}{2}\left(1 + \frac{k_{\rm v}}{\bar{\mu}k_{\rm IR}}\right) + \frac{F_s}{2}\left(\frac{k_{\rm v}}{\bar{\mu}k_{\rm IR}} - \frac{\bar{\mu}k_{\rm IR}}{k_{\rm v}}\right)\left[e^{-k_{\rm v}\tau/(\bar{\mu}k_{\rm IR})} - 1\right]. \tag{12.19}$$

Now if we define $\gamma \equiv k_{\rm IR}/k_{\rm v}$, we can more simply express

$$\pi B(\tau) = \frac{F_{\rm s}}{2}\left(\frac{1}{\bar{\mu}\gamma} - \bar{\mu}\gamma\right)e^{-\tau/(\bar{\mu}\gamma)} + \frac{F_{\rm s}}{2}\left(1 + \bar{\mu}\gamma\right). \tag{12.20}$$

Given that $T = \left(\frac{\pi B}{\sigma}\right)^{1/4}$, we can solve for $T(\tau)$:

$$T = \left[\frac{F_{\rm s}}{2\sigma}\left(\left(\frac{1}{\bar{\mu}\gamma} - \bar{\mu}\gamma\right)e^{-\tau/(\bar{\mu}\gamma)} + 1 + \bar{\mu}\gamma\right)\right]^{1/4}. \tag{12.21}$$

If we assume for simplicity that $\mu \approx 1$, we can write the temperature profile as

$$\boxed{T = \left[\frac{F_{\rm s}}{2\sigma}\left(\left(\frac{1}{\gamma} - \gamma\right)e^{-\tau/\gamma} + 1 + \gamma\right)\right]^{1/4}.} \tag{12.22}$$

Figure 12.1 shows temperature-optical depth profiles from solutions to the double-grey two-stream Equation (12.22). There are three main types of profiles, depending on the value of γ:

1. $\gamma \gg 1$: This is the greenhouse limit, with $k_{\rm IR} \gg k_{\rm v}$. The temperature at depth is much warmer than the effective temperature, and the temperature at high optical depth $\tau \to \infty$ is $\frac{T_e}{2^{1/4}}\left(1+\gamma\right)^{1/4} \approx T_e\left(\frac{\gamma}{2}\right)^{1/4}$. Note also that the temperature profile becomes isothermal at optical depths $\tau \gtrsim \gamma$, as little sunlight penetrates so not much IR flux needs to be transported in radiative equilibrium.

2. $\gamma = 1$: This is an isothermal profile with $T = T_e$.

3. $\gamma \ll 1$: This is the anti-greenhouse limit, with $k_{\rm IR} \ll k_{\rm v}$. In this case, starlight is absorbed at higher altitudes than IR escapes to space, leading to a thermal inversion. The peak temperature of this inversion layer (or stratosphere) is reached where the visible optical depth $\tau_{\rm v} \sim 1$, which corresponds to an IR optical depth of $\tau_{\rm IR} = \tau_{\rm v}\gamma \sim \gamma$. The temperature of the inversion can greatly exceed the effective temperature because these low pressure regions are optically thin to infrared radiation and thus inefficiently radiate to space – requiring them to have a high temperature in radiative equilibrium.

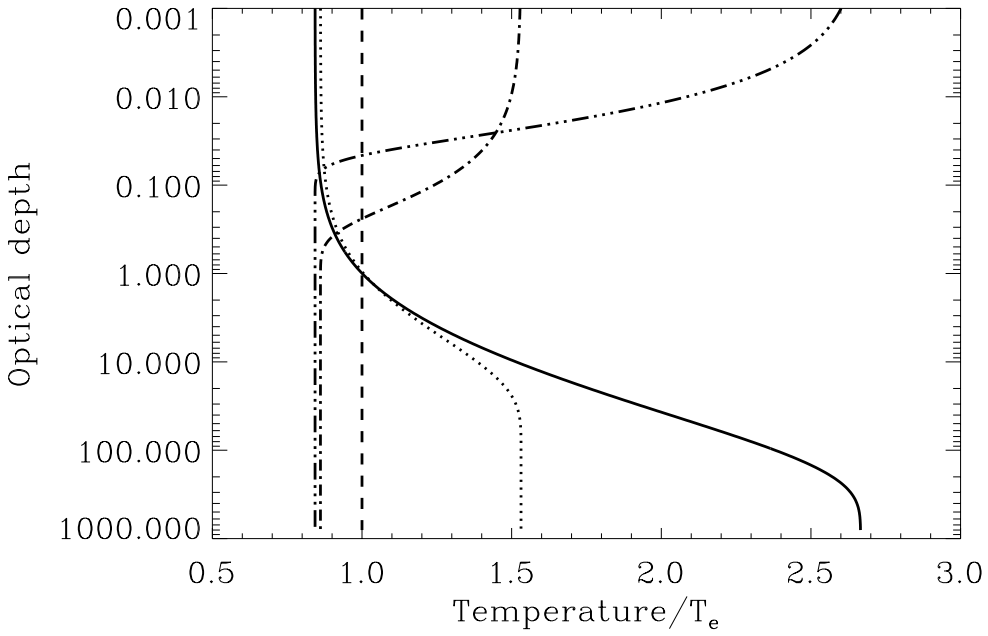


Figure 12.1: Solutions of the double-grey two-stream equations, with temperature normalised by the effective temperature $T_e = \left(\frac{F_s}{\sigma}\right)^{1/4}$. Solid and dotted lines have $\gamma > 1$, dashed-dotted lines have $\gamma < 1$, and the dashed line has $\gamma = 1$.

12.1.1 Application to exoplanet emission spectra

Figure 12.2 shows an example thermal emission spectrum of an ultra-hot Jupiter exoplanet, WASP-18b, from JWST using the NIRISS/SOSS instrument from the optical to near-infrared (Coulombe et al., 2023). These observations (top) find thermal emission features of water vapour, implying that the atmosphere has a thermal inversion and thus is in the $\gamma < 1$ limit with relatively high optical opacity. Retrievals (bottom) indeed find a strong thermal inversion across the photosphere, in agreement with the emission features in the observed spectrum. The optical component of the spectrum has features from a combination of metal oxide absorbers including TiO and VO, which are likely providing the shortwave opacity needed to drive this strong thermal inversion (and enable $\gamma < 1$ locally in the atmosphere).

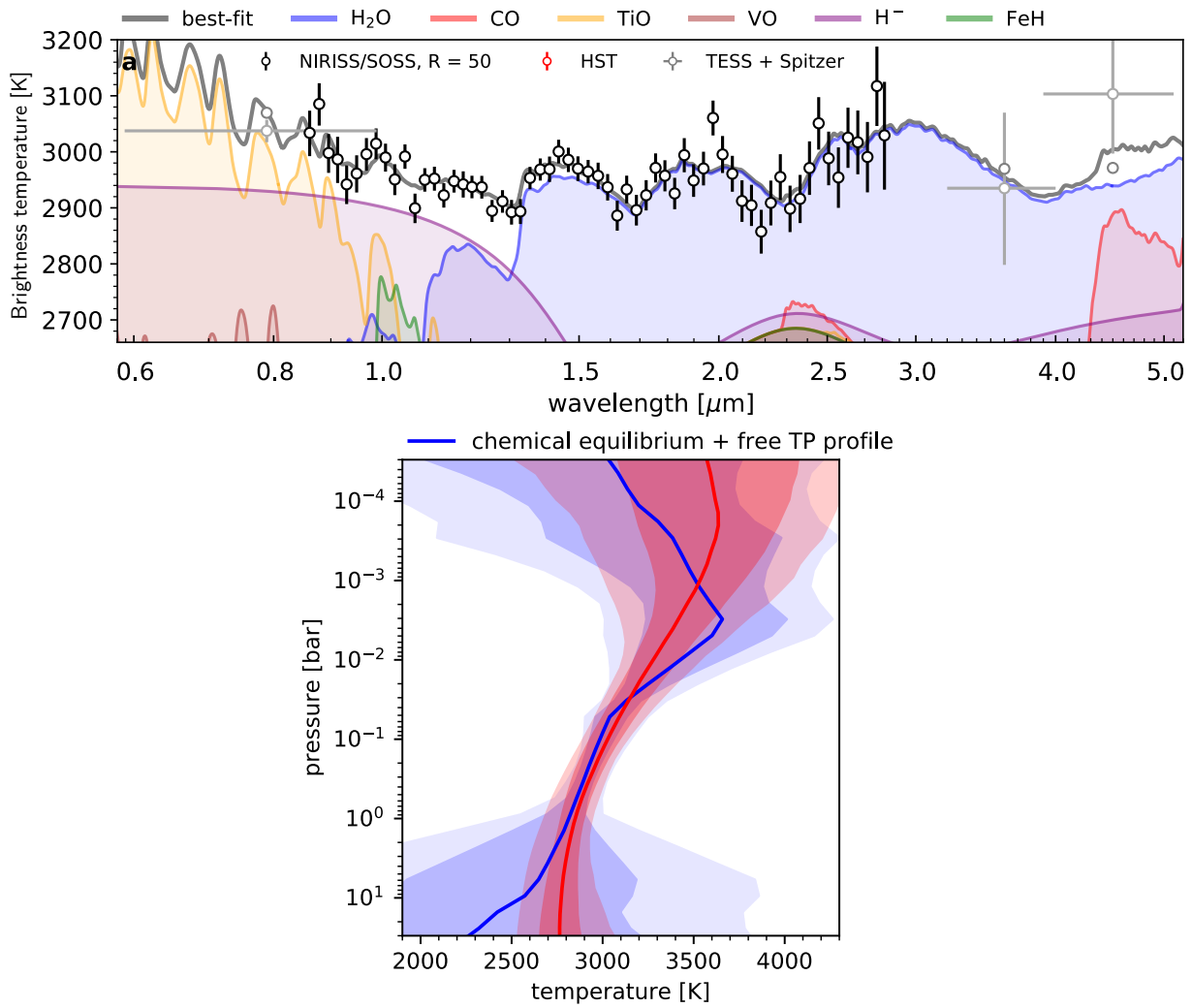


Figure 12.2: Thermal emission spectrum (top) and retrieved temperature-pressure profile (bottom) of WASP-18b from the JWST NIRISS/SOSS observations of Coulombe et al. (2023). Note that spectral features of water vapour are seen in thermal emission, implying an inverted temperature-pressure profile in the $\gamma < 1$ limit.

13 Radiative transfer II: Double-grey, band-grey models

Today we are finishing the discussion of semi-grey and double-grey models. The vast majority of the content in today's lecture is covered in Section 12.

If time remains, we will briefly sketch out band-grey models following the section below, but not derive the solution in full.

13.1 A band semi-grey atmosphere model

Real atmospheres are not grey or double-grey, but have absorption centred on *bands*, which we will discuss in more detail in the next lecture. This means that the absorption coefficients of a real planetary atmosphere are strongly dependent on frequency, and the frequency dependence of the absorption coefficients further depends on temperature, pressure, and atmospheric composition.

For simplicity, let us start by considering the most idealised possible band model. This band has constant absorption coefficient within the band, and zero absorption outside of the band:

$$k_\nu = \begin{cases} k_b & \text{for } \nu_1 < \nu < \nu_2, \\ 0 & \text{otherwise.} \end{cases} \tag{13.1}$$

Such an idealised band model is illustrated in Figure 13.1. Thus, the flux spectrum of a

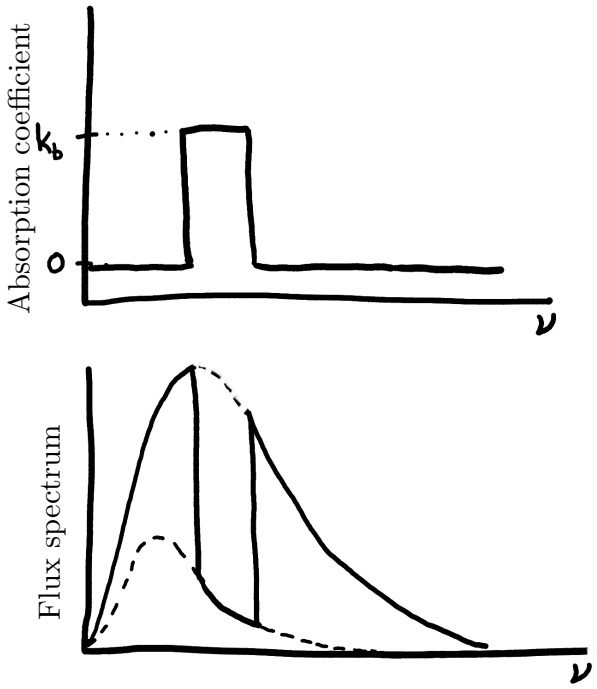


Figure 13.1: Schematic illustration of a band with constant absorption coefficient inside the band and zero outside of the band, and the resulting "chunk" it takes out of the flux spectrum of the planet.

planetary atmosphere absorbing only as this idealised band model would look like a cooler blackbody inside the absorption band, and a second hotter blackbody outside the absorption band. This simple treatment of absorption can further enhance the realism of the simple semi-grey model with no visible opacity we previously considered in Section 11.2.3.

If we integrate the two-stream equations of radiative transfer over the band (from frequency ν_1 to ν_2), we find

$$-\frac{dF_b}{d\tau_b^*} + 2\pi\bar{I}_b = 2\sigma T^4\beta, \tag{13.2}$$

$$2\pi\frac{d\bar{I}_b}{d\tau_b^*} = F_b, \tag{13.3}$$

where here F_b is the net flux integrated over the band and $\bar{I}_b$ is the mean radiance integrated over the band, and τ_b^* is the scaled optical depth in the band. β is the equivalent bandwidth, defined as

$$\beta = \int_{\nu_1}^{\nu_2} \frac{\pi B_\nu}{\sigma T^4} d\nu. \tag{13.4}$$

Thus, we can recover the semi-grey model we previously derived by taking $\beta = 1$. Conversely $\beta = 0$ is the limit of an infinitely narrow band. Figure 13.2 shows β as a function of temperature for the 15 μm CO_2 band, demonstrating that in practice β varies with temperature. However, for simplicity we will keep β fixed in our calculations below.

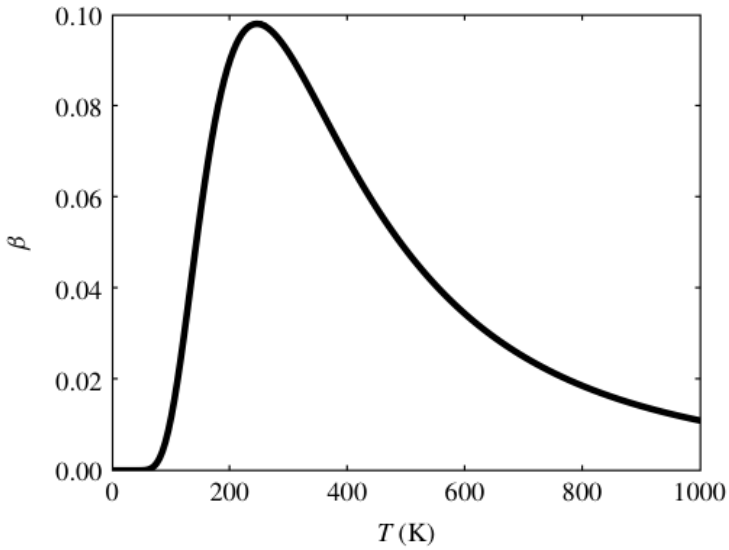


Figure 13.2: Dependence of β on temperature for the 15-μm CO_2 band assuming that it is 2 μm wide, from Caballero et al. (2008).

The boundary conditions are similar to the semi-grey case. First, there is no downward diffuse IR flux at the top-of-atmosphere ($F_b^-(\tau_b^* = 0) = 0$), and the ground at surface temperature $T_{\rm surf}$ radiates as a blackbody, only β of which occurs in the band ($F_b^+ = \beta\sigma T_{\rm surf}^4(\tau_{b,\rm surf}^*)$). These boundary conditions can be written into those using the band net flux F_b and band mean radiance $\bar{I}_b$ as

$$2\pi\bar{I}_b(\tau_b^* = 0) = F_b(\tau_b^* = 0), \tag{13.5}$$

$$2\pi\bar{I}_b(\tau_{b,\rm surf}^*) + F_b(\tau_{b,\rm surf}^*) = 2\beta\sigma T_{\rm surf}^4. \tag{13.6}$$

Top-of-atmosphere radiative equilibrium implies that upward infrared flux equals downward stellar flux, and thus

$$\boxed{F_b + (1-\beta)\sigma T_{\rm surf}^4 = F_s,} \tag{13.7}$$

where the left-hand-side includes both atmospheric contribution from within the band as well as the upward IR flux from the surface. Note that because F_s and β are constant, then F_b is also constant with altitude.

We can then realise that F_b is a constant and thus we can re-write Equation (13.2) as

$$\boxed{2\pi\bar{I}_b = 2\beta\sigma T^4}. \tag{13.8}$$

Next, we can integrate Equation (13.3) from τ_b^* to the top-of-atmosphere (τ_b^*=0) and use the top boundary condition (exercise is left to the interested student) to find

$$\boxed{2\pi\bar{I}_b(\tau_b^*) = F_b(1+\tau_b^*)}. \tag{13.9}$$

Then, we can integrate Equation (13.3) from τ_b^* to the bottom of the atmosphere $\tau_{b,\mathrm{surf}}^*$ and apply the bottom boundary condition to find

$$\boxed{2\pi\bar{I}_b(\tau_b^*) = 2\beta\sigma T_{\mathrm{surf}}^4 - F_b(1-\tau_b^*+\tau_{b,\mathrm{surf}}^*)}. \tag{13.10}$$

Now, using the four boxed equations for four unknowns $T, T_{\mathrm{surf}}, F_b, \bar{I}_b$, we can solve for the surface temperature

$$\sigma T_{\mathrm{surf}}^4 = \frac{F_s(2+\tau_{b,\mathrm{surf}}^*)}{2+(1-\beta)\tau_{b,\mathrm{surf}}^*}, \tag{13.11}$$

and the atmospheric temperature

$$\sigma T^4 = F_s\left[\frac{1+\tau_b^*}{2+(1-\beta)\tau_{b,\mathrm{surf}}^*}\right]. \tag{13.12}$$

Note that as before in the semi-grey case with no visible opacity, there is a temperature discontinuity between the surface and overlying atmosphere.

We can further replace the optical depth in the band with a "Planck-weighted" optical depth defined by $\tau^* = \beta\tau_b^*$ and $\tau_{\mathrm{surf}}^* = \beta\tau_{b,\mathrm{surf}}^*$, which is useful if the optical depth in the band is large even if the overall optical depth is small (e.g., for narrow bands). Thus, we can write the surface temperature as

$$\sigma T_{\mathrm{surf}}^4 = \frac{F_s(2\beta+\tau_{\mathrm{surf}}^*)}{2\beta+(1-\beta)\tau_{\mathrm{surf}}^*}, \tag{13.13}$$

and the atmospheric temperature as

$$\sigma T^4 = F_s\left[\frac{\beta+\tau^*}{2\beta+(1-\beta)\tau_{\mathrm{surf}}^*}\right]. \tag{13.14}$$

Figure 13.3 shows solutions to the band-grey model (Equation 13.14) for varying β and τ_{surf}^*. The temperature profile is warmer for greater β due to the greater line absorption and resulting greenhouse effect. Note also that the surface temperature in this non-grey case at $\tau_{\mathrm{surf}}^* \to \infty$ is $\sigma T_{\mathrm{surf}}^4 = \frac{F_s}{(1-\beta)}$, which is finite and independent of τ_{surf}^* – this is different from the pure semi-grey case, where we found that the surface temperature would tend toward infinity

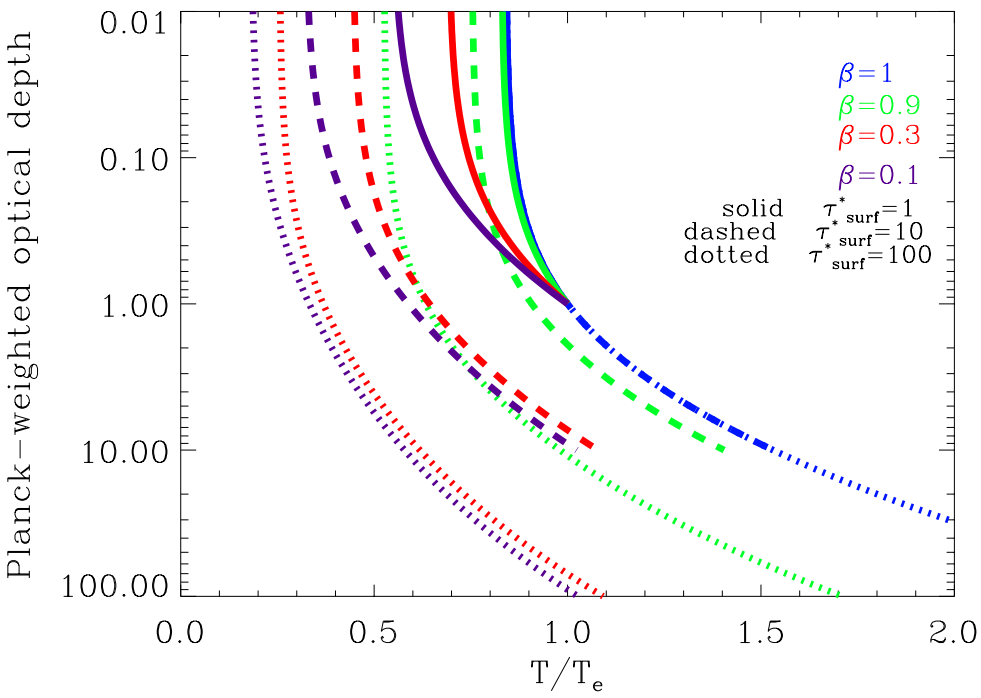


Figure 13.3: Solution to the band semi-grey two stream equations for temperature as a function of Planck-weighted optical depth $\tau^* = \beta\tau_b^*$. Solutions are shown for varying line width parameter β and surface optical depth $\tau^*_{\rm surf}$.

as the surface optical depth increased toward infinity. This is because in the optically thick limit the photons in the absorption band do not contribute significantly to the outgoing IR flux, it is only the emission in the window region that contributes to the top-of-atmosphere energy balance in Equation (13.7). Thus, the surface temperature at high optical depth only depends on the spectral width of the absorption band, not the optical depth in the band.

Finally, note that the skin temperature of the stratosphere in the band-grey limit converges to a value that is dependent on the optical depth of the underlying troposphere (i.e., on $\tau^*_{\rm surf}$). This is quite different than the grey case with $\beta = 1$, where the skin temperature is simply $T_{\rm skin} = T_e/2^{1/4}$. In the band-grey case, the stratospheric temperature goes to zero as $\tau^*_{\rm surf} \rightarrow \infty$. This is because the surface temperature remains finite in the band-grey case, reducing the upward flux in the absorption band – yet the absorption band is the only band where the stratosphere can absorb, making the stratosphere colder for increasing $\tau^*_{\rm surf}$.

14 Radiative transfer III: Spectral line formation

Reading for Day 14: Pierrehumbert Principles of Planetary Climate, Section 4.4.

Today we will go beyond the semi-grey and band-grey models to consider the properties of spectral lines themselves.

14.1 Atomic and molecular transitions

There are three main types of transitions that can occur in planetary atmospheres leading to the absorption or emission of photons: rotational, vibrational, and electronic transitions. In any of these cases, we can consider the emission of a photon as being due to a discrete (*quantized*) transition from an upper to a lower energy state (in the case of emission, as shown in Figure 14.1) or a lower to an upper energy state (in the case of absorption). Note that the

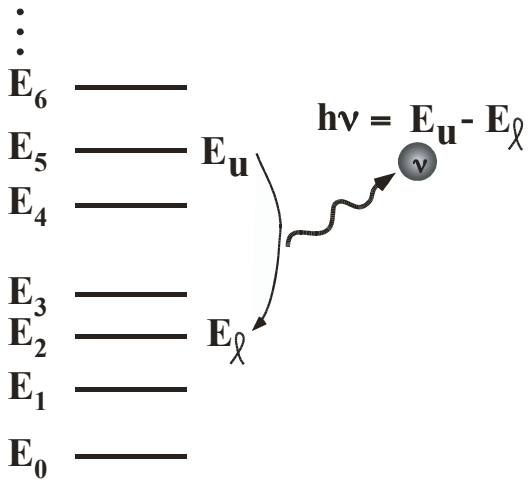


Figure 14.1: Schematic showing an emission of a photon by transition from an upper (high) energy state to a lower energy state. Figure adapted from Pierrehumbert (2010).

quantized nature of these transitions implies that only specific wavelengths or frequencies of light can be absorbed/emitted for a given gaseous species, as we discuss further below.

Electronic transitions (as the name implies) correspond to the change in the energy states of an electron, and can occur for monatomic species as well as molecules. In this case, the change in energy levels $\Delta E = h\nu$, following Planck's relation (as shown in Figure 14.1).

Meanwhile, vibrational and rotational modes occur for molecules, and are due to the relative motions of the atoms in the molecule with respect to one another (for vibrational modes) and the rotation of the molecule itself about a given axis (for rotational modes). Figure 14.2 shows examples of rotational and vibrational modes for a symmetric diatomic molecule (left) and a linear symmetric triatomic molecule (right).

A simple model of vibrational energy levels is that of a simple harmonic oscillator, where the potential energy levels are

$$E_p(x) = \frac{K}{2}(x - x_0)^2, \tag{14.1}$$

where x is the separation of the atoms, x_0 is their equilibrium separation, and K is a constant (like a spring constant). Quantum mechanics (not treated in detail here for sake of time)

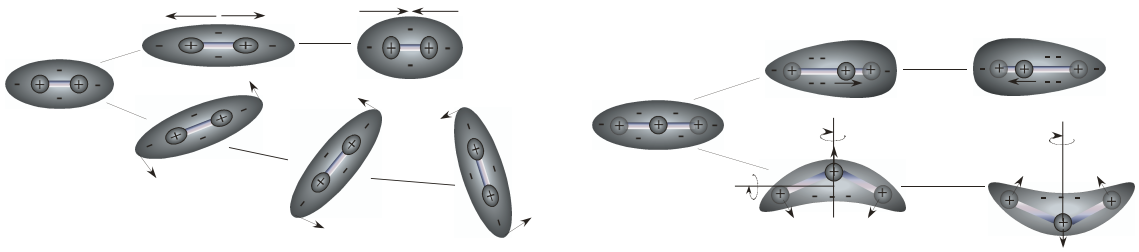


Figure 14.2: Schematic of the rotational and vibrational modes of a symmetric diatomic molecule (e.g., N_2, left) and a linear symmetric triatomic molecule (e.g., CO_2, right). Figures adapted from Pierrehumbert (2010).

shows that the allowed energy levels of this vibrational oscillator are given by

$$E_{\mathcal{V}} = h\nu_0 \left(\mathcal{V} + \frac{1}{2} \right), \tag{14.2}$$

where $\mathcal{V}$ is the vibrational quantum number ($\mathcal{V} = 0, 1, 2...$). Dipolar transitions have $\Delta\mathcal{V} = \pm 1$, corresponding to an energy change $\Delta E_{\mathcal{V}} = \pm h\nu_0$. Typically, vibrational lines occur in the infrared wavelength range (roughly $1 - 20\mu$m).

Rotational energy levels are similarly quantized in the atomic angular momentum L. This varies with rotational quantum number J for a diatomic molecule as

$$L^2 = J(J+1)\hbar^2. \tag{14.3}$$

The corresponding allowed rotational energy levels for a diatomic molecule are

$$E_J = \frac{L^2}{2I} = \frac{J(J+1)\hbar^2}{2I}, \tag{14.4}$$

where I is the moment of inertia of the molecule. This expression is valid for linear rotators (which only have two non-zero moments of inertia) and spherical top molecules (for which all three moments of inertia are equal). Note that then the change in energy for a rotational transition is given by

$$\Delta E_J = \frac{\hbar^2}{I}(J+1). \tag{14.5}$$

In the case of rotational lines, the strength depends on the population of the lower state, which is given by the Boltzmann distribution as

$$N_J = N_0 g_J e^{-\frac{E_J}{k_B T}}. \tag{14.6}$$

Here g_J is the degeneracy of the Jth level. Plugging in for our expression for linear rotators, we find

$$N_J = (2J+1) N_0 e^{-\frac{J(J+1)\hbar^2}{2Ik_B T}}. \tag{14.7}$$

These rotational transitions then have much smaller spacing than vibrational levels, and they typically occur in the far infrared and microwave wavelength ranges (100 - 10^4 μm).

Finally, molecules can also have coupled rotational and vibrational states. This is because the energy of rotational transitions are typically much lower than vibrational transitions, so if vibrational transitions are excited then rotational transitions are also excited. Molecules are not rigid, so rotational states can couple to the vibrational states through the Coriolis and centrifugal forces (Pierrehumbert, 2010), leading to "ro-vibrational" transitions.

Typically, molecular dipole moments are required for significant optically active transitions. This is because, as stated in Pierrehumbert (2010), "one can think of infrared light as providing a large-scale fluctuating electric field and magnetic field which alters the environment in which the molecule finds itself," and consequently the strongest and most active transitions will correspond to interactions between an electromagnetic field and a molecule with net charge. As a result, linear molecules with no net dipole moment do not have observable spectra in isolation, while molecules (linear or otherwise) with a dipole moment can have strong transitions.

A useful rule of thumb is that electronic energy levels require ~ 10 eV to excite, vibrational energy levels require ~ 0.1 eV to excite, and rotational levels require ~ 0.01 eV to excite.

Question for the class – which of these types of transitions (electronic, vibrational rotational) is not relevant for Earth's deep atmosphere? Hint: estimate the thermal energy of gas on Earth in eV, using $k_B = 8.617 \times 10^{-5}$ eV K^{-1}.

Answer: rotational, maybe some vibrational – but rotational can couple to vibrational.

Question for the class – Which of the following molecules will have a rotational spectrum?

1. CN (linear)
2. O_2 (linear)
3. CH_3Cl (symmetric rotator)
4. C_2H_2 (linear)

Answer: CN, CH_3Cl – the thing to realise is that the molecule must be polar.

Another type of transition that leads to a (temporary) dipole moment not discussed above is that due to collisions. When molecules (even those with no permanent dipole moment) collide, they can generate a collision-induced dipole moment – this can occur in dense atmospheres for common species, such as CO_2-CO_2 on Venus, N_2-N_2 on Earth and Titan, and H_2-H_2 on the giant planets (Sanchez-Lavega, 2010).

14.2 Spectral lines and broadening

We now turn to study the properties of spectral lines in planetary atmospheres. Figure 14.3 shows a collection of infrared spectra of planetary atmospheres in our Solar System. Note the richness and diversity of these spectra – the rocky planets each have a strong CO_2 feature, but Earth also has strong ozone absorption. Meanwhile, the giant planets have a panoply of other strong IR absorbers, including ammonia, methane, hydrogen (collision-induced absorption), and Titan's spectra is dominated by organics.

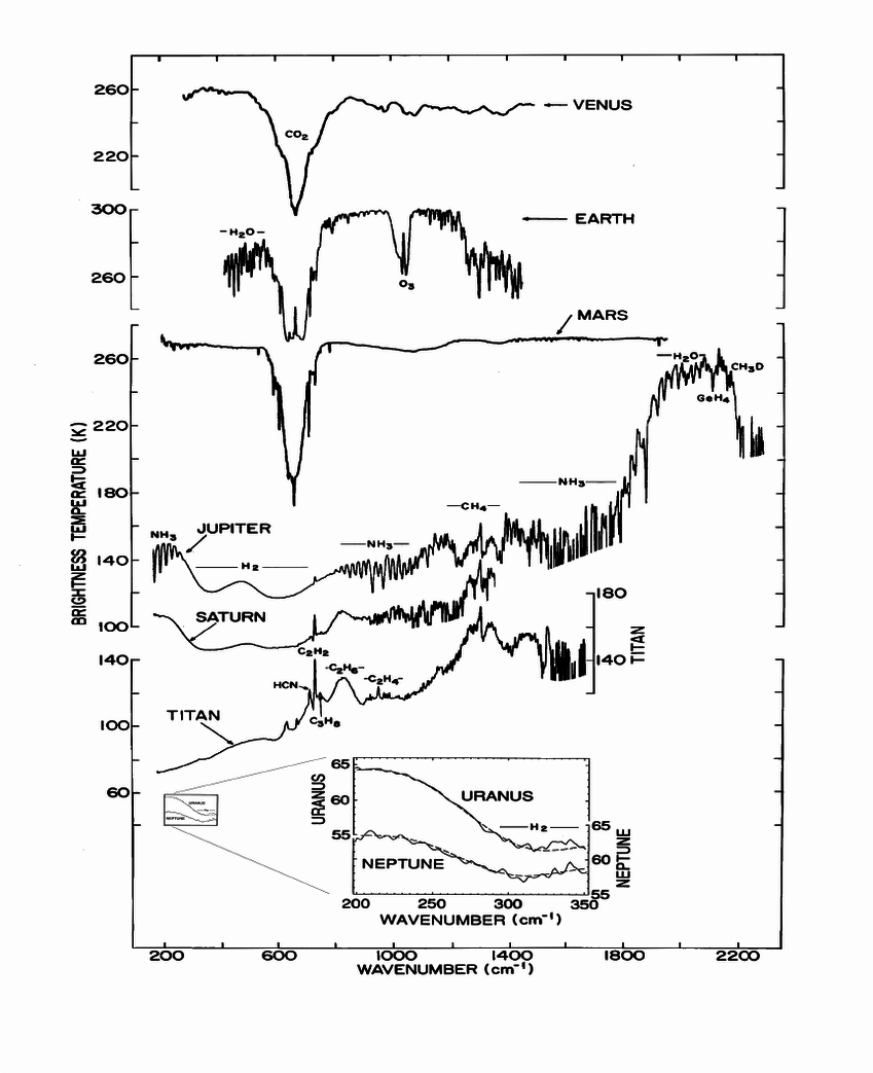


Figure 14.3: Infrared (ro-vibrational) spectra of Solar System objects. Note the prominent CO_2 feature for Venus, Earth, and Mars. Figure adapted from Hanel et al. (1992).

14.2.1 Description of spectral lines

The spectra of molecules is formed by vibration-rotation bands, as shown in the example of CO and CO_2 for Earth atmospheric conditions in Figure 14.4. These molecular spectra transitions are often referred to as having bands in *branches*, named O,P,Q,R,S that correspond to transitions in the rotational quantum number $\Delta J = -2, -1, 0, +1, +2$, with the band structure itself depending on the rotational symmetry of the molecule. Note that these bands (of CO, CO_2, or otherwise) widen at higher temperature, following the Boltzmann distribution as the fraction of states at a given temperature scales as $e^{-E/kT}$, so at higher temperature there are more higher-energy states and the distribution spreads.

Question for the class - Which of the branches P, Q, R does CO have? Why?

Answer: CO does not have a Q branch because it is diatomic, and so it follows $\Delta J = \pm 1$.

The absorption coefficient k_G for a spectral line of a substance G is typically described

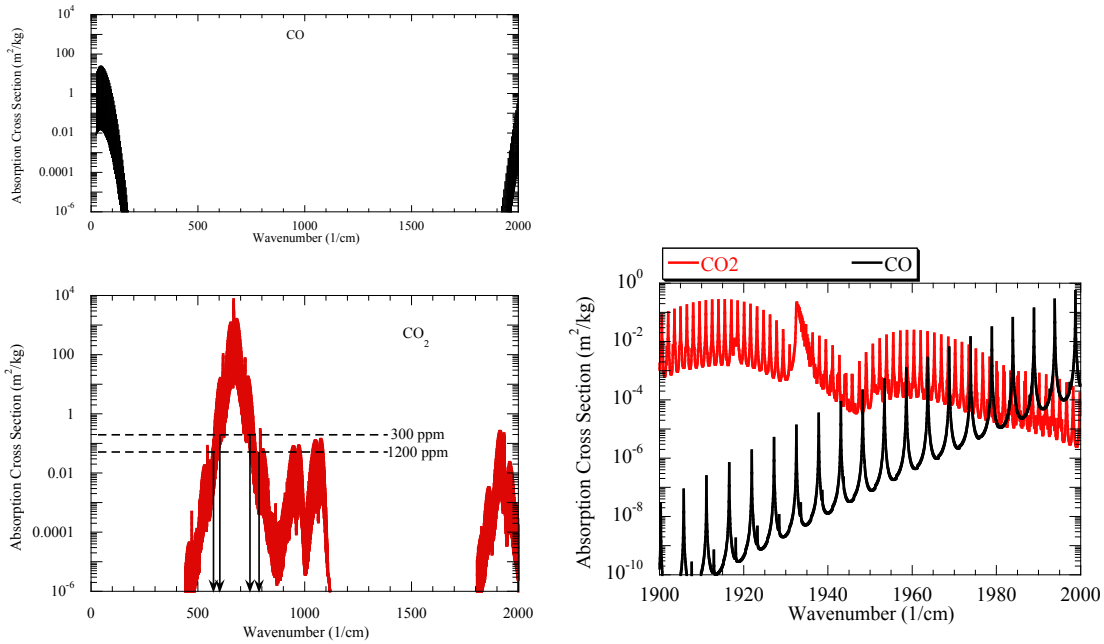


Figure 14.4: Absorption cross sections for CO (top left) and CO_2 (bottom left) as a function of wavenumber at a pressure of 100 mbars and a temperature of 260 K (relevant for Earth's atmosphere). The right-hand side shows a high-resolution slice of these cross sections, demonstrating individual spectral line shapes. Figure adapted from Pierrehumbert (2010).

mathematically as (Pierrehumbert, 2010)

$$k_G(\nu, p, T) = \frac{S}{\gamma} f\left(\frac{\nu - \nu_c}{\gamma}\right), \tag{14.8}$$

where S is the line intensity, ν_c is the frequency at the line centre, γ is the line full-width at half maximum (FWHM), and f is a function of the non-dimensional argument $\Delta\nu/\gamma$ that is normalised such that the integral of f over all frequencies is unity. Note that then, the integral $\int k_G d\nu = S$. The area under a given spectral line remains fixed as the line gets broader/narrower, so that spectral lines get taller if they are narrower (and vice versa). We now turn to discuss ways in which lines can be broadened – there are three main processes: natural broadening, Doppler, broadening, and pressure broadening.

14.2.2 Intrinsic line uncertainty

The intrinsic uncertainty in line width comes from the Heisenberg uncertainty principle, as given that time and energy cannot be precisely known (i.e., $\Delta E \Delta t = \hbar$), then the uncertainty in frequency and time is

$$\Delta\nu\Delta t = \frac{1}{2\pi}. \tag{14.9}$$

Thus, the line broadening in frequency is $\Delta\nu = 1/(2\pi\Delta t)$. This type of broadening is often termed "natural" broadening because it occurs even in the absence of other broadening

processes. Natural broadening is described by the Lorentz profile, here written for the non-dimensional f_N introduced above as

$$f_N(\nu - \nu_0) = \frac{1}{\pi} \frac{\gamma_N}{(\nu - \nu_0)^2 + \gamma_N^2}. \tag{14.10}$$

In the above expression, γ_N is the FWHM from natural broadening and scales as $\gamma \sim 1/(2\pi t)$, where t is the radiative lifetime.

14.2.3 Doppler broadening

Doppler broadening arises from the difference in thermal velocities of atoms and molecules shifting their relative line frequencies. Recall that the Doppler shift for small speed v can be expressed as

$$\nu = \nu_0 \left(1 \pm \frac{v}{c}\right), \tag{14.11}$$

where c is the speed of light. If we assume that the distribution of molecular speeds along the line of sight that leads to the Doppler broadening is given by the Maxwell-Boltzmann distribution, this implies that the speed distribution $P(v)dv \propto \exp\left(-mv^2/(2kT)\right) dv$.

Substituting in the Doppler shift into the Maxwell-Boltzmann relation, we can express the normalised Doppler broadening function f_D as

$$f_D(\nu - \nu_0) = \frac{1}{\gamma_D \sqrt{\pi}} e^{-\frac{(\nu - \nu_0)^2}{\gamma_D}}, \tag{14.12}$$

where the line width

$$\gamma_D = \nu_0 \sqrt{\frac{2k_B T}{mc^2}} = \frac{\nu_0}{c} \sqrt{2RT}. \tag{14.13}$$

Note that here γ_D is the standard deviation (not the FWHM) because the Doppler broadening equation is a Gaussian – to get the actual FWHM, multiply $\gamma_D \sqrt{\ln 2}$.

Figure 14.5 shows the relative shapes of the Doppler and Lorentz line profiles (if it's hard to see the different colours, the Doppler line profile is taller). Note that there is further a line profile that combines the Doppler and Lorentz profiles, known as the Voigt profile. The Voigt profile can be expressed as

$$f_V(\nu - \nu_0) = \frac{1}{\gamma_D \sqrt{\pi}} K(x, y), \tag{14.14}$$

where

$$K(x, y) = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{1}{y^2 + (x - t)^2 \exp(-t^2)} dt, \tag{14.15}$$

with $t = (\nu - \nu')/\gamma_D$, $y = \gamma_N/\gamma_D$, $x = \frac{(\nu - \nu_0)}{\gamma_D}$.

14.2.4 Pressure broadening

Pressure broadening (also known as collisional broadening) is line broadening due to collisions between atoms and/or molecules. The normalised functional form of

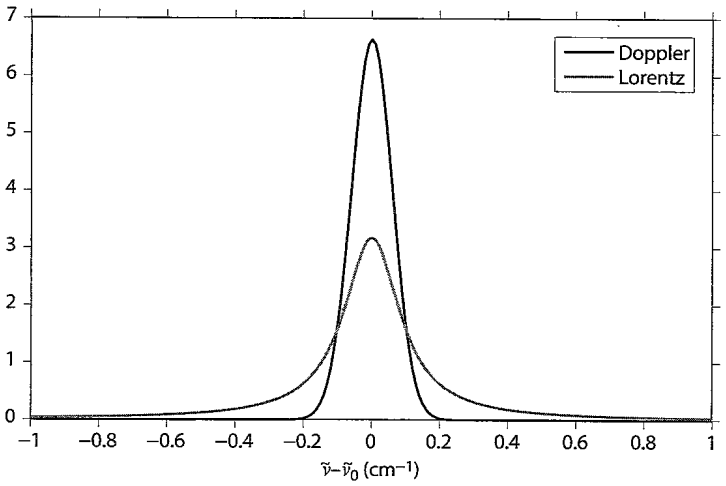


Figure 14.5: Comparison of the line profiles for Doppler broadening and Lorentz (natural) broadening. Figure adapted from Seager (2010).

pressure broadening follows the Lorentz profile (the same general profile as for natural broadening),

$$f_L(\nu - \nu_0) = \frac{1}{\pi} \frac{\gamma}{(\nu - \nu_0)^2 + \gamma^2}. \tag{14.16}$$

Here γ is the half width of the line at half maximum, which scales with pressure and temperature as

$$\gamma = \gamma_0 \frac{p}{p_0} \left(\frac{T_0}{T} \right)^n, \tag{14.17}$$

where the subscript typically refers to standard $T_0 = 273$ K and $p_0 = 1013$ mbar and $n \approx 0.5$. As a result, the absorption coefficients for species undergoing pressure broadening typically scale with pressure, as we discussed earlier in Section 11.2.3.

14.2.5 Transmission function

By integrating the two-stream equations for the upward and downward flux (see Eq. 4.9 of Pierrehumbert, 2010), one can denote the *transmission function*

$$\mathcal{T}_\nu(p_1, p_2) = e^{-|\tau_\nu(p_1) - \tau_\nu(p_2)|}. \tag{14.18}$$

This represents the proportion of incident energy flux at a given frequency that is transmitted through a layer of atmosphere that extends from p_1 to p_2. Though we dealt with the differential form (rather than the integrated form as discussed in more depth in Pierrehumbert, 2010) of the two-stream equations throughout our lectures, the transmission function will be helpful for your radiative transfer problem sheet.

We can thus write the transmission function for an isolated line with half-width Δ as

$$\mathcal{T}(p_1, p_2) = \int_{\nu=\nu_0-\frac{\Delta}{2}}^{\nu=\nu_0+\frac{\Delta}{2}} e^{-\tau(\nu, p_1, p_2)} d\nu. \tag{14.19}$$

In the weak line limit with $\tau \ll 1$ along the entire path,

$$\mathcal{T} \approx \int_{\nu=\nu_0-\frac{\Delta}{2}}^{\nu=\nu_0+\frac{\Delta}{2}} (1-\tau(\nu,p_1,p_2))d\nu = \Delta - Sl_w, \tag{14.20}$$

where S is the line intensity as l_w is the weighted path (here for weak lines), defined as

$$l_w \equiv \frac{1}{g\cos\theta}\int_{p_2}^{p_1} q(p)dp, \tag{14.21}$$

where g is gravity and $q(p)$ is the concentration of the absorber. The quantity Sl_w is known as the *equivalent width* of the line.

In the strong line limit, the transmission is zero near the line centre – thus, all transmission occurs out of the wings. As a result, we can replace the line shape by its far tail form. Assuming a Lorentz line shape, then the optical depth

$$\tau(\nu,p_1,p_2) = -\frac{1}{g\ \cos\theta}\int_{p_1}^{p_2}\frac{S}{\pi}\frac{\delta}{(\nu-\nu_0)^2}q(p)dp = \frac{S}{\pi}\frac{\delta_0}{(\nu-\nu_0)^2}l_p \tag{14.22}$$

where l_p is the pressure-weighted path, defined as

$$l_p \equiv \frac{1}{g\ \cos\theta}\int_{p_2}^{p_1}\frac{p}{p_0}q(p)dp. \tag{14.23}$$

Inserting this into the expression for the transmission function, we find

$$\mathcal{T}(p_1,p_2) \approx \int_{\nu=\nu_0-\frac{\Delta}{2}}^{\nu=\nu_0+\frac{\Delta}{2}} \exp\left[-\frac{S}{\pi}\frac{\delta_0}{(\nu-\nu_0)^2}l_p\right]d\nu \approx \Delta - 2\sqrt{S\delta_0 l_p}. \tag{14.24}$$

In the strong-line transmission function expression above, the term $2\sqrt{S\delta_0 l_p}$ is now the strong-line *equivalent width*. Importantly, we find that the band-averaged transmission is not exponential in path.

15 Radiative transfer III: Scattering

Reading for Day 15: Pierrehumbert Principles of Planetary Climate, Section 5.1-5.2.

We now turn to introduce the effects of scattering on the propagation of electromagnetic radiation through an atmosphere. The full solution to this problem is significantly more difficult than for radiative transfer neglecting scattering, as scattering couples the flux at a given angle with the fluxes from other angles (Pierrehumbert, 2010). A schematic of the complexity of the scattering problem (even in the idealised plane-parallel geometry) is shown in Figure 15.1. As before, we must consider the direct stellar beam as well as

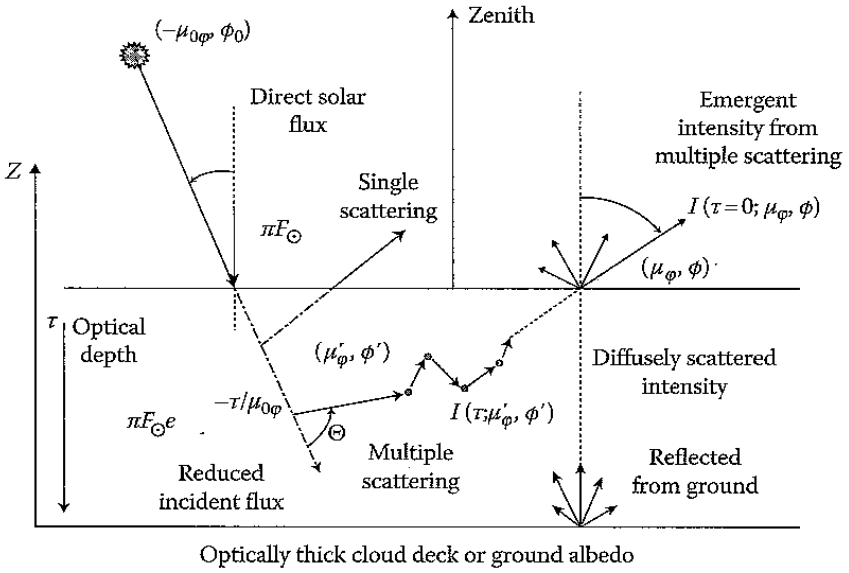


Figure 15.1: Schematic of the plane-parallel scattering problem. Figure adapted from Seager (2010).

emission from the ground and atmosphere. However, now we must also consider single and multiple scattering of the direct stellar beam, as well as reflected light from the ground and atmosphere.

In this lecture, we will introduce key concepts relevant to scattering, and discuss their application to the study of planetary atmospheres.

15.1 Characterising scattering: key quantities

15.1.1 Efficiency and extinction

Let us consider a monochromatic beam of light with flux F travelling in one direction, and its resulting absorption by atmospheric gases as well as scattering by gases and aerosols[29]. We can define the absorption cross section, denoted $\chi_{\rm abs}$, as the rate at which energy is taken out of the beam by absorption. Similarly, we can define the scattering cross section

[29]Here I am using aerosol as a generic term including both condensate clouds, photochemical hazes, dust, or other liquid/solid species that scatter radiation – see `https://www.planetary.org/articles/0324-clouds-and-haze-and-dust-oh-my` for a blog post with more detail on this definition and how it compares to others used in the community.

χ_{scat} as the rate at which energy is scattered into other directions. Thus, the rate of energy absorption is $F\chi_{\text{abs}}$, and the rate of scattering of energy into other directions is $F\chi_{\text{scat}}$.

Note that the absorption and scattering cross sections can be much larger than the actual physical cross-sectional area of a given particle. We can define the scattering efficiency, Q_{scat} as the ratio of the scattering cross section to the actual cross-sectional area of the scattering particle,

$$Q_{\text{scat}} = \frac{\chi_{\text{scat}}}{\pi r^2}, \tag{15.1}$$

where r is the radius of the particle. Similarly, we can define the absorption efficiency as

$$Q_{\text{abs}} = \frac{\chi_{\text{abs}}}{\pi r^2}. \tag{15.2}$$

The total amount of light taken out of the beam is known as the *extinction*, where $\chi_{\text{ext}} = \chi_{\text{abs}} + \chi_{\text{scat}}$ and thus $Q_{\text{ext}} = Q_{\text{abs}} + Q_{\text{scat}}$.

15.1.2 Single-scattering albedo

The single-scattering albedo is the ratio of the flux of the incident beam lost via scattering to the net flux lost by both scattering and absorption. This can be defined either for a particle, or for the medium as a whole. The single scattering albedo for a particle is defined as

$$\omega_{p,0} = \frac{\chi_{\text{scat}}}{\chi_{\text{scat}} + \chi_{\text{abs}}} = \frac{Q_{\text{scat}}}{Q_{\text{scat}} + Q_{\text{abs}}}. \tag{15.3}$$

For a medium as a whole, we can first define the net absorption coefficient

$$\kappa \equiv \sum_i \kappa_i q_i, \tag{15.4}$$

where q_i is the mass concentration of the absorber and $\kappa_i \equiv \frac{\chi_{\text{abs},i}}{m_i}$, where m_i is the mean mass of the substance. The single-scattering albedo for the medium as a whole is then

$$\omega_0 = \frac{\sum_i \frac{q_i \chi_{\text{scat}}}{m_i}}{\kappa + \sum_i \frac{q_i \chi_{\text{scat}}}{m_i}}. \tag{15.5}$$

Note that of course the total single scattering albedo of the medium ω_0 depends both on the concentrations of absorbers and scatterers. For instance, if the mass mixing ratio of a greenhouse gas increases, then ω_0 will decrease even if the cloud particle mass mixing ratio is kept fixed.

15.2 Characterising scattering: directionality

15.2.1 Scattering angles

Figure 15.2 shows the two angles relevant for the problem. First, the propagation angle θ is the angle that the spectral irradiance I is travelling on. Meanwhile, the scattering angle Θ is the angle at which a given particle scatters the beam from an incoming radiance I_{inc} to a scattered radiance I_{sca}.

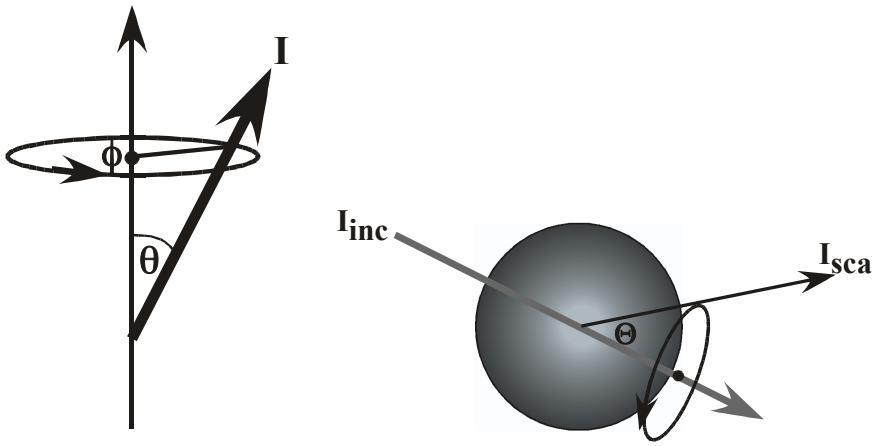


Figure 15.2: Schematic showing definition of propagation angles (left) and scattering angles) right). Figure adapted from Pierrehumbert (2010).

15.2.2 Phase function

The phase function describes the change in direction of the incident intensity relative to the outgoing intensity – effectively, this is the 3D scattering probability. The phase function is a function of the scattering angle Θ and is normalised such that

$$\frac{1}{4\pi}\int_{\Omega} P(\Theta)d\Omega = 1. \tag{15.6}$$

For a spherically symmetric scattering particle, the radiance $I(\vec{\omega})$ is thus related to the incident radiance I_0 as

$$I(\vec{\omega}) = I_0 \frac{\chi_{\text{scat}}}{4\pi r^2} P(\Theta). \tag{15.7}$$

Thus, the direction of the outwelling radiance depends only on the scattering angle Θ relative to the direction of the incident beam. The phase function for isotropic scattering $P(\Theta) = 1$, as the scattered radiation is distributed uniformly over all angles. Some examples of the angular patterns of scattered intensity (i.e., the phase functions) of a molecule and particles with varying sizes is shown in Figure 15.3.

We will discuss the scattering properties of the molecule (following Rayleigh scattering theory) and larger species with varying particle size (following Mie scattering theory) below.

15.2.3 Asymmetry factor

The asymmetry factor defines how anisotropic scattering is, and is defined as

$$g \equiv \frac{1}{4\pi}\int_{\Omega} P(\Theta)\cos\Theta d\Omega. \tag{15.8}$$

Thus, if the scattering is isotropic (such that $P(\Theta) = 1$), then the asymmetry parameter $g = 0$. g varies between -1 and 1, and is positive for forward scattering and negative for backward scattering. If the scattering is purely forward $g = 1$, or if it is purely backward $g = -1$.

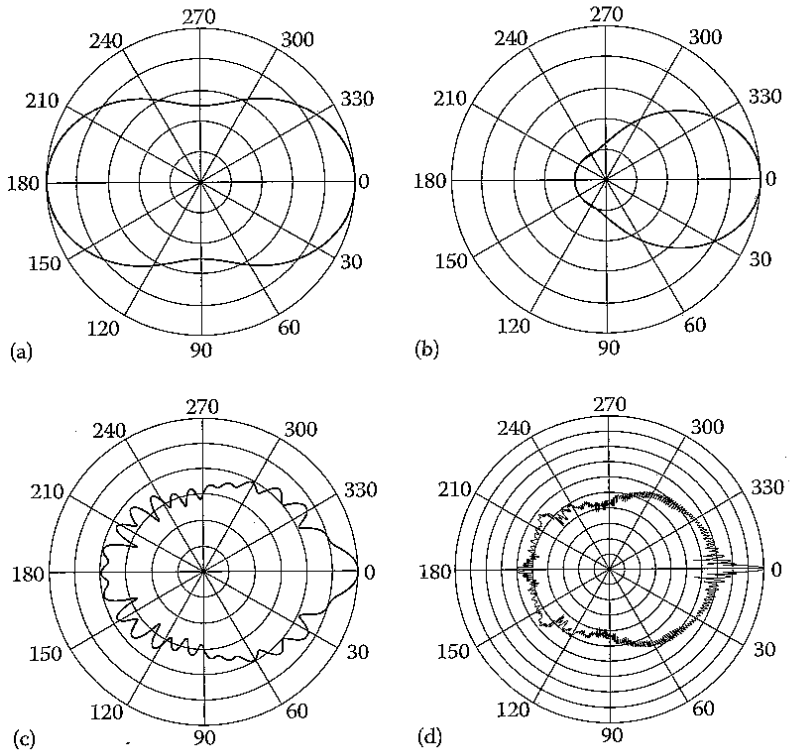


FIGURE 5.12 Angular patterns of the scattered intensity for spherical particles of different sizes corresponding to an incident radiation with a wavelength of 500 nm. The spheres are formed by water (refractive index 1.337724) immersed on air (refractive index 1.0002882). Particle size: (a) molecule (Rayleigh scattering), (b) 0.1 μm, (c) 1 μm, (d) 10 μm. Light is directed from left to right. The radial axes are not to scale (logarithmic representation for (c) and (d)). (Courtesy of Santiago Pérez-Hoyos, UPV-EHU, Bilbao, Spain.)

Figure 15.3: Figure adapted from Seager (2010).

15.3 Rayleigh scattering

Rayleigh scattering theory applies for molecules or any particle that is *much* smaller than the wavelength of light. The Rayleigh scattering cross section (see Goody & Yung, 1989 for a derivation) is

$$\chi_{\rm scat} = \frac{128\pi^5}{3}\frac{\alpha_p^2}{\lambda^4}, \tag{15.9}$$

where λ is wavelength. α_p is the polarizability, which is related to the ratio of the electric field to the induced dipole moment (and is only a weak function of wavelength),

$$\alpha_p = \frac{3}{4\pi n}\left(\frac{m^2-1}{m^2+2}\right), \tag{15.10}$$

where n is the number density of Rayleigh scatterers and m is the average refractive index of these scatterers. In general, the Rayleigh scattering cross section is strongly dependent on wavelength, with short wavelengths scattering much more strongly than longer wavelengths following $\chi_{\rm scat} \propto \lambda^{-4}$.

The phase function for Rayleigh scattering is

$$P(\Theta) = \frac{3}{4}\left(1 + \cos^2\Theta\right). \tag{15.11}$$

The phase function for Rayleigh scattering is shown in Figure 15.4 (as well as above in Figure 15.3).

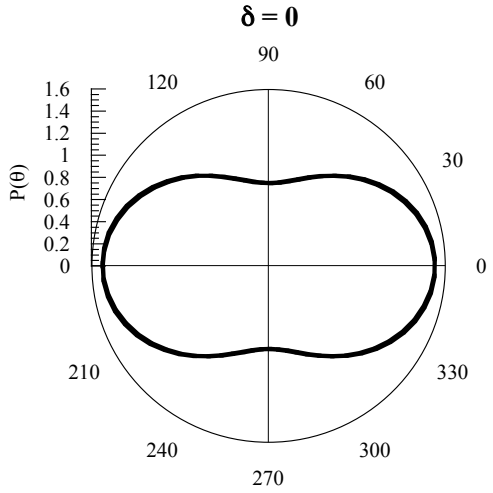


Figure 15.4: Phase dependence of Rayleigh scattering. Figure adapted from Pierrehumbert (2010).

Question for the class: What is the approximate asymmetry parameter for Rayleigh scattering? Hint: study Figure 15.4 carefully.

Answer: Note that Rayleigh scattering is symmetric, with $g = 0$.

15.4 Mie scattering

Mie theory provides the scattering and absorption efficiencies Q_{scat} and Q_{abs} for a particle with any radius r relative to the wavelength of light λ. We will not derive Mie theory here, but assuming that the particle is a sphere with a uniform index of refraction n embedded in a medium with a uniform index of refraction one can solve Maxwell's equations inside and outside the spherical particle and apply a continuity condition at the interface and consider the far field as plane waves to derive the resulting scattering cross section χ_{scat}. This scattering cross-section can then be related to the intensity after scattering as

$$I(\Theta) = I_0 \frac{\chi_{\text{scat}} P(\Theta)}{4\pi r^2}. \tag{15.12}$$

Figure 15.5 shows the Mie scattering efficiency factor as a function of size parameter $x = 2\pi r/\lambda$ and refractive index of the scatterer. Importantly, Mie scattering efficiencies typically peak where $r \sim \lambda$. At $x \to 0$, Mie scattering approaches the Rayleigh limit where the particles are much smaller than the wavelength. At large x, $Q_{\text{scat}} \to 2$.

Mie theory further predicts the phase function of a given particulate scatterer. Phase functions for cloud droplets, aerosols, and molecules (in order of decreasing size) are shown in Figure 15.6. The phase function of a molecule is ~ 1 everywhere, as expected due to the relative symmetry of Rayleigh scattering. Meanwhile, large particles have very forward-peaked scattering, as seen both for the cloud droplet and aerosol. Thus, condensates and

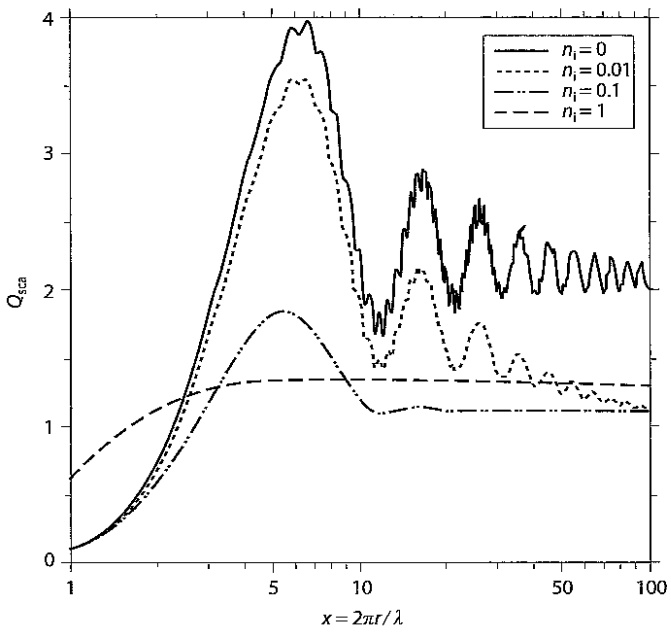


Figure 15.5: Dependence of the Mie scattering efficiency factor $Q_{\rm scat}$ on the size parameter x and the refractive index of the material n_i (line styles). Figure adapted from Seager (2010).

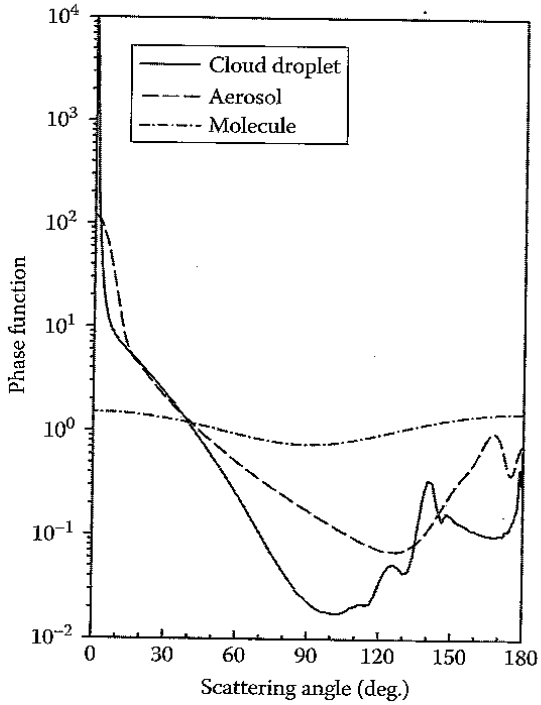


Figure 15.6: Figure adapted from Seager (2010).

FIGURE 5.14 Phase functions (normalized) versus scattering angle computed with the Mie theory for different sizes of spheres when illuminated with a wavelength of 500 nm: cloud droplets (~10 μm), aerosols (~1 μm), molecules (~10^{-4} μm). (Adapted from Hansen, J.E. and Travis, L.D., *Space Sci. Rev.*, 16, 527, 1974. With permission.)

hazes are highly asymmetric scatterers, which must be taken into account in radiative transfer calculations for the impact of aerosols on planetary climate.

References

[1] Bell, T. J., Dang, L., Cowan, N. B., et al. 2021, MNRAS, 504, 3316

[2] Bretherton, C., & Sobel, A. 2003, Journal of the Atmospheric Sciences, 60, 451

[3] Caballero, R., Pierrehumbert, R., & Mitchell, J. 2008, Quarterly Journal of the Royal Meteorological Society, 134, 1269

[4] Coulombe, L.-P., Benneke, B., Challener, R., et al. 2023, Nature, 620, 292

[5] Dang, L., Bell, T. J., Shu, Y. Z., et al. 2025, AJ, 169, 32

[6] de Pater, I., & Lissauer, J. J. 2001, Planetary Sciences (Planetary Sciences, by Imke de Pater and Jack J. Lissauer, pp. 544. ISBN 0521482194. Cambridge, UK: Cambridge University Press, December 2001.)

[7] Esposito, L. W., Knollenberg, R. G., Marov, M. I., Toon, O. B., & Turco, R. P. 1983, in Venus, ed. D. M. Hunten, L. Colin, T. M. Donahue, & V. I. Moroz, 484–564

[8] Fortney, J. J. 2005, MNRAS, 364, 649

[9] Gao, P., Wakeford, H., Moran, S., & Parmentier, V. 2021, Journal of Geophysical Research: Planets, 126, e06655

[10] Goody, R. M., & Yung, Y. L. 1989, Atmospheric radiation : theoretical basis (2nd ed., New York, NY: Oxford University Press, 1989)

[11] Guillot, T. 2010, Astronomy and Astrophysics, 520, A27

[12] —. 2019, arXiv e-prints:1908.02092

[13] Guillot, T., Burrows, A., Hubbard, W., Lunine, J., & Saumon, D. 1996, The Astrophysical Journal Letters, 459, L35

[14] Hammond, M., & Pierrehumbert, R. 2018, The Astrophysical Journal, 869, 65

[15] Hanel, R. A., Conrath, B. J., & Jennings, D. E. 1992, Exploration of the Solar System by Infrared Remote Sensing (Exploration of the Solar System by Infrared Remote Sensing, ISBN 0521326990, Cambridge University Press, 1992.)

[16] Held, I., & Hou, A. 1980, Journal of the Atmospheric Sciences, 37, 515

[17] Holton, J., & Hakim, G. 2013, An introduction to dynamic meteorology, 5th edn. (New York, NY: Academic Press)

[18] Ingersoll, A. P. 1969, Journal of the Atmospheric Sciences, 26, 1191

[19] Kaspi, Y., & Showman, A. 2015, The Astrophysical Journal, 804, 60

[20] Kopparapu, R., Wolf, E., Arney, G., et al. 2017, The Astrophysical Journal, 845, 5

[21] Kreidberg, L. 2017, Exoplanet Atmosphere Measurements from Transmission Spectroscopy and Other Planet Star Combined Light Observations, 100

[22] Mousis, O., Atkinson, D. H., Cavalié, T., et al. 2018, Planet. Space Sci., 155, 12

[23] Peixoto, J., & Oort, A. 1992, Physics of Climate (New York: American Institute of Physics)

[24] Pierrehumbert, R. 2010, Principles of Planetary Climate (Cambridge: Cambridge University Press)

[25] Pierrehumbert, R., & Hammond, M. 2019, Annual Review of Fluid Mechanics, 51, 275

[26] Pierrehumbert, R. T. 2010, Principles of Planetary Climate

[27] Robinson, T. D., & Catling, D. C. 2014, Nature Geoscience, 7, 12

[28] Salby, M. L. 1996, San Diego CA Academic Press Inc International Geophysics Series, 61

[29] Sanchez-Lavega, A. 2010, An introduction to planetary atmospheres, 1st edn. (Boca Raton, FL: CRC Press)

[30] Seager, S. 2010, Exoplanet Atmospheres (Princeton: Princeton University Press)

[31] Seager, S. 2010, Exoplanet Atmospheres: Physical Processes

[32] Showman, A., & Guillot, T. 2002, Astronomy and Astrophysics, 385, 166

[33] Showman, A., & Polvani, L. 2011, The Astrophysical Journal, 738, 71

[34] Showman, A., Tan, X., & Parmentier, V. 2020, Space Science Reviews, 216, 139

[35] Showman, A., Wordsworth, R., Merlis, T., & Kaspi, Y. 2013, Comparative Climatology of Terrestrial Planets, ed. S. Mackwell, A. Simon-Miller, J. Harder, & M. Bullock (Tucson, AZ: University of Arizona Press)

[36] Turbet, M., Fauchez, T. J., Leconte, J., et al. 2023, A&A, 679, A126

[37] Yang, J., Cowan, N., & Abbot, D. 2013, The Astrophysical Journal Letters, 771, L45